Face selection for slowly switching Markov chains

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Abstract

We consider a Markov chain on a finite state space that rarely switches states, and ask which states the most likely vector of occupation counts uses. Staying in a small set costs probability because the chain must never leave it. A larger set is harder to leave, but has more count vectors to share its mass. When the chain makes of order the logarithm of its length switches, these costs compete at the same order. For every start law, the winning sets minimize a sum of their exit rates and dimensions, and a convex hull gives the order in which they win. On each irreducible set with positive start mass, a local limit theorem gives an explicit constant, expressed through spanning trees. This locates crossings up to an error that tends to zero and gives the next correction. A directed three-cycle shows how reducibility changes the logarithmic term. On the complete graph, with the uniform start, every set ties at first order. Explicit constants resolve the tie, and the mode jumps once, from a single state to all states. A nonuniform start or a weak field can instead let an intermediate set win, and a suitable field makes every size win in turn. Near a single state, a different scale appears. A product of Laguerre polynomials locates the crossing of a vector with a few rare counts and the vertex, with a uniform error bound.

Keywords: occupation times of Markov chains; mode; principal Dirichlet eigenvalue; quasi-stationary distribution; local limit theorem; persistent random walk; correlated random walk; Potts chain; spanning trees.
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1 Introduction

Which states does the most likely occupation-count vector of a slowly switching Markov chain use? Fix a generator Q on d states and a start law, and take N steps with transition matrix $I + (T/N)Q$. The occupation counts $n_i = \#\{t \le N : X_t = i\}$ have support $F = \{i : n_i > 0\}$, which we call a face. Write M_F for the largest probability on that exact support and λ_F for its principal exit rate (Definition 2.1). A face is admissible if a path of positive probability has exactly that support. Put $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$.

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1.1 Face selection

Theorem A (first-order face selection)**.** *Fix the rates q_{ij} and the law of X_1, and let $T = \tau \log N$ with $\tau > 0$ fixed. Every admissible face F has $M_F = N^{-\Phi_\tau(F)+o(1)}$. For large N the support of every most likely composition is an admissible face that minimizes Φ_τ among the admissible faces. If the chain can pass between any two states of each such minimizing face without leaving it, then the visit fractions of every most likely composition are within $O(1/\log N)$ of the quasi-ergodic law of its support (Definition 2.1).*

Theorem A is Theorem 3.3(ii)–(iv). Staying in F costs about $e^{-T\lambda_F}$, while sharing its mass among count vectors costs about $N^{-(|F|-1)}$, up to powers of $\log N$. At $T = \tau \log N$ these costs compete. The complete graph is the most degenerate case, since every face ties at first order. Figure 1 shows what the next order decides.

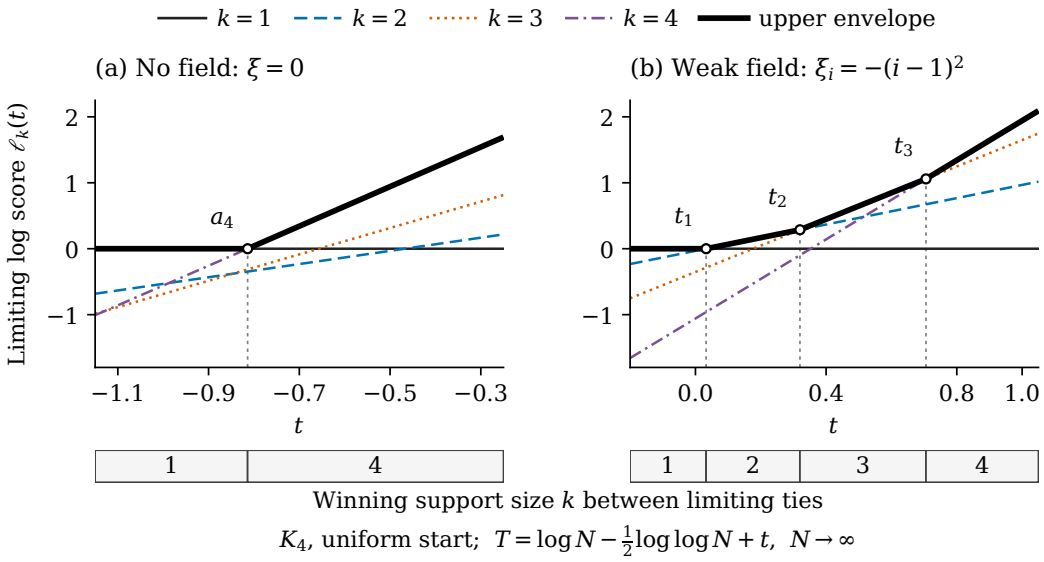


Figure 1: Support selection on the complete graph with four states, unit rates and uniform start, in the window $T = \log N - \frac{1}{2}\log\log N + t$. Thin lines are limiting log face maxima relative to an untilted vertex, and the bold curve is their upper envelope. (a) Without a field the winner jumps from one state to all four. (b) Weighting a composition by $e^{\langle \xi, n\rangle/N}$, with $\xi_i = -(i-1)^2$, lets every size win in turn. The scores and marked crossings are derived in Section 5. Field weights are unnormalized, and marked ties concern the limit, not finite-N equalities.

What is new. Hervé and Ledoux [24] give quantitative density local limits, while occupation large deviations describe exponential costs [19, 23, 4, 32]. Our question compares individual count probabilities across supports with different lattice dimensions. We control their global maxima, obtain relative killed-face estimates and their discrete approximation, and compute constants that resolve first-order ties. The complete graph makes the need for these constants visible.

Scope. The sharp local estimates assume an irreducible face with positive start mass and stay away from its boundary. We do not prove uniform boundary estimates, general sharp expansions on reducible faces, or uniqueness of crossings for arbitrary chains. The first-order support rule still allows reducible faces.

Theorem 3.4 reads the first-order winners from the convex hull of $(\lambda_F, |F|-1)$, and Section 3.1 gives a four-state cycle example. A small-set mode trades the cost of leaving against the number of count vectors sharing its mass. As τ increases the winning size can only grow. The exact three-state example in Figure 2 shows this competition at finite N.

1.2 Second order and the complete graph

First order fixes a crossing only up to a factor $1+o(1)$. To locate it up to $o(1)$ we need the constant in front. Say that a face F satisfies (H) if the chain can move between any two states of F without leaving F, and the start gives F positive mass. For such faces we prove a local limit theorem inside the face (Theorem 4.4), and it gives the following result.

Theorem B (face maxima and crossings)**.** *Fix the rates q_{ij} and the law of X_1.*

(i) Let F satisfy (H)*, with $k=|F|$ states. Uniformly for $1 \le T \le N^{1/3}$,*

$$M_F = N^{-(k-1)}\, T^{(k-1)/2} K_F\, e^{-T\lambda_F}\big(1+O(1/T+T^2/N)\big),$$

where $K_F>0$ is the explicit constant of Definition 4.1 (for $k=1$ it is the start mass of the state).

(ii) Let F and G satisfy (H)*, and suppose that F has $\Delta \ge 1$ more states than G and an exit rate smaller by $\delta>0$. For large N the equation $M_F = M_G$ has a root T in $[1, N^{1/3}]$, and every such root satisfies*

$$\delta T + \frac{\Delta}{2}\log T = \Delta \log N - \log\frac{K_F}{K_G} + O\Big(\frac{1}{\log N}\Big).$$

Part (i) is Theorem 4.6 and part (ii) is Theorem 4.7. The factor $T^{(k-1)/2}$ appears because the visit fractions fluctuate on the scale $T^{-1/2}$ in each of the $k-1$ free directions. Solving the equation in (ii) gives $T = \frac{\Delta}{\delta}[\log N - \frac{1}{2}\log\log N + O(1)]$. So the coefficient $\frac{1}{2}$ of $\log\log N$ inside the bracket, which appears in the crossings of the two-state case and of the complete graph, holds for every such pair, whatever the graph, the rates, the sizes or the start, and without reversibility.

On the complete graph with the uniform start, $Q = J - dI$ with J the all-ones matrix, and the constants are explicit. A face of k states has $K_F = \frac{1}{d}k^{k+1/2}(4\pi)^{-(k-1)/2}$, a mass over a Gaussian volume, and the k^{k-2} spanning trees of the complete graph K_k enter through it. Put $a_k = \frac{1}{2}\log(4\pi) - \frac{k+1/2}{k-1}\log k$, and write V_N for the probability of a vertex. A composition is balanced if its positive coordinates differ by at most 1.

Theorem C (the complete graph)**.** *Let $Q = J - dI$ and let the start be uniform.*

(i) Let F be a face of $2 \le k \le d$ states. For every $N \ge k$ the equation $M_F = V_N$ has exactly one root $T_{N,k}$ in $(0, N/(d-1))$, and the switch probability $T_{N,k}/N$ at the root decreases strictly as N grows. As $N \to \infty$,

$$T_{N,k} + \tfrac{1}{2}\log T_{N,k} = \log N + a_k + \frac{1}{8T_{N,k}} + O\Big(\frac{1}{(\log N)^2}\Big).$$

(ii) For every N and every $0 < T \le N/d$, the most likely compositions with a given support are exactly the balanced ones.

(iii) Since $a_2 > a_3 > \cdots > a_d$, for large N and T up to any fixed multiple of $\log N$ the mode changes exactly once, from the d vertices straight to the balanced compositions with d positive coordinates.

Part (i) is Theorem 5.4, where the term $1/(8T)$ comes from the next term of the local law (Proposition 4.15), which we evaluate by hand for every complete face. Part (ii) is Theorem 5.5 and part (iii) is Corollary 5.6. On the triangle the middle composition $(\frac{N}{3}, \frac{N}{3}, \frac{N}{3})$ overtakes the vertices while the middles of the edges are still below them by the factor $e^{-(a_2-a_3)} = 0.83$. A start law changes the constant of a face through its start mass, and a weak field changes it through the field averaged over the face. Either can let a face of intermediate size win (Example 5.12 and Corollary 5.14). Near a vertex the scale changes. A composition with one large count b and a few rare counts crosses the vertex at switch odds $\sqrt{y/b}\,(1 + o(1))$, where y is the root of a product of Laguerre polynomials (Theorem 6.2).

The rest of Sections 4 to 7 add the following.

- *Ties.* When several faces tie at first order, Theorem 4.8 describes the window of T in which they compete.
- *The constant.* For reversible faces K_F involves the square root of a weighted count of spanning trees (Proposition 4.2). On a face that is not reversible the single count is replaced by two, a count of oriented spanning trees for the flow of the tilted chain and a count of spanning trees for the symmetrized flow (Proposition 4.9).
- *The next term and the role of* (H). Proposition 4.15 adds the next term, of relative order $1/T$, to the local law, to M_F and to the crossing equation. Without (H) the coefficient of $\log\log N$ can change. On a directed 3-cycle it is 1 at one crossing and 0 at the next (Proposition 4.14).
- *A weak field.* On a face satisfying (H), a field of size ξ_i/N per visit to state i multiplies its leading constant by $e^{\langle \xi, \alpha_F^* \rangle}$. If every face tied at first order is irreducible, these factors shift the tie-window lines (Proposition 4.16).
- *Examples.* Section 7 computes all the constants on the 4-cycle and lists the winners on other graphs.

1.3 Related work

This subsection places the results among earlier work on occupation measures and on exact laws of transition counts. Section 3 introduces a rate I_F for the visit fractions inside a face. It is the Donsker–Varadhan rate of the occupation measure of the chain killed on leaving F. Dupuis and Liu give this rate explicitly for reversible jump processes [20, Section 3.1], and on a finite chain their formula agrees with Proposition 3.1(c). On the complete graph it is $d - (\sum_i \sqrt{\alpha_i})^2$ [20, Eq. (3.2)]. Chen and Deng [10] study large deviations for absorbing Markov chains and use them to derive a variational representation of the principal Dirichlet eigenvalue. Guillin, Nectoux and Wu [23] prove a large deviation principle for the occupation measure of a process conditioned to stay in a set. For reversible processes they show that the exit rate of the set is the least value of the Donsker–Varadhan rate over laws on the set (their Cor. 2). In general they prove an inequality between their rate, the Donsker–Varadhan rate and the exit rate, and they conjecture equality (their (2.15)). For a finite chain with Q_F irreducible, the fact that λ_F is the least value of I_F is the variational formula of Donsker and Varadhan [19], which needs no reversibility. Proposition 3.1(a) gives a short proof in our notation. This does not address the conjecture of Guillin, Nectoux and Wu, which concerns their conditional rate. The quasi-ergodic law goes back to Darroch and Seneta [13, 14], the book of Collet, Martínez and San Martín [11] is the standard reference for quasi-stationary and quasi-ergodic laws, and Colonius and Rasmussen [12] treat the reducible case. Dietz and Sethuraman [18] study chains that switch from i to j at time n with probability

$G(i,j)/n$, where every $G(i,j) > 0$. Their chains also make about $\log n$ switches, but spread log-uniformly in time, and their occupation fractions have a limit law with full support on the simplex. In [17] they prove large deviations for chains whose transition matrices converge to a reducible matrix. Bouguet and Cloez [6] study empirical-measure limits and fluctuations for freezing finite-state chains, including the regime with switch probabilities of order $1/n$. In these models the rate changes along the trajectory. Here it is fixed during each path of length N, and changes with N instead.

Exact laws of transition counts go back to Whittle [41], Dawson and Good [15] and Goodman [22]. Polettini [37] and Carugno, Vivo and Coghi [8] derive them from the BEST theorem and the matrix-tree theorem. The expression of Carugno, Vivo and Coghi for the moment generating function is exact at every time, and they use it for large deviations, where every count grows linearly. For two states the composition has the Markov binomial law, and Dekking and Kong [16] find when it has one, two or three modes. Edwards [21] introduced the Markov binomial distribution, for a stationary chain on two outcomes that moves from i to j with probability $\theta\delta_{ij} + (1-\theta)\nu_j$, where ν is its law at each trial, and he wrote the generating function of the number of successes as a product of matrices. Wang and Yang extended this chain to finitely many states, and we follow the account of their work in Johnson, Kotz and Balakrishnan [27, Chapter 35, Section 13.2]. Their chain starts from ν and moves from i to j with probability $\theta\delta_{ij} + (1-\theta)\nu_j$, and the generating function of its vector of category counts is a product of matrices. The complete graph with the uniform start and a stay probability $p \geq 1/d$ is the case where ν is uniform and $\theta = p - T/N$. So its exact law is a special case of this theory, and we do not claim it as new. A start law that is not uniform keeps the uniform switching, so it gives a Wang–Yang chain only when it is uniform. Hsiau [25, Theorem 1] proves compound Poisson limits for sums over multistate chains with one nearly absorbing state, and Wang and Tang [40] prove Poisson-type limits for additive processes on Markov chains. Kassan-Ogly and Filippov [29] solve the Potts chain in a field in the thermodynamic limit. Their paper was not accessible to us, but its abstract and introduction are free online, and we use nothing beyond these. Stadje [38, Theorems 1 and 2] gives the generating functions of the joint numbers of visits of a finite chain, and limit laws when one state becomes nearly absorbing and the chain leaves it with probability of order $1/N$ [38, Theorems 3 and 4]. The other rows of his transition matrix stay fixed, so the counts of the other states stay bounded. We group the words by runs instead (Lemma 2.3), which suits a regime with $O(\log N)$ switches. In continuous time, Brydges, van der Hofstad and König [7] give the joint density of the local times of a chain on the event that its range is a given set, and Huang, Kious, Sidoravicius and Tarrès [26] write it with Bessel functions and spanning trees for symmetric rates. Brydges, van der Hofstad and König use their formula for large deviation upper bounds. Theorem 4.4 gives the leading term of the density, with its constant, in our regime. For a fixed geometrically ergodic chain and a bounded real functional, Kontoyiannis and Meyn [31, Theorem 6.3] give exact large deviation asymptotics of Bahadur–Rao type. Theorem 4.4 is a local form for the whole vector of counts, for a chain that is killed on leaving F and whose switch probabilities shrink with N, uniformly in T and with an explicit constant. Hervé and Ledoux [24] prove a quantitative local limit theorem for densities of finite-state Markov additive processes. Their application to local times assumes irreducibility of the generator and of its one-state-deleted subgenerators. Our continuum estimates are related to this theory. We need relative estimates on killed faces, their discrete approximation, and a comparison of maxima across supports. Bertini, Gabrielli and Landim [4] and Landim [32] expand the level-two rate functions of finite chains whose rates depend on a parameter. They describe the time scales of such chains but do not ask where the mode of the composition sits. Kim and Landim [30] extend this to the joint

rate function of the empirical measure and the flow [3], and they give conditions under which the rate function of the occupation times determines the chain. Remark 4.12 says how the rate I_F of a killed chain relates to their work and to that of Bertini, Faggionato and Gabrielli [2, 3].

The question grew out of Problem 131 of Kagey's Open Problem Collection [28], a two-direction crossing problem. The present paper incorporates the symmetric-model preprint [36].

1.4 Plan of the paper

Section 2 fixes the notation and gives the exact law of the composition as a sum over runs. Section 3 works out four states on a cycle and then proves first-order face selection (Theorems 3.3 and 3.4). Section 4 proves the local limit theorem inside a face, the face maxima, the crossing equation and the tie windows. It then gives a tree formula for the constant, shows on a directed 3-cycle what can change without (H), adds the next term and treats a weak field. Section 5 recovers the two-state walk and treats the complete graph, with a start law and a weak field. Section 6 treats compositions near a vertex of the complete graph. Section 7 computes the 4-cycle with all its constants and lists the winners on other graphs, and Section 8 describes the computer checks and the open questions. Appendix A proves the local law, the uniform upper bound of Lemma 3.2, the first-order bounds for reducible faces and the forest formula of Lemma A.3. Appendix B proves the leading Laplace lemma and states its next term, with the coefficient formula. Appendices C and D hold the proofs for the complete graph and near a vertex. The supplement [34] has the proofs for the table of outcomes and the full proof for the directed 3-cycle, followed by the higher-order Laplace proof and the detailed proof of Proposition 4.15.

We quote a few results of [35] and [36]. From [35] we use the run count, the fact that the polynomials $S_{a,b}$ of Section 2.1 grow toward the middle, the uniform Bessel bound (recalled in (2.1)), the clipped products of its Lemma 3.2 and the expansion of z_N. From [36] we quote, without proof, only results that we do not use, namely its Bessel majorant and its crossover near the boundary of the simplex (Remark 6.3).

2 Setting, notation and the exact law

This section fixes the chain and its notation, says which faces can be the support of a path (Lemma 2.2), and writes the law of the composition as a sum over runs (Lemma 2.3), which every later section uses.

2.1 Setting and notation

The chain. Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$. Its rates are $q_{ij} \geq 0$ for $i \neq j$, the exit rates are $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$, and $q_{\max} = \max_i q_i$. We also write q for the vector (q_i). (The scalar $q = 1 - p$ of the two-state walk appears only where we recall [35], and the scalar q of Proposition 5.1 is a common exit rate.) The start μ is a probability law on $[d]$, and $\mu(F) = \sum_{i \in F} \mu_i$. For $N \geq 2$ and $T > 0$ put

$$h = \frac{T}{N}, \qquad L = \log N, \qquad \tau = \frac{T}{L}.$$

The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + hQ$. So at each step it jumps from i to j with probability hq_{ij}, and N steps look like the continuous-time chain with generator Q run for time T. We assume $hq_{\max} \leq \frac{1}{2}$ unless we say otherwise. The *composition* is the vector of visit counts $n = (n_1, \dots, n_d)$, with $n_i = \#\{t \leq N : X_t = i\}$, so that $n_1 + \cdots + n_d = N$. We write $P_N(n)$ for the probability that the chain has composition

n, $\operatorname{supp} n = \{i : n_i > 0\}$ for its support, and $\alpha = n/N$ for the visit fractions.

Faces. A face is a nonempty set $F \subseteq [d]$, of size $k = k_F = |F|$. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F, and Δ_F° the set of those that are positive on F. We number the states of F as $1, \dots, k$ and use the coordinates $(\alpha_2, \dots, \alpha_k)$ on Δ_F and $(t_2, \dots, t_k)$ on $\{t \in \mathbb{R}^F : t_1 + \cdots + t_k = T\}$. We write $x \cdot y = \langle x, y\rangle = \sum_{i\in F} x_i y_i$ and $x \circ y = (x_i y_i)_{i\in F}$. Let $A_F = (q_{ij})_{i,j\in F}$ with zero diagonal, and $Q_F = A_F - \operatorname{diag}(q_i)_{i\in F}$. Since q_i includes the rates of leaving F, Q_F generates the chain killed when it leaves F. The switching graph G_F has an edge $i \to j$ whenever $i, j \in F$, $i \neq j$ and $q_{ij} > 0$.

Perron roots. A square matrix B is Metzler if its off-diagonal entries are nonnegative, and then its Perron root $\varrho(B)$ is its largest real eigenvalue. If B is irreducible, $\varrho(B)$ is simple and has positive left and right eigenvectors [1].

Definition 2.1 (faces)**.** *Let F be a face. We call F* irreducible *if G_F is strongly connected. In particular every singleton is irreducible. We call F* admissible *if for some N some path $x_1 \cdots x_N$ of positive probability has support exactly F (for $0 < hq_{\max} < 1$ this does not depend on h). We say that F satisfies* (H) *if F is irreducible and $\mu(F) > 0$. The* exit rate *of F is $\lambda_F = -\varrho(Q_F)$, so $\lambda_{\{i\}} = q_i$. If F is irreducible, r_F and l_F are right and left eigenvectors of Q_F for $-\lambda_F$ with positive entries and $\langle l_F, r_F\rangle = 1$, and $\alpha_F^* = l_F \circ r_F$ is the* quasi-ergodic law *of F [13, 14]. The* face maximum *is $M_F = \max\{P_N(n) : \operatorname{supp} n = F\}$, with $\max \emptyset = 0$.*

We call λ_F the exit rate for the following reason. If F is irreducible, then $e^{TQ_F} r_F = e^{-T\lambda_F} r_F$ and r_F is positive. So the chance that the continuous-time chain started at a state of F stays in F up to time T, which is a row sum of e^{TQ_F}, lies between $e^{-T\lambda_F} \min r_F / \max r_F$ and $e^{-T\lambda_F} \max r_F / \min r_F$. If F is reducible, we order the strongly connected components $C_1, \dots, C_r$ of G_F so that every edge between components goes forward. Then Q_F is block triangular with diagonal blocks Q_{C_j}, and $\lambda_F = \min_j \lambda_{C_j}$. In all cases $0 \le \lambda_F \le \min_{i\in F} q_i$, since the rows of Q_F have sums at most 0 and a Perron root is at least every diagonal entry. For a face F satisfying (H), the quasi-ergodic law α_F^* is the long-run law of the visit fractions of the chain conditioned to stay in F, and the best composition on F sits near $N\alpha_F^*$ (Theorem 4.6).

Later sections add their own symbols where they are needed, namely the rate I_F (Section 3) and the tilted chain and the constant K_F (Definition 4.1).

The two-state walk. The persistent random walk on $\mathbb{Z}$ makes its first step left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p and switches with probability $q = 1 - p$. Its directions form the chain of Example 2.5(i). As on a Galton board, [35] calls a composition a *bin*. For two states we identify a composition with its number of right steps. For $a, b \ge 1$ let $S_{a,b}(u) = \sum_w u^{s(w)}$, the sum over the words w with a right steps and b left steps, where $s(w)$ is the number of switches of w. Each such word has probability $\frac{1}{2} p^{N-1} (q/p)^{s(w)}$ for the walk with $N = a + b$ steps. So $S_{a,b}(q/p)$ is the probability of composition a divided by the probability $\frac{1}{2}p^{N-1}$ of an endpoint. For $N \ge 2$ let $S_N = S_{\lfloor N/2\rfloor, \lceil N/2\rceil}$, which belongs to a middle composition. It is a nonzero polynomial with nonnegative coefficients and no constant term, so $S_N(u) = 1$ has exactly one positive root u_N, and $p_N = 1/(1 + u_N)$ is the crossing of the middle and the endpoints. Put $z_N = N(1 - p_N)$ and $c_0 = \frac{1}{2}\log(\pi/8)$. By [35], z_N satisfies the crossing equation $z_N + \frac{1}{2}\log z_N = L + c_0 + 1/(8z_N) + R_N$ with an explicit remainder $R_N = O(1/L^2)$. We also use the uniform Bessel bound of [35, Theorem 3.3(i)]. Put $\psi(z) = z(I_0(z) + I_1(z))$, with the modified Bessel functions I_0 and I_1. In our notation that bound says that for every $N \ge 2$ and every $z > 0$

$$\Big(1 - \frac{z(z+2)}{2N}\Big)\psi(z) \le \frac{N}{2}\, S_N\Big(\frac{z}{N}\Big) \le \psi(z). \tag{2.1}$$

Conventions. Unless we say otherwise, Q, μ and the faces under discussion are fixed,

and every constant, every implied constant in $O(\cdot)$ and every threshold N_0 depends only on them and on the compact set $\mathcal{K}$ when one appears. A bound that holds "uniformly for $1 \le T \le N^{1/3}$" holds for all $N \ge N_0$ and all T in that range with one implied constant. Subscripts as in $O_{\mathcal{K}}$ stress a dependence.

2.2 Admissible faces

This subsection says which faces can be the support of a path. All later statements about face maxima are for these faces.

Lemma 2.2 (admissible faces)**.** *Let F be a face and $C_1, \dots, C_r$ the strongly connected components of G_F.* (a) *F is admissible if and only if the components can be numbered so that $\mu(C_1) > 0$ and, for each $j < r$, $q_{xy} > 0$ for some $x \in C_j$ and $y \in C_{j+1}$.* (b) *An irreducible face is admissible if and only if $\mu(F) > 0$. So* (H) *holds exactly for the irreducible admissible faces.* (c) *If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.*

Proof. We translate paths into walks in G_F. Take a path of positive probability with support F and list the states of its maximal runs. This list is a walk in G_F that starts at a state of $\operatorname{supp}\mu$ and visits every state of F. Conversely, every such walk $v_1, \dots, v_J$ is the list of runs of a path of positive probability with support F, as soon as $N \ge J$. (The path follows the walk and then stays in its last state, and a stay has positive probability.) So F is admissible exactly when G_F has a walk that starts in $\operatorname{supp}\mu$ and visits every state of F.

(a) Suppose that there is such a walk. Once it has left a component C for a state $y \notin C$, it cannot come back to C. (If it did, y could be reached from C and C from y, so y would belong to C.) Hence the walk visits each component in one block of consecutive times, and it passes from one block to the next along a single edge. We number the components in the order of these blocks. The walk starts in C_1, so $\mu(C_1) > 0$, and the edge from block j to block $j+1$ is an edge $x \to y$ with $x \in C_j$ and $y \in C_{j+1}$. This is the condition in (a).

Conversely, suppose that the components are numbered as in (a), with edges $x_j \to y_{j+1}$ from C_j to C_{j+1} for $j < r$. Take $y_1 \in C_1$ with $\mu_{y_1} > 0$ and any $x_r \in C_r$. Since C_j is strongly connected, there is a walk inside C_j from y_j to x_j that visits every state of C_j. We join these r walks by the edges $x_j \to y_{j+1}$. The result is a walk that starts in $\operatorname{supp}\mu$ and visits every state of F.

(b) This is (a) with $r = 1$.

(c) If Q has symmetric support, an edge from one component to another would come with an edge back, and the two components would be the same. So no edge joins two components, and the condition in (a) forces $r = 1$. □

So $M_F > 0$ for large N if F is admissible, and $M_F = 0$ otherwise.

2.3 The exact law

This subsection writes the probability of a composition as a sum over the runs of a path, and gives its continuum form. The upper bound of Lemma 3.2 and the local law of Section 4 both start from these sums.

For $m \in \mathbb{Z}^F_{\ge 1}$ put $M = \sum_{i \in F} m_i$ and $W(m) = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}$, the sum over the words $c = c_1 \cdots c_M$ in F with $c_j \ne c_{j+1}$ in which each $i \in F$ occurs m_i times. Such a word lists the states of the runs of a path, and m_i is the number of runs in state i.

Lemma 2.3 (run expansion)**.** *Let $hq_{\max} \le 1$ and let n have support F. Then, with*

$\binom{n_i-1}{m_i-1} = 0$ *for* $m_i > n_i$ *and* $0^0 = 1$,

$$P_N(n) = \sum_{m \in \mathbb{Z}^F_{\geq 1}} W(m)\, h^{M-1} \prod_{i \in F} \binom{n_i - 1}{m_i - 1} (1 - hq_i)^{n_i - m_i}.$$

Proof. We group the paths with composition n by the word of their runs. The states $c_1 \cdots c_M$ of the maximal runs of a path form a word as in $W(m)$, where m_i is the number of runs in state i. The path starts at c_1, which has probability μ_{c_1}. It has $M-1$ switches, of probabilities $hq_{c_j c_{j+1}}$, and $n_i - m_i$ stays in state i, of probability $1 - hq_i$ each. So every path with the word c and composition n has the same probability, namely $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, h^{M-1} \prod_i (1-hq_i)^{n_i - m_i}$. The number of these paths is $\prod_i \binom{n_i-1}{m_i-1}$, because the lengths of the m_i runs in state i can be any of the $\binom{n_i-1}{m_i-1}$ ways to write n_i as an ordered sum of m_i positive parts. Summing over the words c with given m gives $W(m)$, and summing over m gives the claim. □

Lemma 2.3 is checked in Lean (Section 8). For $F = \{i\}$ the only path is i^N, so uniformly for $1 \leq T \leq N^{1/2}$

$$M_{\{i\}} = \mu_i (1 - hq_i)^{N-1} = \mu_i e^{-q_i T} \big(1 + O(T^2/N)\big). \tag{2.2}$$

For $t \in (0,\infty)^F$ let $H(t) = \sum_m W(m) \prod_{i \in F} t_i^{m_i - 1}/(m_i - 1)!$, which is finite since $W(m) \leq k^M q_{\max}^{M-1}$. The next lemma says that $e^{-\langle q,t\rangle} H(t)$ is the density of the occupation times of the continuous-time chain on the event that it stays in F. It is the continuum form of Lemma 2.3, in which the binomial coefficients become the powers $t_i^{m_i-1}/(m_i-1)!$. Lemma 4.5 below shows that the discrete law is close to it.

Lemma 2.4 (the continuum density)**.** *Let* $(Y_s)_{s \geq 0}$ *be the continuous-time chain with generator* Q *and* $Y_0 \sim \mu$*, let* $\mathcal{O}_i = \int_0^T \mathbf{1}\{Y_s = i\}\, ds$*, and let* E_F *be the event that* Y *stays in* F *during* $[0,T]$ *and visits every state of* F*. Then* $\mathbb{P}(E_F,\ \mathcal{O} \in \cdot\,)$ *has the density* $e^{-\langle q,t\rangle} H(t)$ *in* $(t_2, \ldots, t_k)$ *on* $\{t \in (0,\infty)^F : \sum_i t_i = T\}$*, and* $\mathbb{P}(E_F,\ \mathcal{O}/T \in \cdot\,)$ *has the density*

$$f_T(\alpha) = T^{k-1} e^{-T\langle q,\alpha\rangle} H(T\alpha)$$

in $(\alpha_2, \ldots, \alpha_k)$*. For* n *with support* F *and* $t = hn$*,* $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q,t\rangle} H(t)$*. For* $F = \{i\}$*,* $f_T = \mathbb{P}(E_F) = \mu_i e^{-q_i T}$*.*

Proof. We split E_F by the sequence of states that Y visits, and for each sequence we integrate out the holding times. Up to a null set, E_F is the disjoint union of the events E_c that Y visits exactly $c_1, \ldots, c_M$ in this order during $[0,T]$, over the words c in the sums $W(m)$.

The density of the holding times. Fix c, and let s_j be the time that Y spends in its j-th run during $[0,T]$, so that $s_1 + \cdots + s_M = T$. For $j < M$ the j-th run is complete. Its holding time has the density $q_{c_j} e^{-q_{c_j} s_j}$, and it ends with a jump to c_{j+1}, which has probability $q_{c_j c_{j+1}}/q_{c_j}$. The last run only has to last longer than $s_M = T - \sum_{j<M} s_j$, which has probability $e^{-q_{c_M} s_M}$. We multiply these factors. On the set where $s_j > 0$ for every $j \leq M$, the event E_c has the density

$$\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\; e^{-\langle q,\mathcal{O}\rangle}$$

in $(s_1, \ldots, s_{M-1})$, where $\mathcal{O}_i$ is the sum of the s_j with $c_j = i$.

From holding times to occupation times. This density depends on the holding times only through $\mathcal{O}$, so we change variables. For each state $i \neq c_M$ we replace the last holding time in state i by $\mathcal{O}_i$. This change of variables is triangular with unit diagonal, so

it preserves Lebesgue measure. The new variables are the $k-1$ occupation times $\mathcal{O}_i$ with $i \neq c_M$, and for each state i the other $m_i - 1$ holding times in state i (for $i = c_M$, those before the last run). Now fix $\mathcal{O} = t$. The $m_i - 1$ free holding times in state i are positive and their sum is less than t_i, so they range over a simplex of volume $t_i^{m_i-1}/(m_i-1)!$. (For two states and the word $c = 121$ the only free variable is the first holding time, which ranges over $(0, t_1)$, and the factor is t_1.) Integrating the free holding times out, we find that $\mathbb{P}(E_c,\ \mathcal{O} \in \cdot\,)$ has the density $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q,t\rangle} \prod_{i\in F} t_i^{m_i-1}/(m_i-1)!$ in the coordinates t_i with $i \neq c_M$.

Summing over the words. On $\{\sum_i t_i = T\}$ any $k-1$ of the coordinates t_i give the same Lebesgue measure, so we may use $(t_2, \dots, t_k)$ for every c. Summing over the words c with given m gives $W(m)$, and summing over m gives $e^{-\langle q,t\rangle} H(t)$. The substitution $t = T\alpha$ has Jacobian T^{k-1}, which gives f_T, and $T/N = h$ gives the identity for $t = hn$. For $F = \{i\}$ the event E_F says that Y starts at i and does not jump before time T. □

Example 2.5 (the four models)**.** (i) For $d = 2$, $q_{12} = q_{21} = 1$ and uniform μ the chain is the persistent walk, with $p = 1-h$. So $T = N(1-p)$, the faces $\{1\}$ and $\{2\}$ are the endpoints, and Lemma 2.3 is the run count for two states. (ii) For $Q = J - dI$, with J the all-ones matrix, the chain switches to each other state with probability h and keeps its state with probability $p = 1-(d-1)h$. With uniform μ it is the model of [36], whose switch probability r is our h. A word $x_1 \cdots x_N$ then has probability $\frac{1}{d} h^{N-1} (p/h)^{\#\{s<N:\, x_s = x_{s+1}\}}$, so the path law is the one-dimensional d-state Potts chain with free boundaries. Here $q_i = d-1$ for every i, so Lemma 2.3 reads

$$P_N(n) = p^{N-1} \sum_{a\in F} \mu_a S_n^{(a)}(u), \qquad u = \frac{h}{p}, \tag{2.3}$$

for $p > 0$ and $\operatorname{supp} n = F$, where $S_n^{(a)}(u)$ is the sum of u^j over the words with the counts n that start with a and have j switches. Indeed such a word has probability $\mu_a p^{N-1-j} h^j = \mu_a p^{N-1} u^j$. (iii) For $Q = \Pi - I$, with Π stochastic with zero diagonal, $\lambda_F = 1 - \varrho(\Pi_F)$. This is the chain of a sticky hidden Markov model with switch kernel Π. (iv) Let $d = 4$, number the directions east, north, west and south as $0, 1, 2, 3$ modulo 4, and put $q_{j,j\pm1} = r$, $q_{j,j+2} = s$ and μ uniform. Then X_t is the direction of step t of a planar walk that turns left or right with probability rh each and reverses with probability sh. This gives a planar persistent walk. For $s = 0$ and $r = 1$ this is the 4-cycle of Section 3.1. In (i), (ii) and (iv), Q has symmetric support and μ full support, so all admissible faces satisfy (H) by Lemma 2.2.

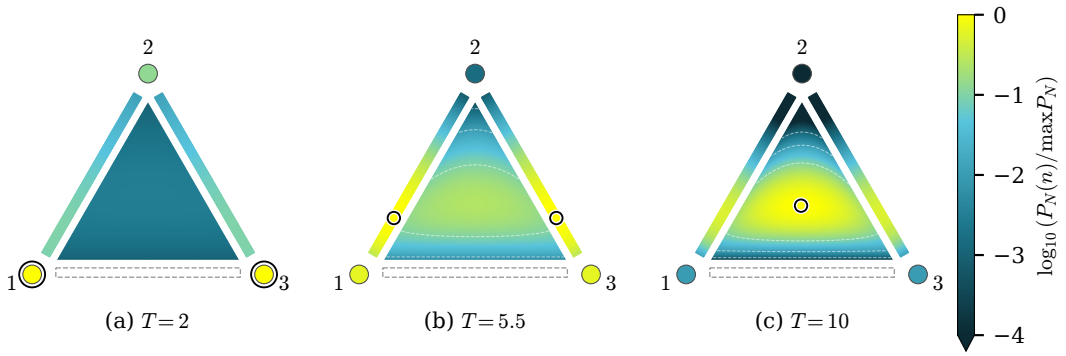
Figure 2 shows exact support competition for the three-state path.

3 First-order face selection

This section proves the first-order rule of the introduction. The mode sits on a face that minimizes Φ_τ (Theorem 3.3), and the winners can be read off a convex hull (Theorem 3.4).

3.1 A worked example with four states on a cycle

Before the general theorem we work out one example by hand. Take $d = 4$ states on a cycle, with rate 1 between neighbors (indices modulo 4), and the uniform start. So at each step the chain moves to each of its two neighbors with probability $h = T/N$ and stays with probability $1 - 2h$. This is the direction of the planar walk that turns but never reverses (Example 2.5(iv) with $r = 1$, $s = 0$). Up to rotation and reflection there are four kinds of faces, namely a vertex, an adjacent pair, an arc of three states and the whole



(a) $T = 2$ (b) $T = 5.5$ (c) $T = 10$

Figure 2: The exact law of the composition for the path $1 - 2 - 3$ (rate 1 on each edge, uniform start, $N = 90$), on the simplex of visit fractions. The faces are drawn apart. The triangle holds the compositions that use all three states, each band holds the compositions that use two states, and each disk is the composition that spends all N steps in one state. The color is the probability of a composition relative to the mode, the thin lines are spaced by a factor of ten, and the rings mark the modes. (a) For small T the mode is at an end of the path. (b) Then it sits on a band, that is, on a pair of neighbors. (c) Then it is inside. The dashed band is the pair $\{1, 3\}$, which the chain cannot use without visiting 2, so it has probability zero. Here λ_F is 1 for an end, $(3 - \sqrt{5})/2$ for a pair of neighbors and 0 for all three states, so to first order the mode moves at $\tau = (1 + \sqrt{5})/2$ and at $\tau = (3 + \sqrt{5})/2$, that is, at $T = 7.3$ and $T = 11.8$. At $N = 90$ it moves near $T = 4.9$ and $T = 8.5$. Corrections of this size are the subject of Section 4.

cycle. (A face such as $\{1, 3\}$ cannot be the support of a path, since the chain cannot pass from 1 to 3 without visiting 2 or 4.) Which kind carries the most likely composition?

For each kind we ask two questions. With what probability does the chain stay in the face for all N steps? Among how many compositions is this probability shared? The best composition of the face gets about the first answer divided by the second. In this subsection T is large and T^2/N is small, the sign $\approx$ means equality up to a bounded factor, and we do not give proofs. The proofs are in Theorems 3.3 and 4.6, and Corollary 7.1 has the constants.

A vertex, say $\{1\}$. The chain stays at state 1 only if it never switches, which has probability $(1 - 2h)^{N-1} \approx e^{-2T}$. So the exit rate is $\lambda_{\{1\}} = 2$. The only composition is $(N, 0, 0, 0)$, so nothing is shared, and $M_{\{1\}} = \frac{1}{4}(1 - 2h)^{N-1} \approx e^{-2T}$.

An adjacent pair, say $\{1, 2\}$. The chain leaves the pair by the switches $1 \to 4$ and $2 \to 3$, while the switches between 1 and 2 keep it inside. So at each step it leaves with probability h, whichever of the two states it is in, and it stays in the pair with probability $(1 - h)^{N-1} \approx e^{-T}$. The exit rate is $\lambda_{\{1,2\}} = 1$, half that of a vertex. Given that the chain stays, it switches between 1 and 2 about T times, so it makes about T runs of length about N/T each. The number a of visits to state 1 is the total length of half of these runs. So a is close to $N/2$, with fluctuations of order $\sqrt{T} \cdot N/T = N/\sqrt{T}$. Hence the mass e^{-T} is shared by about $N/\sqrt{T}$ compositions $(a, N - a, 0, 0)$, and the best one gets

$$M_{\{1,2\}} \approx \frac{e^{-T}}{N/\sqrt{T}} = N^{-1}\, T^{1/2} e^{-T}.$$

Corollary 7.1 confirms this with an exact identity and gives the constant $1/\sqrt{2\pi}$ in front.

An arc of three states, say $\{1, 2, 3\}$. Only the end states 1 and 3 can leave (both to state 4), each at rate 1, and the middle state 2 cannot. So the chain stays longer if it

face F	λ_F	it stays with probability	number of compositions	best composition M_F	exponent $\Phi_\tau(F)$
vertex	2	e^{-2T}	1	e^{-2T}	2τ
adjacent pair	1	e^{-T}	$N/\sqrt{T}$	$N^{-1}T^{1/2}e^{-T}$	$\tau+1$
arc of three	$2-\sqrt{2}$	$e^{-(2-\sqrt{2})T}$	$(N/\sqrt{T})^2$	$N^{-2}\,T\,e^{-(2-\sqrt{2})T}$	$(2-\sqrt{2})\tau+2$
whole cycle	0	1	$(N/\sqrt{T})^3$	$N^{-3}T^{3/2}$	3

Table 1: The four kinds of faces of the 4-cycle with rate 1 between neighbors and the uniform start, up to bounded factors, for T large and T^2/N small. The last column is the exponent of N when $T=\tau\log N$, up to powers of $\log N$.

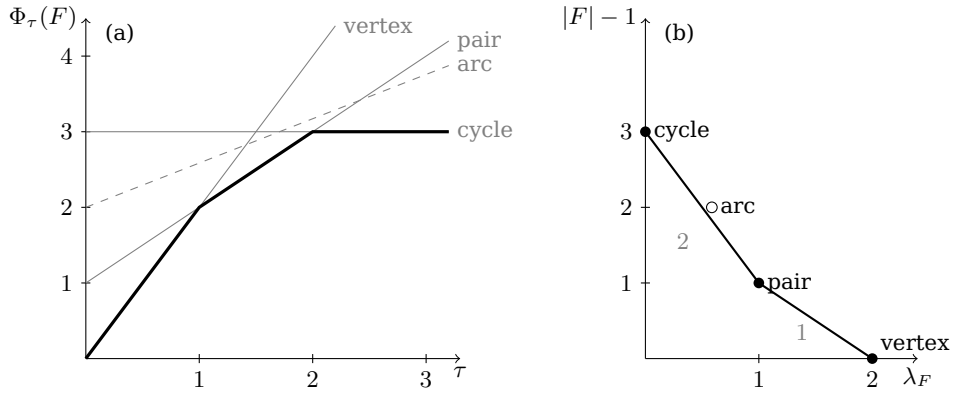


Figure 3: The 4-cycle. (a) The exponents $\Phi_\tau(F)=\tau\lambda_F+|F|-1$ of the four kinds of faces against τ, and their lower envelope (bold). The mode sits on the face whose line is lowest, namely a vertex for $\tau<1$, an adjacent pair for $1<\tau<2$ and the whole cycle for $\tau>2$. The dashed line of the arc of three states is never lowest. (b) The same faces as points $(\lambda_F,|F|-1)$, with the lower-left boundary of their convex hull. The gray numbers are minus the slopes of its edges, the values of τ where the winner changes (Theorem 3.4). The arc (open) lies above the boundary.

prefers the middle state, and we need an eigenvalue to find the exit rate. The chain killed on leaving the arc has the matrix $I+hQ_F$ with $-Q_F=2I-A$, where A is the adjacency matrix of the path $1-2-3$. Its largest eigenvalue is $1-(2-\sqrt{2})h$, with eigenvector $(1,\sqrt{2},1)$. So the chain stays in the arc with probability about $e^{-(2-\sqrt{2})T}$, and $\lambda_{\{1,2,3\}}=2-\sqrt{2}=0.586$. The visit fractions now fluctuate in two directions, so the mass is shared by about $(N/\sqrt{T})^2$ compositions, and $M_{\{1,2,3\}}\approx N^{-2}\,T\,e^{-(2-\sqrt{2})T}$.

The whole cycle. The chain never has to leave, so $\lambda_{[4]}=0$ and the mass is 1. It is shared by about $(N/\sqrt{T})^3$ compositions near $(\frac{N}{4},\frac{N}{4},\frac{N}{4},\frac{N}{4})$. So $M_{[4]}\approx N^{-3}T^{3/2}$.

Comparison. Table 1 collects the four cases, up to bounded factors. Now put $T=\tau\log N$. Then $e^{-\lambda T}=N^{-\tau\lambda}$, while the powers of T are only powers of $\log N$. So, up to such powers, $M_F\approx N^{-\Phi_\tau(F)}$ with the exponent $\Phi_\tau(F)=\tau\lambda_F+|F|-1$ in the last column. The mode sits on the face with the smallest exponent. Figure 3(a) plots the four exponents against τ. The vertices win for $\tau<1$, the adjacent pairs for $1<\tau<2$ and the whole cycle for $\tau>2$. The arcs of three states never win. They beat the pairs only for $\tau>1+\sqrt{2}$, and there the whole cycle beats them, since that needs only $(2-\sqrt{2})\tau>1$. Note also that the whole cycle carries almost all the mass for every $\tau>0$. It carries the mode only for $\tau>2$, because it spreads its mass over about N^3 compositions.

The general rule. For any admissible face F the same two effects give $M_F\approx$

$N^{-(|F|-1)}e^{-T\lambda_F} = N^{-\Phi_\tau(F)}$ up to powers of $\log N$, as announced in Section 1.1. There is a second way to see the factor $N^{-(|F|-1)}$. A path with support F must enter each of the $|F|-1$ states of F where it does not start, and each first entry is a switch of probability about T/N. Every later switch also has probability about T/N, but it multiplies the number of ways to place the runs by a factor of order N, so it loses no power of N. On an irreducible admissible face F, the best composition sits near the quasi-ergodic law α_F^*. Here it is $(\frac{1}{2},\frac{1}{2})$ on a pair, $(\frac{1}{4},\frac{1}{2},\frac{1}{4})$ on an arc (the normalized squared Perron vector) and the uniform law on the cycle. Other visit fractions α lose an extra factor $e^{-T(I_F(\alpha)-\lambda_F)}$, with a rate I_F that we now introduce.

The rest of this section proves the rule. Proposition 3.1 studies the rate inside a face, Theorem 3.3 proves the rule for every Q and every start, and Theorem 3.4 reads the winners off a convex hull as in Figure 3(b). Section 7 returns to the 4-cycle with all constants (Corollary 7.1).

3.2 The rate inside a face

This subsection defines the variational functional I_F and proves the properties that Theorems 3.3 and 4.6 use. For $\alpha \in \Delta_F$ let

$$I_F(\alpha) = \sup_{u\in(0,\infty)^F} \sum_{i\in F} \alpha_i \Big(-\frac{(Q_F u)_i}{u_i}\Big) = \sup_{u\in(0,\infty)^F} \sum_{i\in F} \alpha_i \Big(q_i - \sum_{j\in F\setminus\{i\}} q_{ij}\frac{u_j}{u_i}\Big),$$

and let $q_i^{\mathrm{out}}(F) = \sum_{j\notin F} q_{ij}$, the rate of leaving F from i. For a face satisfying (H), I_F is the Donsker–Varadhan occupation-time rate [19] of the chain killed on leaving F (Section 1.3). Theorem 3.3(i) gives its local form away from the boundary. On a general face we use the variational definition without this probability interpretation. Taking $u = \mathbf{1}$ gives $I_F(\alpha) \geq \sum_i \alpha_i q_i^{\mathrm{out}}(F)$, and dropping the terms with q_{ij} gives $I_F(\alpha) \leq \sum_i \alpha_i q_i$. So I_F lies between the rate of leaving F and the rate of leaving the current state, both averaged over α. On the pair $\{1,2\}$ of Section 3.1, part (c) below gives $I_F(\alpha) = 2 - 2\sqrt{\alpha_1\alpha_2}$, which is 2 at a vertex and has its least value 1 at $(\frac{1}{2},\frac{1}{2})$. We need the properties of I_F without reversibility, and the proofs are short.

Proposition 3.1 (the rate on a face)**.** *Let F be irreducible with $k = |F| \geq 2$.*

(a) The rate satisfies $I_F \geq \lambda_F$ on Δ_F, with equality exactly at $\alpha_F^* = l_F \circ r_F$.

(b) The rate I_F is convex on Δ_F and strictly convex on Δ_F°. For $\alpha \in \Delta_F^\circ$ the supremum defining $I_F(\alpha)$ is attained at a vector $u = r_\alpha$, which is unique up to a positive factor, and α is the stationary law of the generator on F with off-diagonal entries $q_{ij}r_{\alpha,j}/r_{\alpha,i}$.

(c) If there is $\nu \in (0,\infty)^F$ with $\nu_i q_{ij} = \nu_j q_{ji}$ for all $i,j \in F$, then for every $\alpha \in \Delta_F$

$$I_F(\alpha) = \sum_{\{i,j\}\subseteq F,\ i\neq j} \big(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}}\big)^2 + \sum_{i\in F} \alpha_i q_i^{\mathrm{out}}(F).$$

(d) For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $I_F(\alpha) = I_{F'}(\alpha) \geq \lambda_{F'} > \lambda_F$.

Idea of the proof. For $u = r_F$ the quantity $-(Q_F u)_i/u_i$ equals λ_F at every state i. So its average over any α is λ_F, and $I_F \geq \lambda_F$. For the other properties we write $u = e^\psi$. Then the function inside the supremum is concave in ψ and affine in α, and ψ is a maximizer exactly when α is stationary for the chain on F with rates $q_{ij}e^{\psi_j-\psi_i}$. For $u = r_F$ this stationary law is α_F^*, so r_F is optimal there and $I_F(\alpha_F^*) = \lambda_F$. Strict convexity then shows that α_F^* is the only minimizer in the open face, and (d) handles the boundary. Part (d) needs the lower bound on smaller faces, which may have no positive eigenvector. So in the proof a resolvent applied to $\mathbf{1}$ replaces r_F.

Proof. We prove the lower bound in (a) and the value at α_F^* first, and then (b), (d), the rest of (a) and (c), in this order.

Setup. With $u = e^\psi$ we have $I_F(\alpha) = \sup_\psi \Psi_\alpha(\psi)$, where

$$\Psi_\alpha(\psi) = \sum_i \alpha_i q_i - \sum_{i \neq j} \alpha_i q_{ij} e^{\psi_j - \psi_i}$$

and i, j run over F. Each exponential is convex in ψ, so Ψ_α is concave in ψ. It is affine in α, and it does not change when we add a constant to ψ. Let G^ψ be the conservative generator on F (its rows sum to 0) with off-diagonal entries $q_{ij} e^{\psi_j - \psi_i}$. It has the graph G_F, so it is irreducible and has exactly one stationary law. Differentiating in each ψ_i gives $\nabla_\psi \Psi_\alpha = -\alpha^\top G^\psi$. A differentiable concave function attains its maximum exactly at its critical points. So ψ attains $I_F(\alpha)$ if and only if α is the stationary law of G^ψ. We call this the criterion.

The lower bound in (a)*, for every face F'.* We show that $I_{F'} \geq \lambda_{F'}$ on $\Delta_{F'}$ for every face F', irreducible or not. Let $s > \varrho(Q_{F'}) = -\lambda_{F'}$. It is enough to find a positive vector u with $-(Q_{F'}u)_i/u_i > -s$ at every state i. Then $I_{F'}(\alpha) > -s$ for every α, and letting s decrease to $-\lambda_{F'}$ gives the claim. Write $Q_{F'} = B - cI$ with a matrix $B \geq 0$ and a number c. Then $s + c > \varrho(B) \geq 0$, so the series

$$u = (sI - Q_{F'})^{-1}\mathbf{1} = \sum_{\ell \geq 0} (s+c)^{-\ell-1} B^\ell \mathbf{1}$$

converges, and u is positive because the term with $\ell = 0$ is. Since $(sI - Q_{F'})u = \mathbf{1}$, we get $-(Q_{F'}u)_i/u_i = -s + 1/u_i > -s$.

The value at α_F^*. Take $\psi = \log r_F$ and $R = \operatorname{diag}(r_F)$. Then $G^\psi = R^{-1}(Q_F + \lambda_F I)R$, since the off-diagonal entries agree and the rows of the right side sum to 0 by $Q_F r_F = -\lambda_F r_F$. As $\alpha_F^* = l_F \circ r_F$, we get $(\alpha_F^*)^\top G^\psi = l_F^\top (Q_F + \lambda_F I)R = 0$. By the criterion ψ attains the supremum. Its value is λ_F, because $-(Q_F r_F)_i / r_{F,i} = \lambda_F$ at every i. So $I_F(\alpha_F^*) = \lambda_F$.

(b) The function I_F is convex as a supremum of functions that are affine in α. Now let $\alpha \in \Delta_F^\circ$.

The supremum is attained. We normalize $\min_i \psi_i = 0$. Let $\bar\psi = \max_i \psi_i$, and let a be the least positive rate q_{ij} in F. Take a shortest path in G_F from a state where $\psi = 0$ to a state where $\psi = \bar\psi$. It has at most $k-1$ edges and ψ increases by $\bar\psi$ along it, so one of its edges $i \to j$ has $\psi_j - \psi_i \geq \bar\psi/(k-1)$. Keeping only this edge in the second sum of Ψ_α gives

$$\Psi_\alpha(\psi) \leq \sum_i \alpha_i q_i - a\,(\min_i \alpha_i)\, e^{\bar\psi/(k-1)}.$$

The right side tends to $-\infty$ as $\bar\psi \to \infty$. So the supremum is the supremum over a bounded set of normalized ψ, and the continuous function Ψ_α attains it. We write $r_\alpha = e^\psi$ for a maximizer. By the criterion α is the stationary law of G^ψ, which is the last claim of (b).

The maximizer is unique up to a constant. Suppose that ψ and $\psi + \delta$ both attain the supremum. By concavity Ψ_α is constant on the segment $\psi + s\delta$, $0 \leq s \leq 1$. On this segment Ψ_α is a constant minus the sum of the functions $\alpha_i q_{ij} e^{\psi_j - \psi_i + s(\delta_j - \delta_i)}$, and each of these is convex in s. A sum of convex functions is constant only if each of them is affine, and an exponential e^{x+sy} is affine in s only if $y = 0$. As every α_i is positive, this gives $\delta_j = \delta_i$ on every edge of G_F. Since G_F is strongly connected, δ is constant.

Strict convexity. Let $\alpha^0 \neq \alpha^1$ lie in Δ_F°, let $\alpha^s = (1-s)\alpha^0 + s\alpha^1$ with $0 < s < 1$, and let ψ attain $I_F(\alpha^s)$. Since $\Psi_\alpha(\psi)$ is affine in α,

$$I_F(\alpha^s) = (1-s)\Psi_{\alpha^0}(\psi) + s\Psi_{\alpha^1}(\psi) \leq (1-s)I_F(\alpha^0) + sI_F(\alpha^1).$$

Equality would mean that ψ attains both $I_F(\alpha^0)$ and $I_F(\alpha^1)$. By the criterion α^0 and α^1 would both be the stationary law of G^ψ, which is impossible. So the inequality is strict.

(d) Let $\alpha \in \Delta_{F'}$. Only the rows $i \in F'$ enter $\sum_i \alpha_i\big(q_i - \sum_{j\in F\setminus\{i\}} q_{ij}u_j/u_i\big)$, and the terms with $j \in F \setminus F'$ are subtracted. Dropping them leaves the function that defines $I_{F'}(\alpha)$, at the vector $(u_i)_{i\in F'}$. So $I_F(\alpha) \le I_{F'}(\alpha)$. If we put $u_j = \varepsilon$ for $j \in F \setminus F'$ and let $\varepsilon \to 0$, the dropped terms tend to 0, so $I_F(\alpha) = I_{F'}(\alpha)$. Then $I_{F'}(\alpha) \ge \lambda_{F'}$ by the lower bound above. Finally $\lambda_{F'} > \lambda_F$, because the Perron root of an irreducible nonnegative matrix, here $Q_F + cI$ for a large number c, is larger than that of each proper principal submatrix [1, Ch. 2].

(a) The lower bound and the value at α^*_F are proved above, so let $\alpha \ne \alpha^*_F$. If α lies on the boundary of Δ_F, then $\alpha \in \Delta_{F'}$ for the proper face $F' = \operatorname{supp}\alpha$, and (d) gives $I_F(\alpha) > \lambda_F$. If $\alpha \in \Delta_F^\circ$ and $I_F(\alpha) = \lambda_F$, then strict convexity gives $I_F(\frac12(\alpha + \alpha^*_F)) < \lambda_F$, against the lower bound.

(c) We bound the function inside the supremum pair by pair. Since $q_i = \sum_{j\in F\setminus\{i\}} q_{ij} + q_i^{\mathrm{out}}(F)$, this function is $\sum_i \alpha_i q_i^{\mathrm{out}}(F)$ plus, for each pair $\{i,j\}$, the term $\alpha_i q_{ij}(1 - u_j/u_i) + \alpha_j q_{ji}(1 - u_i/u_j)$. By the AM–GM inequality, $\alpha_i q_{ij}u_j/u_i + \alpha_j q_{ji}u_i/u_j \ge 2\sqrt{\alpha_i q_{ij}\alpha_j q_{ji}}$, so the term is at most $(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2$. This gives the upper bound for $I_F(\alpha)$. For the lower bound first let $\alpha \in \Delta_F^\circ$ and take $u_i = \sqrt{\alpha_i/\nu_i}$. Then $\nu_i q_{ij} = \nu_j q_{ji}$ gives $\alpha_i q_{ij}u_j/u_i = \alpha_j q_{ji}u_i/u_j$, so there is equality in the AM–GM inequality for every pair at once. If some α_i vanish, we take $u_i = \varepsilon$ for these i and let $\varepsilon \to 0$. A pair with $\alpha_i = 0 < \alpha_j$ then has the term $\alpha_j q_{ji}(1 - \varepsilon/u_j)$, which tends to its bound $\alpha_j q_{ji}$. A pair with $\alpha_i = \alpha_j = 0$ has the term 0, and the other pairs are as before. □

In words, the most likely way to stay on F is to spend the fractions α^*_F of the time in its states, and the rate at any other fractions is strictly larger. Part (d) says that fractions on the boundary of Δ_F, which miss some states of F, have the rate of the smaller face that they use, and this rate is larger than λ_F. The lower bound in (a) and the value at α^*_F are checked in Lean, and so is the interior form of (c) when Q_F is symmetric. For symmetric Q_F, $l_F = r_F$ and α^*_F is the entrywise square of the Perron vector normalized in ℓ^2. On a face that is not reversible the formula in (c) is only an upper bound (Remark 4.12).

3.3 Face selection

We also need an upper bound that holds on a whole face, including near its boundary, where the local law of Section 4 does not apply. For an irreducible face F put $V_F = \{\theta \in \mathbb{R}^F : \varrho(A_F - \operatorname{diag}\theta) = 0\}$. Note that $q - \lambda_F\mathbf{1} \in V_F$, since $A_F - \operatorname{diag}(q - \lambda_F\mathbf{1}) = Q_F + \lambda_F I$. Each $\theta \in V_F$ gives an exponential bound with the linear rate $\alpha \mapsto \langle q - \theta, \alpha\rangle$, and I_F is the supremum of these linear rates. Indeed, for $u \in (0,\infty)^F$ the vector θ with $\theta_i = (A_F u)_i/u_i$ lies in V_F, since u is a positive null vector of the irreducible Metzler matrix $A_F - \operatorname{diag}\theta$ and only its Perron root has a positive eigenvector, and $\langle q - \theta, \alpha\rangle$ is then the function of u inside the supremum that defines $I_F(\alpha)$. Conversely every $\theta \in V_F$ arises in this way from a positive Perron vector u of $A_F - \operatorname{diag}\theta$.

Lemma 3.2 (a uniform upper bound)**.** *Let F be irreducible with $k = |F| \ge 2$, and let $\theta \in V_F$. There is C_θ, depending on θ, Q and μ, such that for every $n \in \mathbb{Z}^F_{\ge 0}$ with $\operatorname{supp} n = F$ and every h with $hq_{\max} \le \frac12$, the probability that $X_1, \dots, X_{|n|}$ have composition n is at most $C_\theta h^{k-1}\exp\big(-h\langle n, q - \theta\rangle + C_\theta h^2|n|\big)$.*

The proof is in Appendix A. With $\theta = q - \lambda_F\mathbf{1}$ it gives $P_N(n) \le Ch^{k-1}e^{-T\lambda_F + CT^2/N}$ for every n with support F. As $h^{k-1} = T^{k-1}N^{-(k-1)}$, this is the upper half of the rule $M_F \approx N^{-(k-1)}e^{-T\lambda_F}$ of Section 3.1, up to a power of T. The lemma is stated for a path of any length $|n|$, not only N, so it also applies to the blocks of a path.

The next theorem uses the local law (Theorem 4.4) and the face maxima (Theorem 4.6)

of Section 4, which comes later. There is no circularity, since those two theorems are proved there without Theorem 3.3.

Theorem 3.3 (first-order face selection)**.** *Fix Q and μ.*

(i) Let F satisfy (H), let $k = |F|$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There is N_0 such that, uniformly for $N \geq N_0$, $1 \leq T \leq N^{1/3}$ and $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$,

$$\log P_N(n) = -(k-1)L - T\, I_F(n/N) + O_{\mathcal{K}}(1 + \log T).$$

(ii) Let F be admissible with components $C_1, \dots, C_r$, and let k^* be the least size of a component C_j with $\lambda_{C_j} = \lambda_F$. There are κ, T_0 and N_0 such that for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$

$$\log M_F \geq -(k_F - 1)L + (k_F - k^*)\log T - T\lambda_F - \kappa,$$
$$\log M_F \leq -(k_F - 1)L + (k_F - 1)\log T - T\lambda_F + \kappa.$$

If F is not admissible, $M_F = 0$.

(iii) Let $T = \tau L$ with $\tau > 0$ fixed, and let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ. Then $\log \max_n P_N(n) = -L \min_F \Phi_\tau(F) + O(\log L)$, with the minimum over admissible faces, and for large N every global mode $\hat{n}$ has $\operatorname{supp} \hat{n} \in \mathcal{F}_\tau$. This holds for every start μ.

(iv) If in (iii) every face in $\mathcal{F}_\tau$ is irreducible, then every global mode $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/L)$ with $F = \operatorname{supp} \hat{n}$. This is the case when Q has symmetric support or μ has full support (Theorem 3.4(c)).

In words, (i) says that a composition whose fractions α stay in a compact part of the open face has probability $N^{-(k-1)} e^{-T I_F(\alpha)}$ up to a bounded factor and a power of T. Part (ii) says that the best composition on an admissible face has probability $N^{-(k_F-1)} e^{-T\lambda_F}$ times a power of T with exponent between 0 and $k_F - 1$, up to a bounded factor. With $T = \tau L$ the exponent of N is then $-\Phi_\tau(F)$, and part (iii) says that for large N the mode lies on an admissible face that minimizes it among the admissible faces, whatever the start. By (iv), if every such face is irreducible, the mode sits near the quasi-ergodic law of its face.

Idea of the proof. Part (i) comes from the local law of Section 4, which we prove independently of this theorem. Parts (ii) and (iii) are elementary. A face is visited in blocks, one for each strongly connected component (a single block if the face is irreducible). For the upper bound in (ii) we apply Lemma 3.2 to each block, and for the lower bound we write down explicit paths that spend almost all their time in a component of least exit rate. Part (iii) compares the faces, and (iv) follows from Theorem 4.6.

Proof. (i) For $k = 1$ this is (2.2), since $I_{\{i\}} = q_i$. For $k \geq 2$, Lemma 4.5 gives $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O(T^2/N))$ with $T^2/N \leq N^{-1/3}$. We claim that $\log f_T(\alpha) = \frac{k-1}{2} \log T + \log C_F(\alpha) - T I_F(\alpha) + O_{\mathcal{K}}(1)$ (with $C_F(\alpha)$ as in Definition 4.1) for $\alpha \in \mathcal{K}$ and $T \geq 1$. For $T \geq T_1$, with T_1 large, this is Theorem 4.4. For $1 \leq T \leq T_1$ it holds because $f_T(\alpha)$ is positive and continuous on the compact set $[1, T_1] \times \mathcal{K}$ (see the remarks after Theorem 4.4). Finally C_F is positive and continuous on $\mathcal{K}$, so $\log C_F$ is bounded there. So $\log f_T(n/N) = -T I_F(n/N) + O_{\mathcal{K}}(1 + \log T)$, and adding $\log N^{-(k-1)} = -(k-1)L$ gives (i).

(ii) If F is not admissible, $M_F = 0$. Let F be admissible. A path visits each component in one block of consecutive times (proof of Lemma 2.2), and an irreducible face is one component, with $k^* = k_F$. (For irreducible F, Theorem 4.6 later gives the sharper factor $T^{(k_F-1)/2} K_F$, but we do not need it here.) For the upper bound, Lemma 3.2 (or a direct

count if $|C| = 1$) bounds a block of length N_C in the component C by $O(h^{|C|-1}e^{-h\lambda_C N_C})$, and each of the $r-1$ switches between blocks gives a factor h. As $\lambda_C \geq \lambda_F$, this gives $M_F = O(h^{k_F-1}e^{-T\lambda_F})$. For the lower bound the path spends about N/T visits in each component except one of least exit rate, which loses a bounded factor per component, and we divide by the number of compositions. The details are in Appendix A.6.

(iii) We compare the faces by (ii). For large N we have $T_0 \leq T = \tau L \leq N^{1/3}$, so (ii) applies. There $T\lambda_F = \tau L \lambda_F$ and $\log T = O(\log L)$, so both bounds in (ii) read $\log M_F = -L\,\Phi_\tau(F) + O(\log L)$ for every admissible face, while $M_F = 0$ for the others. Since $\max_n P_N(n) = \max_F M_F$ and there are finitely many faces, the first claim follows. For the second claim let $m_\tau = \min_F \Phi_\tau(F)$. An admissible $F \notin \mathcal{F}_\tau$ has $\Phi_\tau(F) \geq m_\tau + \varepsilon_0$ for a fixed $\varepsilon_0 > 0$, so $M_F \leq N^{-m_\tau - \varepsilon_0 + o(1)}$, which is smaller than $\max_n P_N(n) = N^{-m_\tau + o(1)}$ for large N. So the support of a global mode is not such a face. A global mode has positive probability, so its support is admissible, and hence it lies in $\mathcal{F}_\tau$.

(iv) By (iii), for large N a global mode $\hat{n}$ maximizes P_N on its face $F \in \mathcal{F}_\tau$, which is irreducible and admissible and so satisfies (H) by Lemma 2.2(b). So Theorem 4.6 with $T = \tau L \leq N^{1/3}$ gives $\hat{n}/N = \alpha_F^* + O(1/L)$. □

For a reducible face we only have the bounds in (ii), and Proposition 4.14 shows that the power of T can differ from $(k_F - 1)/2$.

3.4 Reading the winners off a convex hull

This subsection turns Theorem 3.3(iii) into a picture. The winners are the points on the lower boundary of a convex hull.

For an admissible face F put $P_F = (\lambda_F, |F| - 1)$, and let S be the set of these points. Then $\Phi_\tau(F) = \tau x + y$ at $(x, y) = P_F$. Picture a line of slope $-\tau$ moving up from far below. The first points of S that it touches are the winners at τ. For each $\tau > 0$ the points of $\operatorname{conv} S$ where $\tau x + y$ is least form a vertex or an edge of $\operatorname{conv} S$, and we call the union of these over all $\tau > 0$ the lower-left boundary of $\operatorname{conv} S$. Figure 3(b) shows it for the 4-cycle.

Theorem 3.4 (geometry of the winners)**.** *Let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ.*

(a) The lower-left boundary of $\operatorname{conv} S$ is a path whose vertices lie in S. Read from right to left, they are $v_i = (\lambda^{(i)}, y^{(i)})$, $0 \leq i \leq m$, with $\lambda^{(0)} > \cdots > \lambda^{(m)}$ and $y^{(0)} < \cdots < y^{(m)}$. Put $\tau_i = (y^{(i)} - y^{(i-1)})/(\lambda^{(i-1)} - \lambda^{(i)})$ for $1 \leq i \leq m$, $\tau_0 = 0$ and $\tau_{m+1} = \infty$. Then $\tau_1 < \cdots < \tau_m$. For $\tau_i < \tau < \tau_{i+1}$, $\mathcal{F}_\tau$ is the set of faces F with $P_F = v_i$. At $\tau = \tau_i$ it is the set of faces whose point lies on the closed edge $[v_{i-1}, v_i]$.

(b) If $0 < \tau < \tau'$, $F \in \mathcal{F}_\tau$ and $G \in \mathcal{F}_{\tau'}$, then $\lambda_F \geq \lambda_G$ and $|F| \leq |G|$. Both inequalities are strict when $\mathcal{F}_\tau \cap \mathcal{F}_{\tau'} = \emptyset$.

(c) If Q has symmetric support, or μ has full support, then every face in $\mathcal{F}_\tau$ is irreducible, for every $\tau > 0$.

In words, as τ grows the winners walk up the lower-left boundary of the hull, from small faces with large exit rates to large faces with small exit rates, and they change at the values τ_i, minus the slopes of its edges.

Idea of the proof. The least value $g(\tau)$ of Φ_τ is a minimum of finitely many affine functions of τ, so it is concave and piecewise affine. Its pieces are the vertices v_i and its kinks are the values τ_i. Part (b) adds two inequalities, and part (c) replaces a reducible face by one of its components.

Proof. (a) We study the least value $g(\tau) = \min_{(x,y)\in S}(\tau x + y)$ for $\tau > 0$. Each point (x, y) of S gives the affine function $\tau \mapsto \tau x + y$, whose slope is x, and different points give different functions. So g is a minimum of finitely many affine functions. It is concave and piecewise affine. Let $\tau_1 < \cdots < \tau_m$ be the points of $(0, \infty)$ where its slope changes, and put $\tau_0 = 0$ and $\tau_{m+1} = \infty$.

Between two kinks. On (τ_i, τ_{i+1}) the function g is affine. The affine function of a point of S is nowhere below g. So if it equals g at one τ inside this interval, it equals g on the whole interval. Hence exactly one point of S attains g there. We call it $v_i = (\lambda^{(i)}, y^{(i)})$, and for $\tau_i < \tau < \tau_{i+1}$ the set $\mathcal{F}_\tau$ consists of the faces F with $P_F = v_i$. The slope of g on this interval is $\lambda^{(i)}$. The slope of a concave function decreases, and it changes at each τ_i, so $\lambda^{(0)} > \cdots > \lambda^{(m)}$. By continuity both v_{i-1} and v_i attain $g(\tau_i)$, namely

$$\tau_i \lambda^{(i-1)} + y^{(i-1)} = \tau_i \lambda^{(i)} + y^{(i)}.$$

This is the formula for τ_i in the statement, and since $\tau_i > 0$ it gives $y^{(i)} > y^{(i-1)}$.

At a kink. The points (x, y) of S that attain $g(\tau_i)$ are those on the line $\tau_i x + y = g(\tau_i)$, which passes through v_{i-1} and v_i. Let (x, y) be one of them. For τ slightly below τ_i we have $\tau x + y \geq g(\tau) = \tau \lambda^{(i-1)} + y^{(i-1)}$, with equality at $\tau = \tau_i$. Subtracting gives $(\tau - \tau_i)(x - \lambda^{(i-1)}) \geq 0$, so $x \leq \lambda^{(i-1)}$. In the same way, τ slightly above τ_i gives $x \geq \lambda^{(i)}$. So (x, y) lies on the closed edge $[v_{i-1}, v_i]$. Conversely every point of S on this edge lies on the line and so attains $g(\tau_i)$.

The boundary. On $\operatorname{conv} S$ the function $\tau x + y$ is least exactly on the convex hull of the points of S where it is least. By the two steps above this hull is the vertex v_i for $\tau_i < \tau < \tau_{i+1}$ and the edge $[v_{i-1}, v_i]$ for $\tau = \tau_i$. So the lower-left boundary is the path $v_0 \cdots v_m$, and its vertices lie in S.

(b) We add two inequalities. Since F wins at τ and G wins at τ', we have $\Phi_\tau(F) \leq \Phi_\tau(G)$ and $\Phi_{\tau'}(G) \leq \Phi_{\tau'}(F)$. Adding them gives $(\tau' - \tau)(\lambda_F - \lambda_G) \geq 0$, so $\lambda_F \geq \lambda_G$. Then the first inequality gives $|F| - |G| \leq \tau(\lambda_G - \lambda_F) \leq 0$. If $\lambda_F = \lambda_G$, the two inequalities give $|F| = |G|$, so $\Phi_{\tau'}(F) = \Phi_{\tau'}(G)$ and $F \in \mathcal{F}_{\tau'}$. So if $\mathcal{F}_\tau$ and $\mathcal{F}_{\tau'}$ are disjoint, then $\lambda_F > \lambda_G$, and the first inequality gives $|F| < |G|$.

(c) With symmetric support every admissible face is irreducible by Lemma 2.2(c). Now let μ have full support, and let F be admissible and reducible. Take a component C of F with $\lambda_C = \lambda_F$. It is irreducible with $\mu(C) > 0$, so it is admissible by Lemma 2.2(b). It has the same exit rate as F and fewer states, so $\Phi_\tau(C) \leq \Phi_\tau(F) - 1$. Hence $F \notin \mathcal{F}_\tau$. □

Lean checks part (b), the fact that a face with a proper subface of the same exit rate never wins, and part (a) when the least exit rate of each size decreases strictly with the size. In that case the vertices of the hull are sizes, and Lean checks which size wins between consecutive kinks and that consecutive vertices tie at τ_i. The general path structure in (a) and the ties along a closed edge are proved only here. A face with small λ_F for its size is a set of states that the chain rarely leaves, a well-separated community. As τ grows the exit rate counts for more, and the mode moves to larger communities. It need not grow around its first vertex, and it can move to a disjoint community (Table 2). For symmetric Q, λ_F is the least value of $\sum_{i<j} q_{ij}(f_i - f_j)^2$, the sum over all pairs in $[d]$, over unit vectors f supported in F (Rayleigh–Ritz), and it decreases strictly as a connected face grows, by Proposition 3.1(d). Since faces of one size share the term $|F| - 1$, only the faces with the least exit rate of each size can win. Figure 4(a) shows S for the 6-cycle.

4 The second-order theory

The first-order rule locates a crossing only up to a factor $1 + o(1)$ in T. To locate it up to $o(1)$ we need the constant in front of $N^{-(k-1)}e^{-T\lambda_F}$ for every face that satisfies

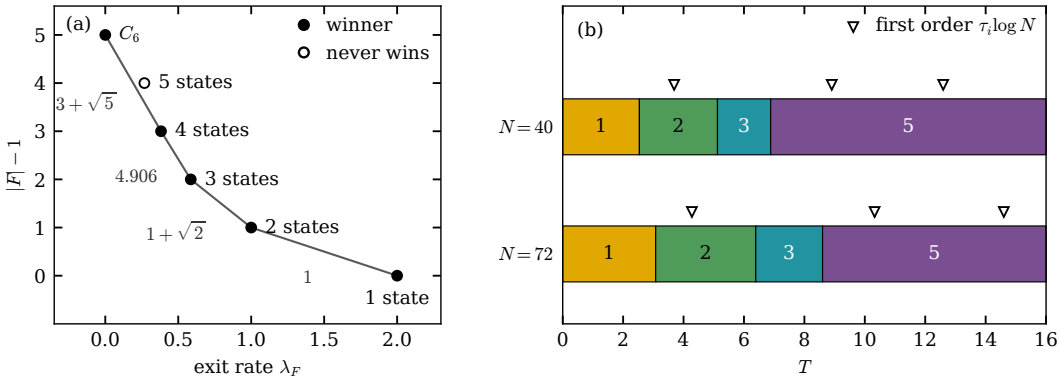


Figure 4: (a) The points $(\lambda_F, |F|-1)$ of the arcs of the 6-cycle C_6 and of C_6 itself, with the lower-left boundary of their hull. The numbers beside the edges are the switch values τ_i of Theorem 3.4. The winners (filled) are the arcs of one to four states and C_6, and the arc of five states (open) never wins. (b) The support size of the exact mode on C_5 with the uniform start, against T, at $N = 40$ and $N = 72$. The mode skips the arc of four states. The triangles mark the first-order switch points $\tau_i \log N$ with $\tau_i = 1$, $1+\sqrt{2}$ and $2+\sqrt{2}$ (Table 2).

(H), together with the power of T. The plan is as follows. Section 4.1 defines the constant K_F. Section 4.2 proves a local limit theorem inside a face and deduces M_F up to a factor $1 + O(1/T + T^2/N)$. Section 4.3 compares two faces. Section 4.4 gives a tree formula for K_F on every irreducible face, Section 4.5 shows what can change without (H), Section 4.6 adds the next term, of relative order $1/T$, and Section 4.7 adds a weak field. In this section gradients and Hessians in α are taken in the coordinates $\alpha' = (\alpha_2, \dots, \alpha_k)$, $|\cdot|$ is the Euclidean norm, and $q = (q_i)_{i \in F}$.

4.1 The tilted chain and the constant

To find the probability of a composition with fractions α we change the rates so that α becomes typical. The optimal vector r_α in the definition of $I_F(\alpha)$ gives a new generator $\widehat{Q}_\alpha$ on F, with the same switching graph, whose stationary law is exactly α. Under it the visit fractions satisfy a central limit theorem with covariance Γ_α/T, and the price of the change of rates is $e^{-TI_F(\alpha)}$ times boundary factors for the start and the end of the path. The constant $C_F(\alpha)$ collects these factors and the Gaussian normalization.

Definition 4.1 (tilt and constants)**.** *Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ let r_α attain the supremum defining $I_F(\alpha)$ (Proposition 3.1(b)), and put $\theta(\alpha) = A_F r_\alpha / r_\alpha$ and $l_\alpha = \alpha / r_\alpha$ (entrywise). The* tilted generator *is $\widehat{Q}_\alpha = R^{-1}(A_F - \operatorname{diag}\theta(\alpha))R$ with $R = \operatorname{diag} r_\alpha$. With $D = \operatorname{diag}\alpha$ let*

$$Z_\alpha = (\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)^{-1} - \mathbf{1}\alpha^\top, \qquad \Gamma_\alpha = DZ_\alpha + Z_\alpha^\top D,$$

and let Σ_α be Γ_α with the row and column of state 1 deleted. Finally let

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}\sqrt{\det \Sigma_\alpha}}, \qquad K_F = C_F(\alpha_F^*).$$

For a single state we put $K_{\{i\}} = \mu_i$. Nothing in Definition 4.1 changes when r_α is scaled. Note that $l_\alpha \cdot r_\alpha = 1$ and $I_F(\alpha) = \langle q - \theta(\alpha), \alpha \rangle$. Since r_α minimizes $u \mapsto \sum_{i \neq j} \alpha_i q_{ij} u_j / u_i$, its derivative in u_j, which is $\sum_i l_{\alpha,i} q_{ij} - l_{\alpha,j}\theta_j(\alpha)$, vanishes at r_α. Hence

$$l_\alpha^\top \big(A_F - \operatorname{diag}\theta(\alpha)\big) = 0. \tag{4.1}$$

Also $(A_F - \operatorname{diag}\theta(\alpha))r_\alpha = 0$ with $r_\alpha > 0$, so $\theta(\alpha) \in V_F$ [1]. So $\widehat{Q}_\alpha$ is an irreducible generator with the same off-diagonal zero pattern as A_F, and by (4.1) its stationary law is α. If $(\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)v = 0$, multiplying by $\alpha^\top$ gives $\alpha^\top v = 0$, so $\widehat{Q}_\alpha v = 0$. Hence v is a multiple of $\mathbf{1}$, and $\alpha^\top v = 0$ gives $v = 0$. Thus Z_α exists, and

$$Z_\alpha \mathbf{1} = 0, \qquad \alpha^\top Z_\alpha = 0, \qquad -\widehat{Q}_\alpha Z_\alpha = I - \mathbf{1}\alpha^\top. \tag{4.2}$$

So $\Gamma_\alpha \mathbf{1} = 0$. As Γ_α is symmetric, its adjugate is then a multiple of $\mathbf{1}\mathbf{1}^\top$, so deleting another state instead of state 1 gives the same $\det \Sigma_\alpha$. At α_F^* the vector $u = r_F$ gives the value $\lambda_F = I_F(\alpha_F^*)$, so $r_{\alpha_F^*}$ is a multiple of r_F, $\theta(\alpha_F^*) = q - \lambda_F \mathbf{1}$, and $l_{\alpha_F^*}$ is the matching multiple of l_F. In the local law below, $\mu \cdot r_\alpha$ accounts for the start of the path, $l_\alpha \cdot \mathbf{1}$ for its free end, and Σ_α for the Gaussian spread of the occupation fractions.

For conductances $c_{ij} = c_{ji}$ on F let $\operatorname{tree}(c) = \sum_{\mathcal{T}} \prod_{\{i,j\} \in \mathcal{T}} c_{ij}$, the sum over the spanning trees $\mathcal{T}$ of the complete graph on F.

Proposition 4.2 (the constant)**.** *Let F be irreducible with $k \geq 2$.*

(a) The maps $\alpha \mapsto \theta(\alpha)$, $I_F(\alpha)$, Σ_α, r_α and l_α (with $\sum_i r_{\alpha,i} = 1$) are real-analytic on Δ_F°, Σ_α is positive definite, and $\nabla^2 I_F(\alpha) = \Sigma_\alpha^{-1}$. Hence $\det \Sigma_\alpha \cdot \det \nabla^2 I_F(\alpha) = 1$ and $C_F(\alpha) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})\sqrt{\det \nabla^2 I_F(\alpha)/(2\pi)^{k-1}}$.

(b) Suppose that $\nu_i q_{ij} = \nu_j q_{ji}$ for $i, j \in F$ and some $\nu \in (0,\infty)^F$. Then r_α is a multiple of $\sqrt{\alpha/\nu}$ and l_α of $\sqrt{\alpha\nu}$, and $\widehat{Q}_\alpha$ is reversible with conductances $c_{ij}(\alpha) = \sqrt{\alpha_i \alpha_j}\, q_{ij} \sqrt{\nu_i/\nu_j}$. Also $I_F(\alpha) = \sum_i q_i \alpha_i - \sum_{i \neq j} c_{ij}(\alpha)$, and

$$\det \Sigma_\alpha = \frac{2^{k-1}(\prod_i \alpha_i)^2}{\operatorname{tree}(c(\alpha))}, \qquad C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(4\pi)^{(k-1)/2}} \, \frac{\sqrt{\operatorname{tree}(c(\alpha))}}{\prod_i \alpha_i}.$$

In words, the Gaussian factor in C_F is, up to a power of 2π, the square root of the determinant of the Hessian of the rate. For a reversible face this determinant is a weighted count of spanning trees divided by $2^{k-1}(\prod_i \alpha_i)^2$.

Proof. The proof of (a) is in Appendix A and uses Lemma A.1.

(b) We guess the vector r_α and check that the guess is a critical point. Put $u = \sqrt{\alpha/\nu}$ and $l = \alpha/u = \sqrt{\alpha\nu}$ (entrywise). As we saw before (4.1), the derivative of $\sum_{i \neq j} \alpha_i q_{ij} u_j/u_i$ in u_j is $\sum_i l_i q_{ij} - l_j (A_F u)_j / u_j$. Here

$$\sum_i l_i q_{ij} = \sum_i \sqrt{\alpha_i \nu_i}\, q_{ij} \qquad \text{and} \qquad \frac{l_j (A_F u)_j}{u_j} = \nu_j \sum_i q_{ji} \sqrt{\alpha_i/\nu_i} = \sum_i \sqrt{\alpha_i \nu_i}\, q_{ij},$$

where the last step uses $\nu_j q_{ji} = \nu_i q_{ij}$. So the derivative vanishes at u. The function is convex in $\log u$, so its critical point u is a minimizer, and the minimizer is unique up to a factor (Proposition 3.1(b)). Hence r_α is a multiple of u and l_α is the matching multiple of l. Then $\alpha_i(\widehat{Q}_\alpha)_{ij} = l_i q_{ij} u_j = c_{ij}(\alpha)$ for $i \neq j$, and $c_{ij}(\alpha) = c_{ji}(\alpha)$ because $q_{ij}\sqrt{\nu_i/\nu_j} = q_{ji}\sqrt{\nu_j/\nu_i}$. So $\widehat{Q}_\alpha$ is reversible with these conductances. Also $\alpha_i \theta_i(\alpha) = l_i (A_F u)_i = \sum_{j \neq i} c_{ij}(\alpha)$, and $I_F(\alpha) = \langle q - \theta(\alpha), \alpha \rangle$ gives the formula for I_F.

For the determinant we express the quadratic form of Σ_α through the Laplacian of the conductances, namely $\mathcal{L} = -D\widehat{Q}_\alpha$. It is symmetric, its off-diagonal entries are $-c_{ij}(\alpha)$, and $\mathcal{L}\mathbf{1} = 0$. Fix $\eta \in \mathbb{R}^F$ with $\eta_1 = 0$, and put $w = Z_\alpha \eta$ and $\zeta = D\eta - (\alpha \cdot \eta)\alpha$. By (4.2), $\mathcal{L} w = D(I - \mathbf{1}\alpha^\top)\eta = \zeta$ and $\alpha \cdot w = 0$. So $\eta^\top \Gamma_\alpha \eta = 2(D\eta)^\top w = 2\zeta^\top w$, where the second step uses $\alpha \cdot w = 0$. Next we remove state 1. Let $v = w - w_1 \mathbf{1}$, so that $v_1 = 0$. Since $\zeta \cdot \mathbf{1} = 0$ and $\mathcal{L}\mathbf{1} = 0$, we have $\zeta^\top w = \zeta^\top v$ and $\mathcal{L} v = \zeta$. We write primes for the coordinates $2, \dots, k$, $D' = \operatorname{diag}\alpha'$, and $\mathcal{L}_{(1)}$ for $\mathcal{L}$ without row and column 1. As $v_1 = 0$, the rows $2, \dots, k$ of $\mathcal{L} v = \zeta$ say that $\mathcal{L}_{(1)} v' = \zeta'$, and as $\eta_1 = 0$ we have $\zeta' = (D' - \alpha'\alpha'^\top)\eta'$.

The matrix-tree theorem [5] gives $\det \mathcal{L}_{(1)} = \operatorname{tree}(c(\alpha))$, which is positive because F is irreducible. So $\mathcal{L}_{(1)}$ is invertible, and

$$\eta'^{\top}\Sigma_\alpha \eta' = 2\zeta'^{\top} v' = 2\eta'^{\top}(D' - \alpha'\alpha'^{\top})\mathcal{L}_{(1)}^{-1}(D' - \alpha'\alpha'^{\top})\eta'$$

for every $\eta' \in \mathbb{R}^{k-1}$. Both matrices are symmetric, hence $\Sigma_\alpha = 2(D' - \alpha'\alpha'^{\top})\mathcal{L}_{(1)}^{-1}(D' - \alpha'\alpha'^{\top})$. Now we take determinants. The matrix determinant lemma gives $\det(D' - \alpha'\alpha'^{\top}) = \prod_{i\ge 2}\alpha_i\,(1 - \sum_{i\ge 2}\alpha_i) = \prod_i \alpha_i$. So $\det \Sigma_\alpha = 2^{k-1}(\prod_i \alpha_i)^2 / \operatorname{tree}(c(\alpha))$, and putting this into Definition 4.1 gives the formula for $C_F(\alpha)$. □

Part (b) needs a reversible face. Proposition 4.9 at the end of this section gives a formula of the same kind for every irreducible face.

Corollary 4.3 (symmetric faces)**.** *Let F satisfy* (H) *with $k \ge 2$, and let Q_F be symmetric with constant row sums. Then $\lambda_F = q_i - \sum_{j\in F\setminus\{i\}} q_{ij}$ for every $i \in F$, α_F^* is uniform on F, and*

$$K_F = \mu(F)\, k^{(k+1)/2} \sqrt{\operatorname{tree}(q_F)}\, (4\pi)^{-(k-1)/2},$$

where $\operatorname{tree}(q_F)$ is $\operatorname{tree}(c)$ for the conductances $c_{ij} = q_{ij}$, $i, j \in F$.

Proof. The positive vector $\mathbf{1}$ is a right and left eigenvector of Q_F, so it belongs to the Perron root. So λ_F is minus the row sum, r_F and l_F are multiples of $\mathbf{1}$, and $\alpha_F^* = \mathbf{1}/k$. At $\alpha = \mathbf{1}/k$, Proposition 4.2(b) with $\nu = \mathbf{1}$ gives $r_\alpha = \mathbf{1}$, $l_\alpha = \alpha$ and $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = \mu(F)$. Also $c_{ij}(\alpha) = q_{ij}/k$, so $\operatorname{tree}(c(\alpha)) = k^{-(k-1)} \operatorname{tree}(q_F)$, and $\prod_i \alpha_i = k^{-k}$. □

4.2 The local law and the face maxima

Inside the face the occupation fractions fluctuate on the scale $T^{-1/2}$ in $k-1$ directions. So a single composition gets the factor $T^{(k-1)/2}$ times the volume $N^{-(k-1)}$ of a lattice cell.

Theorem 4.4 (local law)**.** *Let F satisfy (H) with $k \ge 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\alpha \in \mathcal{K}$ and $T \ge 1$,*

$$f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-T I_F(\alpha)} \big(1 + O_{\mathcal{K}}(1/T)\big).$$

For $F = \{i\}$, $f_T = \mu_i e^{-q_i T}$.

In words, the density of the visit fractions on the event that the chain stays in F is a Gaussian factor $T^{(k-1)/2}$ times the large-deviation factor $e^{-TI_F(\alpha)}$ times the constant $C_F(\alpha)$, with a relative error $O(1/T)$ that is uniform away from the boundary of the face.

Idea of the proof. The proof is in Appendix A. The generating function of the sum over words that defines H is a matrix resolvent, so Cauchy's formula writes H as an integral over a torus with the radii $\theta_2(\alpha), \dots, \theta_k(\alpha)$ (Lemma A.2). All the coefficients are nonnegative, so the integrand is largest at the real point, and a Laplace lemma with a parameter (Lemma B.1) gives the expansion. The Hessian at the real point is computed from the Perron root of a perturbed generator (Lemma A.1), which brings in Σ_α. The hypothesis $\mu(F) > 0$ is needed, since $f_T = C_F = 0$ when $\mu(F) = 0$. Also $f_T(\alpha)$ is continuous and positive in (T, α), as H is a convergent power series with some $W(m) > 0$ by (H).

Lemma 4.5 (discrete and continuum)**.** *Let F satisfy (H) with $k \ge 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$ and $1 \le T \le N^{1/2}$, $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O_{\mathcal{K}}(T^2/N))$.*

In words, the probability of a composition is the continuum density times the volume $N^{-(k-1)}$ of a lattice cell. The proof is in Appendix A. With Theorem 4.4 the lemma replaces the term $O_{\mathcal{K}}(1 + \log T)$ in Theorem 3.3(i) by $\frac{k-1}{2}\log T + \log C_F(n/N) + O_{\mathcal{K}}(1/T + T^2/N)$.

Theorem 4.6 (face maxima)**.** *Let F satisfy (H). If $k \geq 2$, then uniformly for $1 \leq T \leq N^{1/3}$,*

$$M_F = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F} \big(1 + O(1/T + T^2/N)\big),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/T)$. For $k = 1$, $M_{\{i\}}$ is given by (2.2)*. In both cases, uniformly for $1 \leq T \leq N^{1/3}$,*

$$\log M_F = -(k-1)L + \tfrac{k-1}{2} \log T + \log K_F - T\lambda_F + O(1/T + T^2/N). \tag{4.3}$$

In words, the best composition on F sits within $O(N/T)$ of $N\alpha_F^*$, and its probability is the Gaussian peak of the local law at α_F^*. The factor $N^{-(k-1)}$ is the volume of a lattice cell, and $T^{(k-1)/2}$ is the height of a Gaussian density whose spread in each free direction is of order $T^{-1/2}$.

Idea of the proof. Near α_F^* we use the local law and Lemma 4.5, and maximize a Gaussian with a small linear perturbation. Away from α_F^* the rate exceeds λ_F by a fixed amount, and Lemma 3.2, applied with finitely many vectors θ, shows that these compositions are exponentially less likely in T. For bounded T positivity and continuity suffice.

Proof. For $k = 1$ this is (2.2). Let $k \geq 2$. Put $X = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F}$, the value that the theorem claims for M_F. We compare every $P_N(n)$ with X, and we write $x = \alpha' - \alpha_F^{*\prime}$ for the displacement of α from α_F^*. We fix a ball U around α_F^*, bound the compositions outside U, then those inside U, and at the end we locate the maximizer.

The ball U. By Propositions 3.1(a) and 4.2(a), I_F and $\log C_F$ are real-analytic near α_F^*, and I_F has its minimum λ_F there with a positive definite Hessian. So there are a closed ball $U = \{|x| \leq \rho_0\} \subset \Delta_F^\circ$ and positive constants c_U, C_U, C_1 and C_3 such that on U

$$c_U |x|^2 \leq I_F - \lambda_F \leq C_U |x|^2, \qquad \big|\log(C_F/K_F) - \gamma \cdot x\big| \leq C_1 |x|^2,$$

where $\gamma = \nabla \log C_F(\alpha_F^*)$, and such that $\log C_F$ is C_3-Lipschitz on U.

Outside U. We show that a composition with $n/N \notin U^\circ$ has $P_N(n) = X \cdot O(T^{(k-1)/2} e^{-2\eta T})$ for some $\eta > 0$. The set $\Delta_F \setminus U^\circ$ is compact and does not contain α_F^*, and I_F is lower semicontinuous. So Proposition 3.1(a) gives $\eta > 0$ with $I_F \geq \lambda_F + 3\eta$ on this set. Recall that $I_F(\beta) = \sup_{u>0} \langle q - A_F u/u, \beta \rangle$ and that $A_F u/u \in V_F$. So every β in the set has a vector $\theta \in V_F$ with $\langle q - \theta, \beta \rangle > \lambda_F + 2\eta$, and the same θ works for all points near β. By compactness finitely many vectors $\theta_j \in V_F$ are enough, that is, $\max_j \langle q - \theta_j, \beta \rangle > \lambda_F + 2\eta$ for every $\beta \in \Delta_F \setminus U^\circ$. Now let $\operatorname{supp} n = F$ with $n/N \notin U^\circ$, and take j that attains the maximum at $\beta = n/N$. Then $h\langle n, q - \theta_j \rangle > T(\lambda_F + 2\eta)$, and Lemma 3.2 with $\theta = \theta_j$ gives

$$P_N(n) \leq C(T/N)^{k-1} e^{-T(\lambda_F + 2\eta)} = X \cdot O\big(T^{(k-1)/2} e^{-2\eta T}\big).$$

Inside U. By Theorem 4.4 and Lemma 4.5 with $\mathcal{K} = U$, for $\alpha = n/N \in U$,

$$P_N(n) = X \, \tfrac{C_F(\alpha)}{K_F} \, e^{-T(I_F(\alpha) - \lambda_F)} (1 + E), \qquad |E| \leq C_2(1/T + T^2/N). \tag{4.4}$$

For the upper bound we use the bounds on U. For $T \geq 2C_1/c_U$ the logarithm of the two middle factors is at most

$$\gamma \cdot x + C_1 |x|^2 - c_U T |x|^2 \leq \gamma \cdot x - \tfrac{c_U T}{2} |x|^2 \leq \frac{|\gamma|^2}{2 c_U T},$$

where the last step is the maximum over x of a concave quadratic. So $P_N(n) \leq X(1 + O(1/T + T^2/N))$ for $n/N \in U$. The bound outside U is smaller than this for large T, and hence $M_F \leq X(1 + O(1/T + T^2/N))$.

For the lower bound we take a lattice point next to α^*_F. Let n^0 have support F and $|n^0_i - N\alpha^*_{F,i}| < 1$ for all i. It exists for $N \ge N_0$ since every $\alpha^*_{F,i}$ is positive, and its displacement has $|x| \le k/N$. Then $C_F(n^0/N) \ge K_F e^{-kC_3/N}$ and $T(I_F(n^0/N) - \lambda_F) \le C_U k^2 T/N^2$, so (4.4) gives $P_N(n^0) \ge X(1 - O(1/T + T^2/N))$.

Bounded T. Fix T_2 so large that the two bounds above give $M_F/X \in [\frac12, \frac32]$ for $T \ge T_2$. For $1 \le T \le T_2$ the error term $O(1/T)$ is of order 1, so it is enough to show that $M_F/X \in [c, C]$ with constants $0 < c < C$. For such T, $N^{k-1}X$ lies between two positive constants, and so does f_T on U, because it is positive and continuous on $[1, T_2] \times U$. So Lemma 4.5 bounds $N^{k-1}P_N(n)$ above for $n/N \in U$ and bounds $N^{k-1}P_N(n^0)$ below, and the bound outside U is $O(N^{-(k-1)})$. This gives the first display for all $1 \le T \le N^{1/3}$, and (4.3) is its logarithm.

The maximizer. Let $\hat n$ be a maximizer and let $\hat x$ be the displacement of $\hat n/N$. For $T \ge T_3$ with T_3 large, the bound outside U is below $X/2 \le P_N(n^0)$, so $\hat n/N \in U$, and the error in (4.4) has $|E| \le \frac12$. We take logarithms in $P_N(\hat n) \ge P_N(n^0)$ and use (4.4) on the left. There $\log(C_F/K_F) \le C_3|\hat x|$ and $T(I_F - \lambda_F) \ge c_U T|\hat x|^2$, while the right side is at least $\log X - C(1/T + T^2/N)$. So

$$c_U T|\hat x|^2 \le C_3|\hat x| + C_4\Big(\frac1T + \frac{T^2}{N}\Big) \le C_3|\hat x| + \frac{2C_4}{T}.$$

The last step holds because $T \le N^{1/3}$ gives $T^2/N \le 1/T$, and this is the step that needs the exponent $\frac13$. Solving the quadratic inequality gives $|\hat x| \le \big(C_3 + \sqrt{C_3^2 + 8c_U C_4}\big)/(2c_U T) = O(1/T)$. For $1 \le T < T_3$ the claim holds since $|\hat x|$ is bounded. □

4.3 Crossings and tie windows

We now compare two faces. By Theorem 4.6, the difference $\log M_F - \log M_G$ is an explicit increasing function of T up to an error $O(1/T + T^2/N)$, and its zero is the crossing.

Theorem 4.7 (the crossing equation)**.** *Let F and G satisfy (H), with $\Delta = k_F - k_G \ge 1$ and $\delta = \lambda_G - \lambda_F > 0$, and let $D_N(T) = \log M_F - \log M_G$.*

(i) For large N, D_N has a zero in $[\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$.

(ii) For large N, every zero $T_N \in [1, N^{1/3}]$ of D_N satisfies

$$\delta T_N + \frac{\Delta}{2}\log T_N = \Delta L - \log\frac{K_F}{K_G} + O\Big(\frac1L\Big). \tag{4.5}$$

(iii) All such zeros lie in one interval of length $O(1/L)$, and

$$T_N = \frac{\Delta}{\delta}\Big[L - \frac12\log L + b_{FG}\Big] + O\Big(\frac{\log L}{L}\Big), \qquad b_{FG} = -\frac12\log\frac{\Delta}{\delta} - \frac{1}{\Delta}\log\frac{K_F}{K_G}.$$

Proof. We subtract (4.3) for G from (4.3) for F. Uniformly on $[1, N^{1/3}]$,

$$D_N(T) = -\Delta L + \Xi(T) + \log\frac{K_F}{K_G} + O\Big(\frac1T + \frac{T^2}{N}\Big), \qquad \Xi(T) = \delta T + \frac{\Delta}{2}\log T.$$

The function Ξ is increasing, with $\Xi' \ge \delta$.

(i) At $T = \frac{\Delta}{2\delta}L$ the display gives $D_N = -\frac{\Delta}{2}L + O(\log L) < 0$, and at $T = \frac{2\Delta}{\delta}L$ it gives $D_N = \Delta L + O(\log L) > 0$. For fixed N, M_F is a maximum of finitely many polynomials in h, and it is positive because F is admissible. The same holds for G. So D_N is continuous and has a zero in between.

(ii) First we locate a zero $T_N \in [1, N^{1/3}]$ roughly. The error term is $O(1)$ on $[1, N^{1/3}]$, so $\Xi(T_N) = \Delta L - \log(K_F/K_G) + O(1)$. This is larger than $\Xi(\frac{\Delta}{2\delta}L) = \frac{\Delta}{2}L + O(\log L)$, so $T_N > \frac{\Delta}{2\delta}L$ because Ξ is increasing. Also $\delta T_N \le \Xi(T_N)$, so $T_N \le \frac{2\Delta}{\delta}L$. With T_N of order L the error term is $O(1/T_N + T_N^2/N) = O(1/L)$, and $D_N(T_N) = 0$ is (4.5).

(iii) Let T° be the solution of $\Xi(T^\circ) = \Delta L - \log(K_F/K_G)$. By (ii) every zero has $\Xi(T_N) = \Xi(T^\circ) + O(1/L)$, and $\Xi' \ge \delta$, so every zero is within $O(1/L)$ of T°. To solve for T°, note that $T^\circ = \frac{\Delta}{\delta}L + O(\log L)$, so $\log T^\circ = \log L + \log\frac{\Delta}{\delta} + O(\log L/L)$. Putting this into $\delta T^\circ = \Delta L - \frac{\Delta}{2}\log T^\circ - \log(K_F/K_G)$ and dividing by δ gives the explicit form. □

In words, the larger face F pays $N^{-\Delta}$ for spreading and gains $e^{\delta T}$ for staying, and its Gaussian peak gives an extra factor $T^{\Delta/2}$. The crossing balances the three. In the numerical test after Corollary 7.1 the explicit form in (iii) is still off by 0.06 at $N = 10^{14}$, so computations should solve (4.5). We do not prove that the zero of D_N is unique in general.

At a switch value τ^* of Theorem 3.4 several faces minimize Φ_{τ^*}, sometimes more than two, as all faces do at $\tau^* = 1$ for the complete graph (Corollary 5.3). Then the competition plays out in a window $T = \tau^*(L - \frac{1}{2}\log L + t)$ with t of order 1. In this window each tied face has a line $\ell_F(t)$, and the mode follows the highest line.

Theorem 4.8 (tie windows)**.** *Fix $\tau^* > 0$. Let m^* be the least value of Φ_{τ^*} over the admissible faces and $\mathcal{F}^*$ the set of minimizers. Assume that every face in $\mathcal{F}^*$ is irreducible. By Theorem 3.4(c) this holds if Q has symmetric support or μ has full support. Put $T = \tau^*(L - \frac{1}{2}\log L + t)$, and for $F \in \mathcal{F}^*$ let $\ell_F(t) = \log K_F + \frac{k_F - 1}{2}\log\tau^* - \tau^*\lambda_F t$ and $E(t) = \max_{G\in\mathcal{F}^*} \ell_G(t)$. The function E is convex and piecewise linear. We call the finitely many points where its slope changes the* kinks *of E, and we write t_j for them.*

(i) Uniformly for t in compact sets, for $F \in \mathcal{F}^*$, $\log M_F = -m^*L + \frac{m^*}{2}\log L + \ell_F(t) + O(\log L/L)$.

(ii) There are $\varepsilon_1, \varepsilon_0 > 0$ such that for large N and $|T/L - \tau^*| \le \varepsilon_1$, every admissible $F \notin \mathcal{F}^*$ has $M_F \le N^{-\varepsilon_0}\max_{G\in\mathcal{F}^*} M_G$.

(iii) There are $\varepsilon, t_0 > 0$ such that the following holds for large N. If $t_0 \le t \le \varepsilon L$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $[t_0, \infty)$. If $-\varepsilon L \le t \le -t_0$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $(-\infty, -t_0]$. A face of the first kind has the least λ_F in $\mathcal{F}^*$, and a face of the second kind has the largest.

Consequently the following holds for every $\eta > 0$ and large N. If $|t| \le \varepsilon L$ and t is at distance at least η from every kink of E, then every global mode lies on a face whose line equals E near t. So for $|t| \le \varepsilon L$ the line of the face that carries the global mode changes only at $T = \tau^*(L - \frac{1}{2}\log L + t_j) + o(1)$, where t_j runs over the kinks of E, and it changes near each kink. Coincident lines do not distinguish their faces at this order.

In words, inside the window the logarithm of every tied face maximum is the same leading term plus the line $\ell_F(t)$, faces that do not tie are smaller by a power of N, and far out in the window the face with the least (for $t > 0$) or largest (for $t < 0$) exit rate wins. In between, the maximizing line changes only near the kinks of E. Faces with the same line can require a finer comparison.

Idea of the proof. In the window T/L is close to τ^*. For a tied face the terms $-(k_F - 1)L$ and $-T\lambda_F$ of (4.3) add up to $-m^*L + \tau^*\lambda_F(\frac{1}{2}\log L - t)$, because $\tau^*\lambda_F + k_F - 1 = m^*$. The term $\frac{k_F-1}{2}\log T$ is $\frac{k_F-1}{2}\log L$ plus a constant, up to a small error. So the multiples of $\log L$ add up to $\frac{m^*}{2}\log L$. This and $-m^*L$ are the same for all tied faces, and what is left is the line $\ell_F(t)$. This is (i). A face that does not tie has a larger exponent Φ_{τ^*}, and the first-order bounds already make it smaller by a power of N. This is (ii). For (iii), two tied

faces with different exit rates have lines of different slopes, so the gap between their lines grows linearly in $|t|$ and beats every error term once $|t|$ is large.

Proof. Every $F \in \mathcal{F}^*$ is admissible and irreducible, so it satisfies (H) by Lemma 2.2(b). We first put T into (4.3). Write $T = \tau^* L(1+y)$ with $y = (t - \frac{1}{2}\log L)/L$, so that $|y| \le \frac{1}{2}$ for $|t| \le L/4$ and large N. Then $\log T = \log\tau^* + \log L + \log(1+y)$ and $T\lambda_F = \tau^*\lambda_F(L - \frac{1}{2}\log L + t)$, and the error term of (4.3) is $O(1/L)$. Since $\tau^*\lambda_F + k_F - 1 = m^*$, we get, uniformly for $|t| \le L/4$,

$$\log M_F = -m^* L + \tfrac{m^*}{2}\log L + \ell_F(t) + \tfrac{k_F - 1}{2}\log(1+y) + O(1/L) \qquad (F \in \mathcal{F}^*). \tag{4.6}$$

(i) For t in a compact set, $y = O(\log L/L)$, so $\log(1+y) = O(\log L/L)$ in (4.6).

(ii) Let $g > 0$ be the least value of $\Phi_{\tau^*}(F) - m^*$ over the admissible $F \notin \mathcal{F}^*$. (If there is no such face, (ii) says nothing, and we put $g = 1$.) Put $\varepsilon_1 = \min\bigl(\tau^*/2,\, g/(4q_{\max} + 4)\bigr)$. (Any constant below $g/(2q_{\max})$ would do in the second entry. We use that $0 \le \lambda_F \le q_{\max}$ for every face, since the Perron root of Q_F lies between its largest diagonal entry and 0.) The entry $\tau^*/2$ keeps T of order L. Namely, if $|T/L - \tau^*| \le \varepsilon_1$, then $\frac{1}{2}\tau^* L \le T \le \frac{3}{2}\tau^* L$. So for large N we have $T_0 \le T \le N^{1/3}$, which is the range of Theorem 3.3(ii), and $\log T = O(\log L)$. Take an admissible $F \notin \mathcal{F}^*$ and any $G \in \mathcal{F}^*$. The difference $\Phi_\tau(F) - \Phi_\tau(G)$ is at least g at $\tau = \tau^*$, and its slope in τ is $\lambda_F - \lambda_G$, which is at most $q_{\max}$ in absolute value. So $\Phi_\tau(F) - \Phi_\tau(G) \ge g/2$ for $|\tau - \tau^*| \le \varepsilon_1$. With $\tau = T/L$ the upper bound of Theorem 3.3(ii) for F and the lower bound for G give $\log M_F - \log M_G \le -(\Phi_\tau(F) - \Phi_\tau(G))L + O(\log L) \le -\frac{g}{4}L$ for large N. So (ii) holds with $\varepsilon_0 = g/4$.

(iii) Take $\varepsilon = \min(\frac{1}{4}, \varepsilon_1/(2\tau^*))$. Then for $|t| \le \varepsilon L$ and large N we have $|T/L - \tau^*| = \tau^*|y| \le \varepsilon_1$, so (ii) applies and no global mode lies on a face outside $\mathcal{F}^*$. It remains to compare the tied faces. Let $t_0 \ge 1$ be larger than every $|t_j|$, so that E is a single line on $[t_0, \infty)$. Let $F \in \mathcal{F}^*$ have this line, and let $G \in \mathcal{F}^*$ have another line. We show that $M_F > M_G$ for $t_0 \le t \le \varepsilon L$ if t_0 and N are large. Since $\ell_F \ge \ell_G$ on $[t_0, \infty)$, the slopes satisfy $\lambda_F \le \lambda_G$. As both faces are tied,

$$\Delta = k_F - k_G = \tau^*(\lambda_G - \lambda_F) \ge 0.$$

There are two cases.

First let $\Delta = 0$. Then the two lines are parallel, and $\ell_F - \ell_G = \log(K_F/K_G) > 0$. The terms with $\log(1+y)$ in (4.6) are the same for F and G, so $\log(M_F/M_G) = \log(K_F/K_G) + O(1/L) > 0$ for large N.

Now let $\Delta > 0$. Then $\ell_F(t) - \ell_G(t) = \Delta t + \ell_F(0) - \ell_G(0)$, and (4.6) gives

$$\log\frac{M_F}{M_G} = \Delta t + \ell_F(0) - \ell_G(0) + \frac{\Delta}{2}\log(1+y) + O\Bigl(\frac{1}{L}\Bigr).$$

For large N and $t \ge 1$ we have $|\log(1+y)| \le 2|y| \le (2t + \log L)/L \le t/2$. So the right side is at least $\frac{3}{4}\Delta t - C$, which is positive for $t \ge t_0$ once t_0 is large.

So for $t_0 \le t \le \varepsilon L$ every global mode lies on a face whose line is E on $[t_0, \infty)$. The slope of ℓ_F is $-\tau^*\lambda_F$, and E follows the line of largest slope on the far right, so such a face has the least λ_F in $\mathcal{F}^*$. The case $t \le -t_0$ is the same with the signs reversed.

The consequence. Fix $\eta > 0$ and let $S_\eta = [-t_0, t_0] \setminus \bigcup_j (t_j - \eta, t_j + \eta)$, a finite union of closed intervals without kinks. Take $G \in \mathcal{F}^*$. The function $E - \ell_G$ is convex, piecewise linear and nonnegative, and its slope changes only at kinks of E. So its zero set is empty, or a single kink of E, or an interval whose finite endpoints are kinks of E. Hence each interval of S_η lies inside the zero set or is disjoint from it. In the first case $\ell_G = E$ on the interval, and in the second $E - \ell_G \ge g_\eta$ there for some $g_\eta > 0$. Now let $t \in S_\eta$

and let $F \in \mathcal{F}^*$ have $\ell_F(t) = E(t)$. By (i), every $G \in \mathcal{F}^*$ whose line is not E near t has $\log(M_F/M_G) \geq g_\eta - O(\log L/L) > 0$, and by (ii) the faces outside $\mathcal{F}^*$ are smaller still. This is the claim for $t \in S_\eta$. For $t_0 \leq |t| \leq \varepsilon L$ it is (iii). Since η is arbitrary, the line of the face that carries the mode can change only at $t = t_j + o(1)$. It does change near each kink, because E is a different line on the two sides of a kink. □

4.4 A tree formula for every face

The next proposition extends Proposition 4.2(b) from reversible faces to every irreducible face. It uses two counts of spanning trees in place of one, namely tree as before and the count arb of spanning trees oriented toward a root, which is defined just before Lemma A.3. Its proof rests on Lemma A.3 in Appendix A.

Proposition 4.9 (the constant on a general face)**.** *Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ put*

$$f_{ij}(\alpha) = l_{\alpha,i}\, q_{ij}\, r_{\alpha,j} \qquad (i \neq j \text{ in } F),$$

which does not change when r_α is scaled.

(a) The weights $f(\alpha)$ are the stationary flow of the tilted chain, that is, $f_{ij}(\alpha) = \alpha_i(\widehat{Q}_\alpha)_{ij}$, and $\sum_{j\neq i} f_{ij}(\alpha) = \sum_{j\neq i} f_{ji}(\alpha) = \alpha_i\theta_i(\alpha)$ for every i. The number $\mathrm{arb}_i(f(\alpha))$ does not depend on the root i, and

$$\det \Sigma_\alpha = \frac{(\prod_i \alpha_i)^2\ \mathrm{tree}\big(f(\alpha) + f(\alpha)^\top\big)}{\mathrm{arb}(f(\alpha))^2}$$

and

$$C_F(\alpha) = \frac{(\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1})}{(2\pi)^{(k-1)/2}}\,\frac{\mathrm{arb}(f(\alpha))}{\prod_i \alpha_i\,\sqrt{\mathrm{tree}(f(\alpha)+f(\alpha)^\top)}}.$$

(b) Also $\mathrm{arb}(f(\alpha))/\prod_i \alpha_i = \sum_{i\in F}\det\big(\mathrm{diag}\,\theta(\alpha) - A_F\big)_{(i)}$. At $\alpha = \alpha_F^*$ this is $\prod_\rho(\rho - \lambda_F)$, the product over the other $k-1$ eigenvalues ρ of $-Q_F$, counted with multiplicity. So with $l_F\cdot r_F = 1$ and $f_{ij} = l_{F,i}\,q_{ij}\,r_{F,j}$,

$$K_F = \frac{(\mu\cdot r_F)(l_F\cdot\mathbf{1})}{(2\pi)^{(k-1)/2}}\,\frac{\mathrm{arb}(f)}{\prod_i \alpha^*_{F,i}\,\sqrt{\mathrm{tree}(f+f^\top)}} = \frac{(\mu\cdot r_F)(l_F\cdot\mathbf{1})}{(2\pi)^{(k-1)/2}}\,\frac{\prod_\rho(\rho-\lambda_F)}{\sqrt{\mathrm{tree}(f+f^\top)}}.$$

(c) In the reversible case of Proposition 4.2(b), $f(\alpha)$ is symmetric and equals $c(\alpha)$, and (a) *reduces to Proposition 4.2(b).*

(d) Let $F = \{1,\dots,k\}$ be a directed cycle, that is, $q_{i,i+1} = a_i > 0$ with indices modulo k, and $q_{ij} = 0$ for all other $i \neq j$ in F. Then for every $\alpha \in \Delta_F^\circ$

$$\det\Sigma_\alpha = \frac{k\,(\prod_i \alpha_i)^2}{\big(\prod_i \alpha_i a_i\big)^{(k-1)/k}}.$$

If moreover $q_i = a_i$ for all i, then $\alpha^*_{F,i} = \varphi^*/a_i$ with $\varphi^* = (\sum_i 1/a_i)^{-1}$, and

$$\det\Sigma_{\alpha^*_F} = \frac{k\,(\varphi^*)^{k+1}}{\prod_i a_i^2}, \qquad K_F = \frac{\mu(F)\,\prod_i a_i}{(2\pi)^{(k-1)/2}\,\sqrt{k}\,(\varphi^*)^{(k+1)/2}}.$$

In words, up to the factor $(2\pi)^{-(k-1)/2}/\prod_i\alpha_i$, the Gaussian factor in C_F is a count of oriented spanning trees for the stationary flow of the tilted chain, divided by the square root of a count of spanning trees for the symmetrized flow. For the directed 3-cycle of Proposition 4.14 part (d) gives $\varphi^* = b/(1+2b)$ and $K_F = (1+2b)^2/(2\sqrt{3}\,\pi b)$ again.

Idea of the proof. Lemma A.3 applies to the tilted generator $\widehat{Q}_\alpha$, whose stationary law is α. Divided by $\prod_i \alpha_i$, the count of oriented trees is the sum of the principal minors of order $k-1$ of $-\widehat{Q}_\alpha$, and these minors do not change under the diagonal conjugation that defines $\widehat{Q}_\alpha$. At α^*_F the sum of these minors is the elementary symmetric polynomial of degree $k-1$ in the eigenvalues of $-Q_F - \lambda_F I$, namely the product of the numbers $\rho - \lambda_F$ over the other eigenvalues ρ of $-Q_F$. (Up to the sign $(-1)^{k-1}$ this is the derivative of the characteristic polynomial of $-Q_F$ at its root λ_F.)

Proof. (a) By Definition 4.1, $(\widehat{Q}_\alpha)_{ij} = q_{ij} r_{\alpha,j}/r_{\alpha,i}$ for $i \neq j$, and $\alpha_i = l_{\alpha,i} r_{\alpha,i}$. So $\alpha_i (\widehat{Q}_\alpha)_{ij} = f_{ij}(\alpha)$. The diagonal entry of $\widehat{Q}_\alpha$ is $-\theta_i(\alpha)$. So $\widehat{Q}_\alpha \mathbf{1} = 0$ gives $\sum_{j\neq i} f_{ij}(\alpha) = \alpha_i \theta_i(\alpha)$, and $\alpha^\top \widehat{Q}_\alpha = 0$ gives $\sum_{j\neq i} f_{ji}(\alpha) = \alpha_i \theta_i(\alpha)$. Now apply Lemma A.3 to $G = \widehat{Q}_\alpha$ and $\pi = \alpha$, for which $\Sigma = \Sigma_\alpha$. Part (iii) is the formula for $\det \Sigma_\alpha$, and Definition 4.1 then gives $C_F(\alpha)$.

(b) We have $-\widehat{Q}_\alpha = R^{-1}(\operatorname{diag}\theta(\alpha) - A_F)R$ with R diagonal, and a conjugation by a diagonal matrix does not change principal minors. So Lemma A.3(i) gives the first claim. At α^*_F we have $\theta(\alpha^*_F) = q - \lambda_F \mathbf{1}$, so $\operatorname{diag}\theta(\alpha^*_F) - A_F = -Q_F - \lambda_F I$. For a $k \times k$ matrix M the sum of the principal minors of order $k-1$ is the elementary symmetric polynomial of degree $k-1$ in the eigenvalues of M. The eigenvalues of $-Q_F - \lambda_F I$ are the numbers $\rho - \lambda_F$, where ρ runs over the eigenvalues of $-Q_F$. Exactly one of them is 0, because the Perron root is a simple eigenvalue. So the symmetric polynomial is the product of the other $k-1$ numbers. Finally $r_{\alpha^*_F}$ and $l_{\alpha^*_F}$ are multiples of r_F and l_F with $l_{\alpha^*_F} \cdot r_{\alpha^*_F} = 1$, and (a) gives the two forms of K_F.

(c) By Proposition 4.2(b), $\alpha_i (\widehat{Q}_\alpha)_{ij} = c_{ij}(\alpha)$, so $f(\alpha) = c(\alpha)$ is symmetric. Then $\operatorname{arb}(f(\alpha)) = \operatorname{tree}(c(\alpha))$, because every spanning tree has exactly one orientation toward a given root. Also $\operatorname{tree}(2c) = 2^{k-1} \operatorname{tree}(c)$, because a spanning tree has $k-1$ edges. So (a) gives $\det \Sigma_\alpha = 2^{k-1} (\prod_i \alpha_i)^2 / \operatorname{tree}(c(\alpha))$.

(d) Only the pairs $(i, i+1)$ carry flow, so by (a) the flow $f_{i,i+1}(\alpha)$ out of i equals the flow $f_{i-1,i}(\alpha)$ into i. Hence all $f_{i,i+1}(\alpha)$ are equal to one number φ. Their product is $\prod_i l_{\alpha,i}\, a_i\, r_{\alpha,i+1} = \prod_i \alpha_i a_i$, so $\varphi = (\prod_i \alpha_i a_i)^{1/k}$. The only spanning tree oriented toward a root i with nonzero weight is the cycle without the edge that leaves i. So $\operatorname{arb}(f(\alpha)) = \varphi^{k-1}$. The symmetrized flow has conductance φ on each of the k edges of the undirected cycle. For $k \geq 3$ the spanning trees are the cycle without one edge, so $\operatorname{tree}(f(\alpha) + f(\alpha)^\top) = k\varphi^{k-1}$. For $k = 2$ the two edges of the cycle are the same pair, the conductance is 2φ, and the count is again $k\varphi^{k-1}$. Now (a) gives $\det \Sigma_\alpha = k(\prod_i \alpha_i)^2 / \varphi^{k-1}$. If $q_i = a_i$, then Q_F is a generator, so $\lambda_F = 0$, $r_F = \mathbf{1}$ and $l_F = \alpha^*_F$ is the stationary law. Its flow $l_{F,i} a_i$ is constant, which gives $\alpha^*_{F,i} = \varphi^*/a_i$ and $\varphi = \varphi^*$ at α^*_F. Then $\prod_i \alpha^*_{F,i} = (\varphi^*)^k / \prod_i a_i$, and $(\mu \cdot r_F)(l_F \cdot \mathbf{1}) = \mu(F)$. Putting this into the formulas above gives $\det \Sigma_{\alpha^*_F}$ and K_F. □

On the complete graph the tree sum of Proposition 4.2(b) has a closed form. This is classical, and we include a short proof.

Lemma 4.10 (weighted Cayley formula)**.** *For $x_1, \dots, x_k > 0$ and the conductances $c_{ij} = x_i x_j$,*

$$\operatorname{tree}(c) = x_1 x_2 \cdots x_k \, (x_1 + \cdots + x_k)^{k-2}.$$

With every $x_i = 1$ this is Cayley's count k^{k-2} of the spanning trees of the complete graph K_k.

Proof. Put $X = \sum_i x_i$. The Laplacian of the conductances is $\operatorname{diag}(Xx_i) - xx^\top$, so without its first row and column it is $\operatorname{diag}(Xx_2, \dots, Xx_k) - x' x'^\top$ with $x' = (x_2, \dots, x_k)$. The matrix determinant lemma gives the determinant $X^{k-1} \prod_{i\geq 2} x_i \, (1 - \sum_{i\geq 2} x_i/X) = X^{k-2} \prod_i x_i$,

and this is $\mathrm{tree}(c)$ by the matrix-tree theorem [5]. (For integer conductances this is the multigraph case, and both sides are polynomials in the c_{ij}.) □

Example 4.11 (the complete face)**.** Let F be a face of $k \geq 2$ states with $q_{ij} = 1$ for all $i \neq j$ in F, and any exit rates, for example a face of $Q = J - dI$, where $q_i = d - 1$. Then F is reversible with $\nu = \mathbf{1}$, so by Proposition 4.2(b) we may take $r_\alpha = l_\alpha = \sqrt{\alpha}$ (entrywise), and the conductances $c_{ij}(\alpha) = \sqrt{\alpha_i \alpha_j}$ are products. Put $A = \sum_{i \in F} \sqrt{\alpha_i}$. Lemma 4.10 gives $\mathrm{tree}(c(\alpha)) = (\prod_i \alpha_i)^{1/2} A^{k-2}$, so for every $\alpha \in \Delta_F^\circ$

$$C_F(\alpha) = \frac{(\mu \cdot \sqrt{\alpha})\, A^{k/2}}{(4\pi)^{(k-1)/2}\, (\prod_i \alpha_i)^{3/4}} = \frac{(\mu \cdot \sqrt{\alpha})\, A}{\sqrt{\det(2\pi \Sigma_\alpha)}}, \qquad I_F(\alpha) = \sum_{i \in F} q_i \alpha_i - (A^2 - 1).$$

The second form of $C_F(\alpha)$ uses $\det \Sigma_\alpha = 2^{k-1} (\prod_i \alpha_i)^2 / \mathrm{tree}(c(\alpha))$. For $Q = J - dI$ the rate is $I_F(\alpha) = d - A^2$, the rate of Dupuis and Liu [20, Eq. (3.2)]. At the center $\alpha = \mathbf{1}/k$ we have $A^2 = k$ and $C_F = \mu(F) k^{k-1/2} (4\pi)^{-(k-1)/2}$, which is Corollary 4.3 with $\mathrm{tree}(q_F) = k^{k-2}$. With the uniform start, $dC_F(\alpha)$ is the constant $D(\alpha)$ of [36, Theorem 3.3 and Proposition 3.6], written there as a mass over a Gaussian volume. Section 5 uses this example.

Remark 4.12 (the rate through flows)**.** Let F be irreducible with $k \geq 2$, and let $\alpha \in \Delta_F^\circ$ and $f = f(\alpha)$. By Proposition 4.9(a) and $I_F(\alpha) = \langle q - \theta(\alpha), \alpha \rangle$,

$$I_F(\alpha) = \sum_{i \in F} \alpha_i q_i - \sum_{i \neq j} f_{ij},$$

so the rate is the drop in the total jump rate, from the original chain at α to the tilted one. Also $f_{ij} f_{ji} = \alpha_i \alpha_j q_{ij} q_{ji}$, since the factors $r_{\alpha,j}/r_{\alpha,i}$ cancel. As $\sqrt{f_{ij} f_{ji}} = \sqrt{\alpha_i q_{ij} \alpha_j q_{ji}}$, the right side $R_F(\alpha)$ of the formula in Proposition 3.1(c), which is defined on every face, satisfies

$$R_F(\alpha) - I_F(\alpha) = \sum_{\{i,j\}} \bigl(\sqrt{f_{ij}} - \sqrt{f_{ji}}\bigr)^2 \geq 0,$$

the sum over the unordered pairs in F. It vanishes exactly when f is symmetric. This is the case on a reversible face by Proposition 4.2(b). Conversely, if f is symmetric, then $\nu_i = \alpha_i / r_{\alpha,i}^2$ satisfies $\nu_i q_{ij} = \nu_j q_{ji}$, so the face is reversible. So a face that is not reversible lets a net flow circulate, and its rate is smaller than the reversible formula.

The killing only adds a linear term. Writing $Q_F u$ out in the definition of I_F gives $I_F(\alpha) = \sum_{i \in F} \alpha_i q_i^{\mathrm{out}}(F) + I_F^0(\alpha)$, where I_F^0 is the Donsker–Varadhan rate of the chain on F with the rates q_{ij}, $i, j \in F$, and no killing. So statements about I_F^0 apply to I_F. The rate $I_F^0(\alpha)$ is the least value of $\sum_{q_{ij} > 0} \bigl(J_{ij} \log \frac{J_{ij}}{\alpha_i q_{ij}} - J_{ij} + \alpha_i q_{ij} \bigr)$ over the balanced flows J on F, the cost of Bertini, Faggionato and Gabrielli [3] for the empirical measure together with the empirical flow (see [30, (1.3) to (1.7)]). The minimizing flow exists and is unique [2, Theorem 1.6], and it is $\alpha_i q_{ij} r_{\alpha,j} / r_{\alpha,i}$ by [30, Lemma 5.7], which is $f_{ij}(\alpha)$.

4.5 Reducible faces

This subsection shows by an example that the coefficient $\frac{1}{2}$ can fail on a face that is not irreducible.

Remark 4.13 (scope of the $\frac{1}{2}$)**.** The coefficient $\frac{1}{2}$ of $\log \log N$ in Theorems 4.7 and 4.8 is due to the power $T^{(k-1)/2}$ in Theorem 4.6, so it needs only (H). At a kink of E where the line of G gives way to that of F as t increases, $\Delta = \tau^* \delta$ and $t_j = b_{FG}$. So every change in a tie window is a crossing of Theorem 4.7. The best composition of a reducible face can visit a state only once, with no Gaussian window in that direction. Then the $\frac{1}{2}$ can fail, and in the next example it is 1 and then 0.

Proposition 4.14 (a directed 3-cycle)**.** *Let $d = 3$, $q_{12} = b > 2$, $q_{23} = q_{31} = 1$, all other rates 0, and $\mu_1 = 1$.*

(i) The admissible faces are $\{1\}$, $\{1,2\}$ (reducible, $\lambda = 1$) and $\{1,2,3\}$ (irreducible, $\lambda = 0$). Exactly $M_{\{1\}} = (1-bh)^{N-1}$ and $M_{\{1,2\}} = bh(1-h)^{N-2}$, and the latter is attained only at $n = (1, N-1, 0)$. For the full face, $\alpha^* = (1,b,b)/(1+2b)$, $\det \nabla^2 I = (1+2b)^4/(3b^2)$ and $K_{\{1,2,3\}} = (1+2b)^2/(2\sqrt{3}\,\pi b)$.

(ii) The equation $M_{\{1\}} = M_{\{1,2\}}$ has exactly one root $T_N^{(1)}$ with $bh < 1$. It satisfies $(b-1)T + \log T = L - \log b + O(T^2/N)$, so $T_N^{(1)} = \frac{1}{b-1}[L - \log L + \log \frac{b-1}{b}] + O(\log L/L)$, and the coefficient of $\log\log N$ in the bracket is 1.

(iii) Every root $T \in [L/2, 2L]$ of $M_{\{1,2\}} = M_{\{1,2,3\}}$ satisfies $T = L + \log \frac{2\sqrt{3}\,\pi b^2}{(1+2b)^2} + O(1/L)$, so the coefficient of $\log\log N$ is 0. These roots lie in an interval of length $O(1/L)$.

(iv) Both are changes of the global mode. For large N and $T \le N^{1/3}$, the mode is $(N,0,0)$ for $T < T_N^{(1)}$, it is $(1, N-1, 0)$ between $T_N^{(1)}$ and the smallest root in (iii), and it lies on the full face above the largest root in (iii).

In words, the face $\{1,2\}$ is reducible, and its best composition $(1, N-1, 0)$ visits state 1 once and then stays in state 2. Its maximum is about $b(T/N)e^{-T}$, with the power T^1 where an irreducible pair would have $T^{1/2}$. This changes the coefficient of $\log\log N$ at both crossings.

Sketch of the proof. (i) From state 1 the chain can only move along $1 \to 2 \to 3 \to 1$, so the admissible faces are the three listed. A path with support $\{1,2\}$ is $1^{n_1}2^{n_2}$, of probability $(1-bh)^{n_1-1}\,bh\,(1-h)^{n_2-1}$, which is largest at $n_1 = 1$ because $b > 1$. On the full face $\lambda = 0$ and α^* is the stationary law, and the AM–GM inequality gives $I(\alpha) = \sum_i q_i\alpha_i - 3(b\alpha_1\alpha_2\alpha_3)^{1/3}$, whose Hessian at α^* gives $K_{\{1,2,3\}}$ by Proposition 4.2(a). Proposition 4.9(d) gives the same value. (ii) The logarithm of $M_{\{1,2\}}/M_{\{1\}}$ increases strictly in h, and its expansion $\log b + \log T - L + (b-1)T + O(T^2/N)$ locates the root. (iii) The exact $M_{\{1,2\}}$ is compared with Theorem 4.6 for the full face, where $k = 3$ gives the power T^1, the same power as in $M_{\{1,2\}}$, so $\log T$ cancels. (iv) The three face maxima are compared on the ranges $T < T_N^{(1)}$, $T_N^{(1)} < T \le L/2$, $[L/2, 2L]$ and $[2L, N^{1/3}]$, with Lemma 3.2 for the full face on the second range. The full proof is in Section S2 of the supplement. The exact parts of (i) and the uniqueness of the root in (ii) are checked in Lean.

4.6 The next term

The local law, the face maxima and the crossing equation each have a next term, of relative order $1/T$. For a face F that satisfies (H) with $k \ge 2$ we write $\Sigma_F = \Sigma_{\alpha_F^*}$ and $\gamma_F = \nabla \log C_F(\alpha_F^*)$, in the coordinates α'. The function c and the constants c_F below are not related to the conductances $c_{ij}(\alpha)$ of Proposition 4.2.

Proposition 4.15 (the next term)**.** *Let F satisfy* (H)*.*

(i) Let $k \ge 2$. There is a real-analytic function c on Δ_F°, given by (B.2)*, such that for every compact $\mathcal{K} \subset \Delta_F^\circ$, uniformly for $\alpha \in \mathcal{K}$ and $T \ge 1$,*

$$f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-TI_F(\alpha)} \Big(1 + \frac{c(\alpha)}{T} + O_{\mathcal{K}}\Big(\frac{1}{T^2}\Big)\Big).$$

(ii) Put $c_F = c(\alpha_F^*) + \frac{1}{2}\gamma_F^\top \Sigma_F \gamma_F$ if $k \ge 2$, and $c_F = 0$ if $k = 1$. For large N, uniformly for $1 \le T \le N^{1/3}$,

$$\log M_F = -(k-1)L + \frac{k-1}{2}\log T + \log K_F - T\lambda_F + \frac{c_F}{T} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big). \qquad (4.7)$$

(iii) Let G also satisfy (H)*, with Δ, δ and D_N as in Theorem 4.7. For large N, every zero $T_N \in [1, N^{1/3}]$ of D_N satisfies*

$$T_N\Big(\Delta L - \frac{\Delta}{2}\log T_N - \delta T_N - \log\frac{K_F}{K_G}\Big) = c_F - c_G + O\Big(\frac{1}{L}\Big). \tag{4.8}$$

(iv) Let $F = \{i, j\}$ with $q_{ij} = q_{ji} > 0$, $q_i = q_j$ and $\mu_i = \mu_j > 0$. Then $c_F = -1/(8q_{ij})$. Let $Q = J - dI$, and let F have $k \geq 2$ states with μ constant and positive on F. Then $c_F = -(k-1)/8$.

In words, the local law has a relative correction $c(\alpha)/T$, where $c(\alpha)$ is the Gaussian mean of a polynomial in the Taylor coefficients of the torus integrand at its peak, divided by the value of the amplitude of the integrand there. The quadratic model shifts its maximizing composition from $N\alpha_F^*$ by $N\Sigma_F\gamma_F/T$ toward larger C_F, and this gains $\frac{1}{2}\gamma_F^\top\Sigma_F\gamma_F/T$ in the logarithm. The crossing equation (4.5) then gains the term $(c_G - c_F)/T_N$ on its right side, with an error $O(L^{-2})$.

Idea of the proof. The coefficient formulas are in Appendix B, and the proof is in Section S4 of the supplement. For (i) we carry the Laplace lemma one step further (Lemma B.2). The terms of order $T^{-1/2}$ and $T^{-3/2}$ are odd and integrate to 0, so the error is $O(T^{-2})$. For (ii) the logarithm of the local law near α_F^* is a concave quadratic plus the linear term $\gamma_F \cdot x$, and we complete the square. The maximizer is within $O(1/T)$ of α_F^*, and rounding the quadratic-model maximizer preserves the claimed remainder. Part (iii) subtracts (4.7) for the two faces. In (iv) the value for a symmetric pair comes from a second expansion of the same probability, and the value for the complete graph from an evaluation of the formula (B.2) by hand.

The term c_F/T is asymptotically resolved when $T \to \infty$ and $T^3/N \to 0$, in particular at the crossings $T \asymp \log N$.

The two values in (iv) are proved in different ways. For a symmetric pair we compare with a second expansion of M_F, which uses only Proposition 5.1, the Bessel bound (2.1) of [35] and the classical expansion of $I_0 + I_1$. These give

$$\log M_{\{i,j\}} = -L + \tfrac{1}{2}\log T + \log K_{\{i,j\}} - T\lambda_{\{i,j\}} - \frac{1}{8q_{ij}T} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big), \tag{4.9}$$

for large N and uniformly for $1 \leq T \leq N^{1/3}$, without the Laplace lemma. For the complete graph we evaluate (B.2) by hand at the center of the face, for every k (Section S4 of the supplement). The center is a symmetric point, so $\gamma_F = 0$ and $c_F = c(\alpha_F^*)$. The torus integrand has a closed form there, and a Gaussian vector with exchangeable coordinates reduces the Wick sums of (B.1) to a few moments. For the pair, (B.2) gives the same value $-1/(8q_{ij})$, which checks both computations, and for the full faces of K_3 and K_4 the script `verify_second_order.py` evaluates (B.2) exactly by computer algebra. For other faces c_F has to be computed from (B.2), γ_F and Σ_F, as we do for the 4-cycle in Section 7.

4.7 A weak field

Give each visit to state i the extra weight $e^{\xi_i/N}$, for a fixed vector $\xi \in \mathbb{R}^d$. Over N steps these weights multiply to $e^{\langle\xi,n\rangle/N}$, a factor of order 1, which is the order of the constants K_F. So such a field cannot change the first-order rule, but it can change the winner inside a tie window. For a composition n put

$$w_\xi(n) = e^{\langle\xi,n\rangle/N}P_N(n), \qquad Z_{N,\xi} = \sum_n w_\xi(n), \qquad \widetilde{P}_{N,\xi}(n) = \frac{w_\xi(n)}{Z_{N,\xi}},$$

and for a face F let $M_F^\xi = \max\{w_\xi(n) : \operatorname{supp} n = F\}$. Write $\|\xi\| = \max_i|\xi_i|$. On the complete graph with the uniform start, $\widetilde{P}_{N,\xi}$ is the law of the counts of the Potts chain in a field of size ξ_i/N at each site in state i [36].

Proposition 4.16 (a weak field)**.** *Fix Q, μ and a compact set $\mathcal{X} \subset \mathbb{R}^d$.*

(a) $\widetilde{P}_{N,\xi}$ is the law of the composition when the law of the path $X_1, \dots, X_N$ is multiplied by $\exp(N^{-1}\sum_{s=1}^{N} \xi_{X_s})$ and normalized. For every face F, $e^{-\|\xi\|} M_F \le M_F^{\xi} \le e^{\|\xi\|} M_F$.

(b) Let F satisfy (H)*. Uniformly for $\xi \in \mathcal{X}$ and $1 \le T \le N^{1/3}$,*

$$M_F^{\xi} = e^{\langle \xi, \alpha_F^* \rangle} M_F \big(1 + O(1/T + T^2/N)\big),$$

and every maximizer $\hat{n}$ of w_ξ on F satisfies $\hat{n}/N = \alpha_F^* + O(1/T)$.

(c) Fix ξ and $\tau^* > 0$, and let m^*, $\mathcal{F}^*$ and ℓ_F be as in Theorem 4.8, with every face in $\mathcal{F}^*$ irreducible. Then Theorem 4.8 holds for the modes of $\widetilde{P}_{N,\xi}$, with M_F replaced by M_F^{ξ} and ℓ_F by $\ell_F^{\xi} = \ell_F + \langle \xi, \alpha_F^* \rangle$. Thus it is the constants $K_F e^{\langle \xi, \alpha_F^* \rangle}$ that determine whether two field-shifted lines coincide.

(d) If Q is irreducible with stationary law π, then $Z_{N,\xi} \to e^{\langle \xi, \pi \rangle}$ as $N \to \infty$ with $T \to \infty$ and $T \le N^{1/3}$.

In words, on a face satisfying (H) a weak field shifts the line by the field averaged over its quasi-ergodic law, since the best composition of F sits within $O(N/T)$ of $N\alpha_F^*$. On the complete graph α_F^* is uniform on F, and the tilt is the average of ξ over F (Corollary 5.14). By (a) the first-order rule of Theorem 3.3(iii) holds for $\widetilde{P}_{N,\xi}$ without change, and by (b) so does the location in Theorem 3.3(iv).

Proof. (a) A path $x_1 \cdots x_N$ has probability $\mu_{x_1} \prod_{s<N} (I + hQ)_{x_s x_{s+1}}$, and $\sum_s \xi_{x_s} = \langle \xi, n \rangle$ depends on the path only through its composition n. So the multiplied law, summed over the paths with composition n, is $w_\xi(n)$, and normalizing gives $\widetilde{P}_{N,\xi}$. Since n/N is a probability vector, $|\langle \xi, n \rangle|/N \le \|\xi\|$, which gives the bounds.

(b) For a singleton the assertion is exact. For $k \ge 2$, use the proof of Theorem 4.6 with the amplitude $C_F^{\xi}(\alpha) = e^{\langle \xi, \alpha \rangle} C_F(\alpha)$. Number the states of F as $1, \dots, k$, put $x = \alpha' - \alpha_F^{*\prime}$ and let $\hat{\xi} = (\xi_i - \xi_1)_{2 \le i \le k}$. Then

$$\log \frac{C_F^{\xi}(\alpha)}{e^{\langle \xi, \alpha_F^* \rangle} K_F} = \log \frac{C_F(\alpha)}{K_F} + \hat{\xi} \cdot x.$$

Thus the linear coefficient γ_F in that proof is replaced by $\gamma_F + \hat{\xi}$, while its quadratic rate bound is unchanged. On the compact field set $\mathcal{X}$ the added coefficient is uniformly bounded. Outside the fixed ball U, the field weight changes the old upper bound by at most $e^{\|\xi\|}$, so the exponential gap still excludes those compositions. Inside U, the same quadratic upper bound and the same lattice lower bound give

$$M_F^{\xi} = e^{\langle \xi, \alpha_F^* \rangle} N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F} \big(1 + O(1/T + T^2/N)\big),$$

and the quadratic location argument gives $|x| = O(1/T)$ at every maximizer. For bounded T the error estimate follows from (a), and the location estimate follows from boundedness of the simplex. Dividing by the unweighted face maximum proves (b), uniformly on $\mathcal{X}$.

(c) The proof of Theorem 4.8 uses (4.3) for the faces in $\mathcal{F}^*$ and the bounds of Theorem 3.3(ii) for the other faces. By (b), (4.3) holds for M_F^{ξ} with $\log K_F$ replaced by $\log K_F + \langle \xi, \alpha_F^* \rangle$, so (4.6) and part (i) of that theorem hold with ℓ_F^{ξ}. By (a), the bounds of Theorem 3.3(ii) hold for M_F^{ξ} with κ replaced by $\kappa + \|\xi\|$, so part (ii) holds. Part (iii) and the consequence use only (4.6) and part (ii), so their proofs apply. Finally the modes of $\widetilde{P}_{N,\xi}$ are the maximizers of w_ξ, and $\max_n w_\xi(n) = \max_F M_F^{\xi}$.

(d) Since $e^{\langle \xi, n\rangle/N}$ lies between $e^{-\|\xi\|}$ and $e^{\|\xi\|}$, by bounded convergence it is enough to show that $n/N \to \pi$ in probability. As Q is irreducible, the face $[d]$ satisfies (H) with $\lambda_{[d]} = 0$, $r_{[d]} = \mathbf{1}$ and $\alpha^*_{[d]} = l_{[d]} = \pi$. Every proper face F has $\lambda_F > 0$, because the Perron root of the irreducible matrix $Q + cI$ (for a large number c) is larger than that of each proper principal submatrix [1]. A face has at most $(N+1)^{|F|-1}$ compositions, so by Theorem 3.3(ii), for $T \geq T_0$, the compositions with a proper support have total probability at most $\sum_F (N+1)^{|F|-1} e^{\kappa} N^{-(|F|-1)} T^{|F|-1} e^{-T\lambda_F} \to 0$. Now take a ball U around π in $\Delta^\circ_{[d]}$. The step outside U in the proof of Theorem 4.6 bounds every composition with full support and $n/N \notin U$ by $X \cdot O(T^{(d-1)/2} e^{-2\eta T}) = O(N^{-(d-1)} T^{d-1} e^{-2\eta T})$, and there are at most $(N+1)^{d-1}$ of them. So $\mathbb{P}(n/N \notin U) \to 0$. □

5 The two-state walk and the complete graph

This section applies the second-order theory to the two simplest chains, the two-state walk and the complete graph, which is the Potts chain. On the complete graph every face ties at first order, and the theory of Section 4 with a few exact facts resolves the tie, also with a start law and a weak field.

5.1 The two-state walk

We use the polynomials $S_{a,b}$ and S_N, the root u_N, the crossing p_N, $z_N = N(1-p_N)$ and $c_0 = \frac{1}{2}\log(\pi/8)$ of Section 2.1. The first result shows that, at every N, the comparison of a symmetric pair with one of its states is exactly the two-state walk.

Proposition 5.1 (an exact pair identity)**.** *Let $i \neq j$ with $q_{ij} = q_{ji} = a$, $q_i = q_j = q$ and $\mu_i = \mu_j > 0$, and let $F = \{i, j\}$. For every $N \geq 2$, $1 \leq m \leq N-1$ and $h > 0$ such that $I + hQ$ is stochastic and $qh < 1$,*

$$\frac{P_N(me_i + (N-m)e_j)}{P_N(Ne_i)} = S_{m,N-m}(u), \qquad u = \frac{ah}{1-qh}.$$

So the balanced compositions, with $m = \lfloor N/2 \rfloor$ or $m = \lceil N/2 \rceil$, are the most likely ones with support F, $M_F = M_{\{i\}} S_N(u)$, and $M_F > M_{\{i\}}$ exactly when $u > u_N$.

Proof. We group the paths by their number of switches. A path with this composition is a word in i, j with m letters i. Each stay has probability $1 - qh$ and each switch ah, so a word with s switches has probability $\mu_i (1-qh)^{N-1-s}(ah)^s = \mu_i (1-qh)^{N-1} u^s$, while $P_N(Ne_i) = \mu_i (1-qh)^{N-1}$. Sum over these words and divide. This gives $\sum_w u^{s(w)} = S_{m,N-m}(u)$, which is the identity. By [35, Proposition 4.1] the polynomials $S_{m,N-m}$ grow coefficientwise as m moves toward the middle, and $S_{m,N-m} = S_{N-m,m}$. So the balanced m give the largest $S_{m,N-m}(u)$. Finally S_N increases strictly and equals 1 only at u_N (Section 2.1). □

Lean checks the identity for a chain with two states. For $d \geq 3$ the paths with this composition never leave F, so the same computation applies, but this case is not in Lean.

Corollary 5.2 (the two-state walk)**.** *Let $d = 2$, $q_{12} = q_{21} = 1$ and $\mu = (\frac{1}{2}, \frac{1}{2})$, the walk of [35] with $T = N(1-p)$. Then $K_{\{1\}} = K_{\{2\}} = \frac{1}{2}$ and $K_{\{1,2\}} = \sqrt{2/\pi}$, so $b_{FG} = c_0$ for $F = \{1,2\}$ and $G = \{1\}$. For every $N \geq 2$, $M_{\{1,2\}} = M_{\{1\}}$ holds exactly at $T = z_N$, and $z_N + \frac{1}{2}\log z_N = L + c_0 + \frac{1}{8 z_N} + O(L^{-2})$.*

Proof. Corollary 4.3 with $k = 2$, $\mathrm{tree}(q_F) = 1$ and $\mu(F) = 1$ gives $K_{\{1,2\}}$, and then $b_{FG} = -\log(K_{\{1,2\}}/K_{\{1\}}) = c_0$. Proposition 5.1 with $a = q = 1$ gives $M_{\{1,2\}} = M_{\{1\}} S_N(h/(1-h))$, and $h/(1-h) = u_N$ exactly when $p = p_N$. So for every N the only solution of $M_{\{1,2\}} = M_{\{1\}}$ is $T = z_N$, and by Theorem 4.7(i) it lies in $[L/2, 2L] \subset [1, N^{1/3}]$ for large N. By

Proposition 4.15(iv), $c_{\{1,2\}} = -\frac{1}{8}$, and a vertex has $c_{\{1\}} = 0$. So (4.8) with $\Delta = \delta = 1$ reads $z_N(L - \frac{1}{2}\log z_N - z_N + c_0) = -\frac{1}{8} + O(1/L)$, and dividing by z_N gives the equation. □

This is the crossing equation of [35] up to its explicit remainder R_N, which [35] bounds at every N.

5.2 The complete graph

From now on in this section $Q = J - dI$. The chain switches to each other state with probability h (the r of [36]) and keeps its state with probability $p = 1 - (d-1)h$. The following symbols of [36] are local to this section and to Appendix C. Put

$$\sigma = p - h, \qquad u = \frac{h}{p}, \qquad v = \frac{h}{\sigma} = \frac{u}{1-u}, \qquad V_N = \frac{p^{N-1}}{d}.$$

So u is the odds of a switch to one given state against no switch, and V_N is the probability of a vertex Ne_i for the uniform start, that is, $M_{\{i\}}$. For a composition n we write $S_n(u)$ for the sum of u^j over the words with the counts n, where j is the number of switches of the word, and $W_n(j)$ for the number of these words with j switches. For a composition with some zero coordinates, S_n is the polynomial of its positive coordinates, which does not depend on d. By (2.3), the uniform start has

$$P_N(n) = V_N S_n(u) \qquad (0 < p \le 1), \tag{5.1}$$

and so does every start that is constant on $\operatorname{supp} n$, with V_N replaced by $\mu_a p^{N-1}$ for $a \in \operatorname{supp} n$. A composition is *balanced* if its positive coordinates differ by at most 1. For the uniform start all faces of k states have the same face maximum M_F, and all balanced compositions with k positive coordinates have the same probability. For $p \ge 1/d$ this probability is M_F, by Theorem 5.5 below. (The preprint [36] writes $C_{N,k}$ for it.) Put

$$a_k = \frac{1}{2}\log(4\pi) - \frac{k+\frac{1}{2}}{k-1}\log k, \qquad D_k = e^{-(k-1)a_k} = \frac{k^{k+1/2}}{(4\pi)^{(k-1)/2}} \qquad (k \ge 2),$$

and $D_1 = 1$. Numerically $a_2 = -0.4674$, $a_3 = -0.6571$ and $a_4 = -0.8139$.

Corollary 5.3 (the complete graph)**.** *A face F of k states has $\lambda_F = d - k$ and α_F^* uniform, and if $\mu(F) > 0$ then $K_F = \frac{\mu(F)}{k} D_k$. So $\Phi_1(F) = d - 1$ for every face, and all faces tie at $\tau = 1$. With the uniform start, the crossing of a face of $k \ge 2$ states with a vertex has $\Delta = \delta = k - 1$ and $b_{FG} = a_k$. If μ is constant and positive on F, then $c_F = -(k-1)/8$.*

Proof. Each Q_F is symmetric with row sums $-(d-k)$, and the complete graph on F has k^{k-2} spanning trees by Lemma 4.10. So Corollary 4.3 gives λ_F, α_F^* and $K_F = \mu(F) k^{(k+1)/2} k^{(k-2)/2} (4\pi)^{-(k-1)/2} = \frac{\mu(F)}{k} D_k$. A vertex has $K = \frac{1}{d}$ and $\lambda = d - 1$ for the uniform start, so $b_{FG} = -\frac{1}{k-1}\log(dK_F) = -\frac{1}{k-1}\log D_k = a_k$. The last claim is Proposition 4.15(iv). The arithmetic that gives b_{FG} from K_F here and in Corollary 5.2 is checked in Lean. □

By Example 4.11 at the center, $D_k = dK_F = k/\sqrt{\det(2\pi\Sigma_k)}$ with $\Sigma_k = \Sigma_{\mathbf{1}/k} = (2/k^2)(I - J/k)$ in the coordinates α'. So D_k is the mass k of a face, in units of a vertex, over the Gaussian volume of the fluctuations of its fractions. The preprint [36, Proposition 3.7] obtains it also by summing the exact law over the whole face. Here Σ_α is the occupation covariance defined in Definition 4.1. Example 4.11, Theorem 4.4 and Lemma 4.5 give every composition whose fractions are bounded away from the boundary

of its face. Let $\alpha = n/N$ have k positive coordinates, all at least ε, and put $A = \sum_i \sqrt{\alpha_i}$. Uniformly for such compositions and for $1 \le T \le N^{1/3}$,

$$\frac{P_N(n)}{V_N} = D(\alpha)\,\frac{T^{(k-1)/2} e^{(A^2-1)T}}{N^{k-1}}\Big(1 + O_\varepsilon\Big(\frac{1}{T} + \frac{T^2}{N}\Big)\Big), \qquad D(\alpha) = \frac{A^{k/2+1}}{(4\pi)^{(k-1)/2}(\prod_i \alpha_i)^{3/4}}, \tag{5.2}$$

for the uniform start. The relation between $D(\alpha)$ and $C_F(\alpha)$ is given in Example 4.11, and $V_N = \frac{1}{d} e^{-(d-1)T}(1 + O(T^2/N))$. This is [36, Theorem 3.3], and at the center $D(\mathbf{1}/k) = D_k$. By the Cauchy–Schwarz inequality $A^2 \le k$, with equality only at the center.

The next theorem holds at every N in its exact parts, and its asymptotic parts come from Section 4.

Theorem 5.4 (balanced crossings)**.** *Let μ be uniform and let F be a face of $2 \le k \le d$ states.*

(i) For every $N \ge k$ there is exactly one $p = p_{N,k} \in (0,1)$ with $M_F = V_N$. The face maximum M_F is larger than V_N below $p_{N,k}$ and smaller above it, and so is the probability of a balanced composition of F. The crossing $p_{N,k}$ is strictly increasing in N, $p_{k,k} = 1/(1 + (d-1)(k!)^{-1/(k-1)})$, and every crossing has $u \le (k!)^{-1/(k-1)} < 1$, so $p_{N,k} > 1/d$.

(ii) For large N, uniformly for $1 \le T \le N^{1/3}$,

$$\log \frac{M_F}{V_N} = (k-1)\Big[T - L + \frac{1}{2}\log T - a_k\Big] - \frac{k-1}{8T} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big).$$

With $T = L - \frac{1}{2}\log L + t$, the ratio M_F/V_N tends to $e^{(k-1)(t-a_k)}$, uniformly for t in compact sets.

(iii) The value $T_{N,k} = N(1-p_{N,k})/(d-1)$ of T at the crossing satisfies

$$T_{N,k} + \frac{1}{2}\log T_{N,k} = L + a_k + \frac{1}{8T_{N,k}} + O(L^{-2}),$$

and

$$T_{N,k} = L - \frac{1}{2}\log L + a_k + \frac{\frac{1}{4}\log L - \frac{1}{2}a_k + \frac{1}{8}}{L} + O\Big(\frac{(\log L)^2}{L^2}\Big). \tag{5.3}$$

(iv) $a_2 > a_3 > a_4 > \cdots$.

For $d = 2$, $T_{N,2}$ is the crossing z_N of Corollary 5.2, and $a_2 = c_0$. The proof is in Appendix C. The exact part uses that S_n has nonnegative coefficients and the first nonzero coefficient $W_n(k-1) = k!$, from the words that use each state in a single run. The next coefficient $W_n(k) = k!(k-1)(N-k)/2$ grows with N, which moves the root. The asymptotic part is Proposition 4.15 with the value $c_F = -(k-1)/8$ that we compute by hand in Appendix B.2. So the term $1/(8T)$, which [36] found through a Bessel majorant, is the next term of the local law.

For $p \ge 1/d$ the face maxima are the balanced compositions, at every N. Once the number of runs of each state is fixed, splitting the visits evenly gives the most ways to choose the run lengths.

Theorem 5.5 (exact balancing)**.** *Let $1/d \le p < 1$, and let μ be constant and positive on a face F. Moving one unit from a coordinate b of a composition with support F to another coordinate a, where $n_b \ge n_a + 2$ and $n_a \ge 1$, strictly increases the probability of the composition. So the most likely compositions with support F are exactly the balanced ones, and they all have the same probability.*

For an integer $n \geq 1$ put

$$B_n(w) = \sum_{m=1}^{n} \binom{n-1}{m-1} \frac{w^m}{m!} = \frac{w}{n} L_{n-1}^{(1)}(-w), \tag{5.4}$$

where $L_{n-1}^{(1)}$ is the Laguerre polynomial [33, Eq. 18.5.12]. The proof in Appendix C rests on the log-concavity of $B_n(w)$ in n for $w > 0$. The condition $p \geq 1/d$ holds at every crossing by Theorem 5.4(i), and for $T \leq N/d$.

Corollary 5.6 (exactly one mode change)**.** *Let μ be uniform, and fix $d \geq 2$ and $C > 1$. There is N_0 such that for $N \geq N_0$ and every p with $0 < T \leq CL$ the following hold.*

(i) If $p > p_{N,d}$, the global modes are exactly the d vertices.

(ii) If $p < p_{N,d}$, they are exactly the balanced compositions with d positive coordinates.

(iii) If $p = p_{N,d}$, they are these two classes and nothing else.

Moreover $p_{N,2} < p_{N,3} < \cdots < p_{N,d}$ and $p_{N,d} - p_{N,k} = (d-1)(a_k - a_d + o(1))/N$. So all the crossings of balanced compositions with the vertices lie in a p-interval of length $(d-1)(a_2 - a_d + o(1))/N$, and as p decreases the balanced compositions with d positive coordinates cross first.

So for large N the mode jumps once, from the vertices straight to the center of the simplex, and no proper face holds it in between. On the triangle the middle composition $(\frac{N}{3}, \frac{N}{3}, \frac{N}{3})$ overtakes the vertices while the middles of the edges are still below them by the factor $e^{-(a_2 - a_3)} = 0.827$. By Theorem 5.5 every global mode is a vertex or a balanced composition, and Theorem 5.4(ii) compares these. The proof is in Appendix C. In the tie window at $\tau^* = 1$ this is Theorem 4.8, whose envelope E has a single kink, at $t = a_d$.

Remark 5.7 (an intermediate face can be a mode at finite N)**.** Corollary 5.6 does not hold for every N. Take $d = 3$, $N = 4$ and $p = 10/17$, so $u = 7/20$. For an edge center and a full center, counting the words gives

$$\frac{P_4(2,2,0)}{V_4} = 2u + 2u^2 + 2u^3 = \frac{4123}{4000} > 1, \qquad \frac{P_4(2,1,1)}{V_4} = 6u^2 + 6u^3 = \frac{3969}{4000} < 1.$$

Here $p > 1/3$, so by Theorem 5.5 every global mode is a vertex or a balanced composition, and the three edge centers are the only global modes. Also $0 < T = 14/17 < \log 4$, so this p lies in the range of Corollary 5.6 for every $C > 1$. Hence $N_0 > 4$ when $d = 3$. These values are checked in Lean.

5.3 A start

By Corollary 5.3 the start enters K_F only through $\mu(F)$. It changes the constants by an amount of order 1, the order at which the faces are separated, so a start can decide which face wins.

Corollary 5.8 (crossings with a start)**.** *Let F be a face of $k \geq 2$ states with $\mu(F) > 0$, and let i be a state with $\mu_i > 0$. Put $a_k(\mu) = a_k + \frac{1}{k-1} \log \frac{k\mu_i}{\mu(F)}$. For large N the equation $M_F = M_{\{i\}}$ has a root in $[1, N^{1/3}]$, and every such root satisfies $T + \frac{1}{2} \log T = L + a_k(\mu) + O(1/L)$. With $T = L - \frac{1}{2} \log L + t$, $\log(M_F/M_{\{i\}}) = (k-1)(t - a_k(\mu)) + o(1)$, uniformly for t in compact sets.*

Proof. By Corollary 5.3, $\log(K_F/K_{\{i\}}) = \log D_k + \log \frac{\mu(F)}{k\mu_i} = -(k-1)a_k(\mu)$, and $\Delta = \delta = k-1$. So Theorem 4.7 gives the root and its equation. For the last claim, Theorem 4.8(i) at $\tau^* = 1$ gives $\log(M_F/M_{\{i\}}) = \ell_F(t) - \ell_{\{i\}}(t) + O(\log L/L)$, and $\ell_F(t) - \ell_{\{i\}}(t) = \log(K_F/\mu_i) + (k-1)t$. □

So a start changes only the constant of each crossing [36, Theorem 4.4]. It lowers a_k by the logarithm of the average start mass on the face against the start mass of the vertex, $\log(\mu(F)/(k\mu_i))$, shared among the $k-1$ new states. If μ is constant on F, then $P_N(n) = \mu_a p^{N-1} S_n(u)$ for every n with support F and $a \in F$, so Theorems 5.4 and 5.5 hold on F with V_N replaced by $\mu_a p^{N-1}$. Otherwise balancing can fail.

Remark 5.9 (balancing can fail with a start)**.** Let $d = 2$, $N = 4$ and $\mu = (\frac{1}{2}+\varepsilon, \frac{1}{2}-\varepsilon)$ with $0 < \varepsilon \le \frac{1}{2}$. Counting the switches in the four words with counts $(3,1)$ and the six with counts $(2,2)$ gives $P_4(3,1) = p^3(u + 2\mu_1 u^2)$ and $P_4(2,2) = p^3(u + u^2 + u^3)$. So $P_4(3,1) - P_4(2,2) = p^3 u^2(2\varepsilon - u)$, and the unbalanced composition is more likely when $u < 2\varepsilon$.

Each face now gets a line in the window coordinate t, and the highest line wins.

Corollary 5.10 (global modes with a start)**.** *Order the start masses as $\mu_{(1)} \ge \cdots \ge \mu_{(d)}$, put $\bar\mu_k = \mu_{(1)} + \cdots + \mu_{(k)}$ and $T = L - \frac{1}{2}\log L + t$, and define the lines*

$$\ell_1^\mu(t) = \log \mu_{(1)}, \qquad \ell_k^\mu(t) = \log \frac{\bar\mu_k}{k} + (k-1)(t - a_k) \quad (2 \le k \le d).$$

Fix a compact set $\mathcal{J}$ of t on which $\max_k \ell_k^\mu(t)$ is attained only at $k = k^*$. For large N and all $t \in \mathcal{J}$, every global mode has exactly k^* positive coordinates, and its support S has $\mu(S) = \bar\mu_{k^*}$. If $k^* = 1$, the global modes are exactly the vertices i with $\mu_i = \mu_{(1)}$. If $k^* \ge 2$ and μ is constant on the support S of a global mode, then the global modes with support S are exactly the balanced compositions of S. Every global mode n satisfies $n/N = \mathbf{1}_S/k^* + O(1/L)$.

Proof. By Lemma 2.2(b) and (c) the admissible faces are the faces with $\mu(F) > 0$, and they satisfy (H). By Corollary 5.3 they all tie at $\tau^* = 1$, so Theorem 4.8 applies with $m^* = d - 1$. Its line for a face F of k states is $\ell_F(t) = \log K_F - (d-k)t$, and $\ell_F(t) + (d-1)t = \log(\mu(F)/k) + (k-1)(t - a_k)$ for $k \ge 2$ and $\log \mu_i$ for $F = \{i\}$. Among the faces of size k the largest of these lines belongs to the faces with $\mu(F) = \bar\mu_k$, and it is ℓ_k^μ. The faces of size k^* with $\mu(F) = \bar\mu_{k^*}$ all have the same line. By the assumption and by compactness, every other face has a line below it by a fixed amount on a neighborhood of $\mathcal{J}$. So no kink of the envelope E lies within some distance $\eta > 0$ of $\mathcal{J}$, and the consequence in Theorem 4.8 says that for large N and all $t \in \mathcal{J}$ every global mode lies on a face of size k^* with $\mu(F) = \bar\mu_{k^*}$. If $k^* = 1$, such a mode is a vertex Ne_i with $\mu_i = \mu_{(1)}$, and all of these have the probability $\mu_{(1)} p^{N-1}$. If $k^* \ge 2$ and μ is constant on S, then $p > 1/d$ for large N, and Theorem 5.5 gives the claim on balanced compositions. Finally a global mode with support S maximizes P_N among the compositions with support S, so Theorem 4.6 gives $n/N = \alpha_S^* + O(1/T)$ with $\alpha_S^* = \mathbf{1}_S/k^*$. □

Remark 5.11 (the order of the winners)**.** The winning size is nondecreasing in t. Indeed, if k_1 attains $\max_k \ell_k^\mu$ at t_1 and k_2 attains it at $t_2 > t_1$, adding the two inequalities $\ell_{k_1}^\mu(t_1) \ge \ell_{k_2}^\mu(t_1)$ and $\ell_{k_2}^\mu(t_2) \ge \ell_{k_1}^\mu(t_2)$ gives $(k_2 - k_1)(t_2 - t_1) \ge 0$. Put $t^* = a_d + \frac{1}{d-1}\log(d\mu_{(1)})$, where ℓ_1^μ and ℓ_d^μ cross. Since the slope $k-1$ of ℓ_k^μ lies strictly between the slopes 0 and $d-1$, a size $2 \le k \le d-1$ can be the unique winner at some t only if $\ell_k^\mu(t^*) > \ell_1^\mu(t^*)$. For the uniform start $t^* = a_d$ and $\ell_k^\mu(t^*) - \ell_1^\mu(t^*) = (k-1)(a_d - a_k) < 0$, in line with Corollary 5.6.

Example 5.12 (the start decides which face wins)**.** Let $d = 3$ and $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$, and put $t_{\mathrm{ef}} = \log\frac{3}{2} + 2a_3 - a_2 = \frac{1}{2}\log\frac{32\pi}{243}$. Fix t and let N be large. If $t < a_2$, the global modes are $(N,0,0)$ and $(0,N,0)$. If $a_2 < t < t_{\mathrm{ef}}$, they are exactly the balanced compositions of the edge $\{1,2\}$. If $t > t_{\mathrm{ef}}$, every global mode has three positive coordinates. The window $(a_2, t_{\mathrm{ef}}) = (-0.4673558, -0.4412978)$ has width $\log\frac{16}{9\sqrt{3}} = 0.0260580$, while with the uniform start no proper face wins (Corollary 5.6).

Proof. Here $\bar{\mu}_1 = \frac{1}{2}$ and $\bar{\mu}_2 = \bar{\mu}_3 = 1$, so the lines of Corollary 5.10 are $\ell_1^\mu = \log\frac{1}{2}$, $\ell_2^\mu = \log\frac{1}{2} + t - a_2$ and $\ell_3^\mu = \log\frac{1}{3} + 2(t - a_3)$. Hence $\ell_2^\mu > \ell_1^\mu$ exactly when $t > a_2$, $\ell_2^\mu > \ell_3^\mu$ exactly when $t < t_{\mathrm{ef}}$, and $\ell_3^\mu > \ell_1^\mu$ exactly when $t > t^*$, where $t^* = a_3 + \frac{1}{2}\log\frac{3}{2}$ is the point t^* of Remark 5.11. By the definition of a_k, $t_{\mathrm{ef}} - a_2 = \log\frac{3}{2} + 2(a_3 - a_2) = 4\log 2 - \frac{5}{2}\log 3 = \log\frac{16}{9\sqrt{3}}$. This is positive, since $16^2 = 256 > 243 = (9\sqrt{3})^2$. Also $t^* - a_2 = \frac{1}{2}(t_{\mathrm{ef}} - a_2)$. So $a_2 < t^* < t_{\mathrm{ef}}$, and the unique winning size is 1 for $t < a_2$, 2 for $a_2 < t < t_{\mathrm{ef}}$ and 3 for $t > t_{\mathrm{ef}}$. Now apply Corollary 5.10 with $\mathcal{J} = \{t\}$. The vertices with $\mu_i = \frac{1}{2}$ are 1 and 2. The only edge with $\mu(S) = 1$ is $\{1, 2\}$, and μ is constant on it. So for $a_2 < t < t_{\mathrm{ef}}$ every global mode is a balanced composition of $\{1, 2\}$, and then all the balanced compositions of $\{1, 2\}$ are global modes. □

The proof uses only Theorems 4.6 and 4.8 and exact balancing on the edge, through Corollary 5.10. The comparison of the lines and the value of the width are checked in Lean. Dynamic programs over all compositions at $N = 90$, 300 and 1002 show the three phases at finite N, with the window moved to larger t (Section 8). The next remark explains the shift.

Remark 5.13 (the window at finite N)**.** Since μ is constant on $\{1, 2\}$, Proposition 5.1 with $a = 1$ and $q = 2$ gives $P_N(n_1, n_2, 0)/P_N(N, 0, 0) = S_{n_1,n_2}(u)$. So for every $N \geq 2$ the balanced compositions of the edge cross the vertices exactly at $u = u_N$, the crossing odds of the two-state walk, that is, at $T = Nu_N/(1+2u_N)$. This differs from $z_N = Nu_N/(1+u_N)$ by $O(L^2/N)$. Near $t = a_2$ the lines of the full face and of the edges through 3 stay below those of the vertices, so for large N this crossing is the lower end of the window. For the two ends at finite N we use $z = Nu = T/p$. A heuristic calculation, which we do not prove, puts them at $z - L + \frac{1}{2}\log z = a_2 + \frac{1}{8z} + \frac{z^2}{2N} + \cdots$ and $t_{\mathrm{ef}} + \frac{1}{12z} + \frac{3z^2}{2N} + \cdots$. Only $a_2 + \frac{1}{8z}$ is proved, by Theorem 5.4(iii) and $z - T = O(L^2/N)$. The terms z^2/N come from a discrete term $-\frac{(k-1)^2}{2}\frac{z^2}{N}$ in the logarithm of M_F/V_N. Exact computations give the widths 0.0843, 0.0324, 0.0223 and 0.0214 in $z = Nu$ at $N = 300$, 3000, 30000 and 300000, and the heuristic gives 0.0802, 0.0321, 0.0226 and 0.0217 (Section 8). At fixed N these are also widths in $z - L + \frac{1}{2}\log L$. The limiting window in the coordinate t has width $t_{\mathrm{ef}} - a_2$ (Example 5.12).

5.4 A weak field

Proposition 4.16 applies to the complete graph with the uniform start. There α_F^* is uniform, so the field shifts the line of a face S by the average $\bar{\xi}_S = \frac{1}{|S|}\sum_{i\in S}\xi_i$ of the field over S. In the Potts chain the field ξ_i/N acts at each site in state i.

Corollary 5.14 (weak-field modal phase diagram)**.** *Let μ be uniform, put $T = L - \frac{1}{2}\log L + t$ and define the face scores*

$$\ell_S(t,\xi) = \begin{cases} \xi_i, & S = \{i\}, \\ (|S|-1)(t - a_{|S|}) + \bar{\xi}_S, & |S| \geq 2. \end{cases}$$

Uniformly for (t, ξ) in compact sets,

$$\log \max_n \frac{e^{\langle \xi, n\rangle/N} P_N(n)}{V_N} \longrightarrow \max_{\varnothing \neq S \subseteq [d]} \ell_S(t,\xi). \tag{5.5}$$

For fixed (t, ξ) and all large N, every global mode of $\widetilde{P}_{N,\xi}$ has a support that maximizes the limiting score. If the maximizing support is unique, all global modes have exactly that support, and if it has $|S| \geq 2$ states they satisfy $n/N = \mathbf{1}_S/|S| + O(1/L)$. Equal limiting scores do not imply that both supports hold modes at finite N.

If the fields are sorted as $\xi_1 \geq \cdots \geq \xi_d$, the maximum of the scores is the upper envelope of the d lines

$$\ell_1(t) = \xi_1, \qquad \ell_k(t) = (k-1)(t - a_k) + \frac{1}{k}\sum_{i=1}^{k} \xi_i, \quad 2 \leq k \leq d. \tag{5.6}$$

Away from ties, the winning support size is nondecreasing in t. If the fields are strictly ordered, the winning supports are the nested sets $\{1, \dots, k\}$. Every support size $k \in \{2, \dots, d-1\}$ is the unique winning size on an open interval of t for suitable fixed fields. So proper faces can hold the mode under a field of order $1/N$ per step.

Proof. Step 1. The best composition on a support. Let S have $k \geq 2$ states. By Theorem 5.4(ii), $\log(M_S/V_N) = (k-1)(t - a_k) + o(1)$, uniformly for t in a compact set. Proposition 4.16(b), with α_S^* uniform, gives $\log(M_S^\xi/V_N) = \ell_S(t, \xi) + o(1)$, uniformly for (t, ξ) in compact sets. A vertex has $M_{\{i\}}^\xi = e^{\xi_i} V_N$ exactly. There are finitely many supports, which proves (5.5).

Step 2. The modes. A support whose score is below the maximum by a fixed amount has M_S^ξ smaller than the largest face maximum for large N, so it holds no global mode. A global mode with support S maximizes w_ξ on S, and Proposition 4.16(b) gives its location.

Step 3. The lines. Among the supports of size k the largest $\bar{\xi}_S$ is the average of the k largest fields, which gives (5.6). The lines have slopes $k - 1$, so the comparison in Remark 5.11 shows that the winning size is nondecreasing away from ties. If the fields are strictly ordered, the unique best support of size k is $\{1, \dots, k\}$.

Step 4. Every intermediate size. To realize a given $2 \leq k < d$, set $\xi_1 = \cdots = \xi_k = 0$ and $\xi_{k+1} = \cdots = \xi_d = -H$, and take $t > a_k$ so close to a_k that $t < a_{k-1}$ if $k > 2$. Then the score of $\{1, \dots, k\}$ is positive, the largest vertex score is 0, and every face with $2 \leq j < k$ states scores at most $(j-1)(t - a_j) < 0$, by Theorem 5.4(iv). A face with $j > k$ states scores at most $(j-1)(t - a_j) - H(j-k)/j$, and another face with k states at most $(k-1)(t - a_k) - H/k$, so both lose once H is large. The comparisons are strict, so they hold on an open interval of t, and small changes make the fields strictly ordered without removing the phase. □

The lines (5.6) are the lines ℓ_F^ξ of Proposition 4.16(c) plus $(d-1)t + \log d$, since $K_F = D_{|F|}/d$ for the uniform start.

Figure 1 in the introduction shows how the field changes the single jump into a sequence of phases, one for each support size.

Example 5.15 (a persistent edge phase on the triangle)**.** Let $d = 3$, let μ be uniform and let $\xi = (0, 0, -H)$. If $H > 6(a_2 - a_3)$, then for every fixed t with

$$a_2 < t < \frac{H}{3} + 2a_3 - a_2$$

and all large N, all global modes of $\widetilde{P}_{N,\xi}$ lie on the edge $\{1, 2\}$. Without a field no proper face holds a mode in the window (Corollary 5.6).

Proof. The scores of Corollary 5.14 are 0 for the vertices 1 and 2, $-H$ for the vertex 3, $t - a_2$ for the edge $\{1, 2\}$, $t - a_2 - H/2$ for the other two edges and $2(t - a_3) - H/3$ for the full face. For t in the interval, $t - a_2 > 0$, and $t - a_2 > 2(t - a_3) - H/3$ exactly when $t < H/3 + 2a_3 - a_2$. So the edge $\{1, 2\}$ is the unique maximizer, and the interval is not empty exactly when $H > 6(a_2 - a_3)$. Corollary 5.14 gives the claim. □

One field can make every support size win in turn, from one state to all d.

Corollary 5.16 (one field realizes the full face hierarchy)**.** *Let μ be uniform, fix $d \geq 2$ and $H > 3/4$, and let $\xi_i = -H(i-1)^2$ for $1 \leq i \leq d$. In particular $\xi_i = -(i-1)^2$ works in every fixed dimension. Put*

$$t_k = \log D_k - \log D_{k+1} + \frac{H(4k-1)}{6} \quad (1 \leq k < d),$$

and $t_0 = -\infty$, $t_d = +\infty$. Then $t_1 < \cdots < t_{d-1}$, and for every $k \in \{1, \dots, d\}$ the support $\{1, \dots, k\}$ is the unique winning support on the nonempty interval $t_{k-1} < t < t_k$. So for each fixed t in this interval and all large N, every global mode of $\widetilde{P}_{N,\xi}$ has exactly those k positive coordinates.

Proof. Since $\frac{1}{k}\sum_{i=1}^{k}(i-1)^2 = (k-1)(2k-1)/6$ and $-(k-1)a_k = \log D_k$, the lines (5.6) are $\ell_k(t) = (k-1)t + \log D_k - H(k-1)(2k-1)/6$, also for $k = 1$, and $\ell_{k+1}(t) - \ell_k(t) = t - t_k$. Extend $\log D_k$ to the smooth function $c(x) = (x + \frac{1}{2})\log x - \frac{x-1}{2}\log(4\pi)$ for $x \geq 1$. Then $0 < c''(x) = (x - \frac{1}{2})/x^2 \leq \frac{1}{2}$, and integrating twice over unit intervals gives $\log D_{k+2} - 2\log D_{k+1} + \log D_k \leq \frac{1}{2}$. Hence

$$t_{k+1} - t_k = \frac{2H}{3} - (\log D_{k+2} - 2\log D_{k+1} + \log D_k) > 0.$$

For $t_{k-1} < t < t_k$, summing the differences $\ell_{j+1}(t) - \ell_j(t) = t - t_j$ shows that line k strictly exceeds every other line. The fields are strictly decreasing, so the unique best support of size k is $\{1, \dots, k\}$, and Corollary 5.14 gives the conclusion for finite N. Most of this arithmetic is checked in Lean. □

6 Near a vertex

Theorem 4.4 treats compositions whose positive visit fractions stay away from zero. Near a vertex of the complete graph the crossing has another scale. We keep the setting and the notation of Section 5.2, namely $Q = J - dI$, the uniform start, p, u, v, and $S_n(u) = P_N(n)/V_N$ by (5.1). Let one count b be large and let the other positive counts $a_1, \dots, a_\ell$ be small compared with b, with $\ell \geq 1$ fixed. We put $|a| = a_1 + \cdots + a_\ell$ and write $S_{(b,a)}$ for S_n with $n = (b, a_1, \dots, a_\ell)$. These rare counts are not the constants a_k of Section 5.2, which do not appear here. As S_n does not depend on the start, $P_N(n) \leq (\max_i \mu_i)p^{N-1}S_n(u)$ for every start, so upper bounds on S_n hold for every start.

A rare run away from the ends needs two switches, which gives a factor u^2, and it can go in about b places. So the rare counts enter through bu^2. For fixed rare counts the crossing moves to odds u of order $b^{-1/2}$, that is, to T of order $\sqrt{N}$ instead of $\log N$. Use B_a from (5.4) and put

$$\mathcal{I}(x) = \sum_{j \geq 0} \frac{x^j}{j!\,(j+1)!} = \frac{I_1(2\sqrt{x})}{\sqrt{x}},$$

where I_1 is the modified Bessel function [33, Eq. 10.25.2], and $\mathcal{I}(0) = 1$. The Laguerre form in (5.4) makes the next lemma a quantitative form of the limit of Laguerre polynomials to Bessel functions.

Lemma 6.1 (a uniform Laguerre–Bessel inequality)**.** *For every integer $a \geq 1$ and real $w > 0$,*

$$0 \leq 1 - \frac{B_a(w)}{w\,\mathcal{I}(aw)} \leq \frac{w}{2}. \tag{6.1}$$

The proof is in Appendix D. Put

$$\mathcal{B}(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \qquad \kappa(y) = \frac{y\,\mathcal{B}'(y)}{\mathcal{B}(y)}.$$

The polynomial $\mathcal{B}$ has positive coefficients, $\mathcal{B}(0) = 0$ and $\mathcal{B}(1) \geq 1$, so $\mathcal{B}(y_a) = 1$ has exactly one positive root y_a, and $y_a \leq 1$. The polynomial $S_{(b,a)}$ has nonnegative coefficients, zero constant term and a positive coefficient at u^ℓ, so $S_{(b,a)}(u) = 1$ has exactly one positive root $u_{b,a}$, the crossing odds of the composition $(b, a_1, \dots, a_\ell)$ with a vertex.

Theorem 6.2 (the vertex law)**.** *Fix $\ell \geq 1$ and put*

$$c_\ell = \ell + \sqrt{\ell} + 1, \qquad \varepsilon = \sqrt{\frac{|a|+1}{b}}, \qquad \delta = 16 c_\ell \varepsilon.$$

For all positive counts $b, a_1, \dots, a_\ell$ with $\delta \leq 1/2$,

$$(1-\delta)\sqrt{y_a/b} \ \leq u_{b,a} \leq (1+\delta)\sqrt{y_a/b}, \qquad \mathcal{B}(y_a) = 1. \tag{6.2}$$

In particular

$$\left| \frac{u_{b,a}}{\sqrt{y_a/b}} - 1 \right| \leq 16 c_\ell \sqrt{\frac{|a|+1}{b}} \qquad \text{whenever } \sqrt{\frac{|a|+1}{b}} \leq \frac{1}{32 c_\ell}, \tag{6.3}$$

so $u_{b,a} = \sqrt{y_a/b}\,(1 + o(1))$ as $(|a|+1)/b \to 0$. The constants are not optimized.

For fixed rare counts the bounds of Lemma D.2 give $S_{(b,a)}(\sqrt{y/b}) \to \mathcal{B}(y)$ as $b \to \infty$, so the crossing is near $\sqrt{y_a/b}$, and the theorem shows that this stays correct uniformly as long as $|a| = o(b)$. For $d = 2$ and $\ell = 1$ its leading term $\sqrt{y_a/b}$ is that of the fixed-edge expansion of [35, Theorem 4.10], where a is fixed. The proof is in Appendix D. It rests on two facts. First, $\log \mathcal{B}$ is concave, and for $0 < y \leq 1$ its elasticity $\kappa(y)$ lies between $\max\{\ell, \frac{1}{2}\sqrt{|a|y}\}$ and $\ell + \sqrt{\ell |a| y}$ (Lemma D.1). Second, $S_{(b,a)}(u) = (1-u)^{|a|}\mathbb{E}[\mathcal{B}(v\mathcal{X})]$ exactly, where $v = u/(1-u)$ and $\mathcal{X}$ is a gamma variable plus a compound binomial sum with mean $2 + (b-1)u$ (Lemma D.2). Jensen's inequality then gives $S_{(b,a)}(u) \geq (1-u)^{|a|}\mathcal{B}(bu^2)$, which exceeds 1 at $u = (1+\delta)\sqrt{y_a/b}$. The tangent line of $\log \mathcal{B}$ at bu^2 and the moment generating function of $\mathcal{X}$ give an upper bound that is below 1 at $u = (1-\delta)\sqrt{y_a/b}$.

Remark 6.3 (the boundary of the simplex)**.** Version 1 of the preprint [36] also treats compositions whose small coordinates are not near a vertex, and we quote these results without using them. It gives a Bessel majorant for every positive count vector, with relative error $O(L^2/N)$ when $T = O(L)$ [36, Theorem 6.2]. When at least two coordinates are of order N, a small coordinate starts to matter once it has size of order $N/(\log N)^2$ [36, Remark 6.3 and Theorem C.1]. Under the hypotheses of that theorem, if every rare count is $o(N/(\log N)^2)$, their sizes change the crossing expansion only by $o(1)$, while their number ℓ still changes its leading and logarithmic coefficients [36, Corollary C.3].

7 The 4-cycle and other graphs

In this section Q is symmetric. So the admissible faces are the connected faces with positive start mass, and they satisfy (H) (Lemma 2.2). By Theorem 3.4(c), Theorem 4.8 applies at every tie. We call the faces in $\mathcal{F}_\tau$ the *winners* at τ. We first complete the worked example of Section 3.1 with all its constants, and then list the outcomes for other graphs.

Corollary 7.1 (the 4-cycle cascade)**.** *Let Q be the 4-cycle C_4 with $q_{i,i\pm1} = 1$ (indices mod 4), and let μ be uniform.*

(a) The admissible faces are the vertices, the adjacent pairs, the arcs of three states and C_4, with exit rates 2, 1, $2 - \sqrt{2}$ and 0. The winners are the vertices for $\tau < 1$, the adjacent pairs for $1 < \tau < 2$ and C_4 for $\tau > 2$. The arcs of three states never win.

(b) The constants are $K_{\mathrm{vertex}} = \frac{1}{4}$, $K_{\mathrm{pair}} = 1/\sqrt{2\pi}$, $K_{\mathrm{arc3}} = (4+3\sqrt{2})/(4\pi)$ *and* $K_{C_4} = 8/\pi^{3/2}$, *and* $\det \Sigma = 1/512$ *on* C_4.

(c) For every $N \geq 2$ *and* $0 < h < \frac{1}{2}$, $M_{\mathrm{pair}} = M_{\mathrm{vertex}} S_N(u)$ *with* $u = h/(1-2h)$. *So the face maxima of a vertex and an adjacent pair cross exactly at* $u = u_N$, *the crossing of [35]. For large* N *this crossing satisfies* $T + \frac{1}{2}\log T = L + c_0 + O(1/L)$.

(d) For large N, *every crossing* $T_N \in [1, N^{1/3}]$ *of the face maxima of an adjacent pair and* C_4 *satisfies* $T_N + \log T_N = 2L - \log(8\sqrt{2}/\pi) + O(1/L)$.

(e) Put $b = -\frac{1}{2}\log 2 - \frac{1}{2}\log(8\sqrt{2}/\pi)$. *In the tie window at* $\tau = 1$ *the face that carries the global mode changes from the vertices to the adjacent pairs only at* $T = L - \frac{1}{2}\log L + c_0 + o(1)$. *In the tie window at* $\tau = 2$ *it changes from the adjacent pairs to* C_4 *only at* $T = 2(L - \frac{1}{2}\log L + b) + o(1)$.

In words, the face maxima of a vertex and of an adjacent pair are equal exactly when the odds $h/(1-2h)$ equal the odds u_N at the crossing of the two-state walk, and this holds at every N. For large N this is where the mode moves from the vertices to the adjacent pairs. The mode then moves from the pairs to the whole cycle at a crossing whose equation has the coefficient $1 = \frac{\Delta}{2}$ of $\log T$, since the cycle has two more states than a pair.

Proof. (a) A proper face that is not an arc is disconnected, so it is not admissible. For an adjacent pair $-Q_F$ has rows $(2,-1)$ and $(-1,2)$, with least eigenvalue 1. For an arc of three states $-Q_F = 2I - A$, where A is the adjacency matrix of the path on three states, and $r = (1/\sqrt{2}, 1, 1/\sqrt{2})$ is a positive eigenvector with eigenvalue $2-\sqrt{2}$. Also $Q\mathbf{1} = 0$. So Φ_τ equals 2τ, $\tau+1$, $(2-\sqrt{2})\tau + 2$ and 3, and comparing the first, second and fourth gives the winners. Since $2-\sqrt{2} > \frac{1}{2}$, the 3-arc has $\Phi_\tau > \frac{1}{2}(\tau+1) + \frac{1}{2}\cdot 3$, which is at least the smaller of the exponents of the pair and of C_4. So it never wins.

(b) A vertex has $K = \mu_i = \frac{1}{4}$. The pair and C_4 have symmetric Q_F with constant row sums, so Corollary 4.3 applies, with $\mu(F) = \frac{1}{2}$ and one spanning tree for the pair, and $\mu(F) = 1$ and four spanning trees for C_4. On C_4 the law α^* is uniform and every edge has conductance $\frac{1}{4}$, so Proposition 4.2(b) gives $\det \Sigma = 2^3 \cdot 4^{-8}/(4 \cdot 4^{-3}) = 1/512$. The 3-arc has unequal row sums, and we use Proposition 4.2(b) with $\nu = \mathbf{1}$. Here $\alpha^* = r \circ r/|r|^2 = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ and $r_\alpha = l_\alpha = \sqrt{\alpha^*}$, so $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = \frac{1}{4}(1+1/\sqrt{2})^2 = (3+2\sqrt{2})/8$. The only spanning tree is the path, with $\mathrm{tree}(c) = c_{12}c_{23} = \frac{1}{8}$, and $\prod_i \alpha_i^* = \frac{1}{32}$. So $K_{\mathrm{arc3}} = \frac{3+2\sqrt{2}}{8} \cdot \frac{32}{2\sqrt{2}} \cdot \frac{1}{4\pi} = \frac{4+3\sqrt{2}}{4\pi}$.

(c) The identity is Proposition 5.1 with $a = 1$ and $q = 2$. So the two face maxima are equal only at $u = u_N$, and by Theorem 4.7(i) this crossing lies in $[L/2, 2L]$ for large N. As $K_{\mathrm{pair}}/K_{\mathrm{vertex}} = 2\sqrt{2/\pi} = e^{-c_0}$, Theorem 4.7(ii) with $\Delta = \delta = 1$ gives the equation.

(d) This is Theorem 4.7 with $F = C_4$, G an adjacent pair, $\Delta = 2$, $\delta = 1$ and $K_F/K_G = 8\sqrt{2\pi}/\pi^{3/2} = 8\sqrt{2}/\pi$.

(e) At $\tau = 1$ the tied faces are the vertices and the adjacent pairs, and at $\tau = 2$ they are the adjacent pairs and C_4. Faces of one type have the same line. We write $T = \tau(L - \frac{1}{2}\log L + t)$ as in Theorem 4.8, and let E be the upper envelope of the lines in that theorem. By Remark 4.13 the kink of E is at $t = b_{FG}$, which is c_0 at $\tau = 1$ by (c) and b at $\tau = 2$ by (d). So Theorem 4.8 gives (e). □

The finite parts of (a) are checked in Lean. Lean also checks that a closed form for the constants of all arcs, which we do not use, gives the values of K_{pair} and K_{arc3} above.

Proposition 4.15 gives the next term c_F/T of the face maxima, with the explicit coefficient $c_F = c(\alpha_F^*) + \frac{1}{2}\gamma_F^\top \Sigma_F \gamma_F$, where c is given by (B.2). For an adjacent pair, Proposition 4.15(iv) gives $c_{\mathrm{pair}} = -\frac{1}{8}$. For the 3-arc and for C_4 we evaluated the coefficient

exactly by computer algebra (the script `verify_second_order.py`), and the values are $c_{\mathrm{arc3}} = \frac{5}{4}\sqrt{2} - 2$ and $c_{C_4} = -\frac{9}{16}$. So these two constants are exact evaluations of the formula of Proposition 4.15, done by computer and not by hand. With the evaluated constant c_{C_4}, Proposition 4.15(iii) gives the next term at the crossings in (d), namely

$$T_N\Big(2L - \log T_N - T_N - \log\frac{8\sqrt{2}}{\pi}\Big) = c_{C_4} - c_{\mathrm{pair}} + O\Big(\frac{1}{L}\Big) = -\frac{7}{16} + O\Big(\frac{1}{L}\Big).$$

Theorem 4.7 also applies to a face that never carries the mode, and this is a good test of the implicit equation (4.5). Take the 3-arc F against a vertex G, so that $\Delta = 2$, $\delta = \sqrt{2}$ and $\log(K_F/K_G) = \log((4+3\sqrt{2})/\pi) = 0.964591$. At $N = 10^{14}$ a numerical computation of the face maxima, over the symmetric compositions $(a, N-2a, a)$, puts the crossing at $T_N = 42.26334$, where $\Delta L - \frac{\Delta}{2}\log T_N - \delta T_N = 0.959070$. The root of the implicit equation is within 0.004 of T_N, while the explicit form in Theorem 4.7 gives 42.2059. Also $T_N(0.959070 - 0.964591) = -0.2333$, close to the limit $c_{\mathrm{arc3}} - c_{\mathrm{vertex}} = -0.2322$ that Proposition 4.15(iii) gives with the evaluated constant c_{arc3} (a vertex has $c_{\mathrm{vertex}} = 0$).

Remark 7.2 (the outcomes table)**.** Table 2 lists the outcomes for other graphs. Each row comes from Theorem 3.4 applied to the exit rates of the connected faces. So a row needs two things, namely the exit rates and the points that lie on the lower-left boundary of the hull. The second row has a short proof. By Rayleigh–Ritz with the unit vector f that equals $(d-1)^{-1/2}$ off x and 0 at x, we get $\lambda_{[d]\setminus\{x\}} \le \sum_{y\neq x} q_{xy}/(d-1) = q_x/(d-1)$, with equality exactly when f is an eigenvector of $-Q$ restricted to $[d]\setminus\{x\}$. Its entry at y is $q_{yx}(d-1)^{-1/2}$, so this happens exactly when q_{yx} does not depend on y, that is, $q_{yx} = q_x/(d-1)$. With unit rates on a connected graph, take x of least degree δ. The vertices with exit rate δ and the face $[d]$ tie at $\tau_0 = (d-1)/\delta$, and there a component C of $[d]\setminus\{x\}$ of least exit rate has $\lambda_C = \lambda_{[d]\setminus\{x\}}$ (the restricted matrix is block diagonal) and $\Phi_{\tau_0}(C) \le \tau_0\lambda_{[d]\setminus\{x\}} + d - 2 \le d - 1$. Equality needs every q_{yx} to equal $\delta/(d-1)$, which for unit rates means that the graph is complete. So on an incomplete graph the connected face C beats both at τ_0, and on K_d the first row applies. The first row is proved by Corollary 5.3. There $\Phi_\tau(F) = \tau(d-k) + k - 1 = (d-1)\tau + (k-1)(1-\tau)$ for a face of k states, which is least at $k = 1$ for $\tau < 1$ and at $k = d$ for $\tau > 1$, and the same for all k at $\tau = 1$. It is also proved in Lean 4 (v4.33.1 with Mathlib v4.33.1), by `complete_vertex_wins`, `complete_full_wins` and `complete_tie`. The rows marked "Lean only" are computer-verified, that is, proved in Lean and not in the text. They are C_d by `cycle_cascade` and `arcCost_unique_min`, the paw by `paw_phase1` to `paw_phase4`, the house by `house_phase1` to `house_phase4`, and K_4 plus a vertex by `kfv_phase1` to `kfv_phase3`, with the lemmas listed in MAP.md (Section 8). On C_d two steps are by hand. A proper face that is not an arc is disconnected, hence not admissible, and the growth of the switch values follows from $2 - 2\cos x = x^2 + O(x^4)$. The rows P_d and Θ_ε are proved in Section S1 of the supplement, for every $d \ge 3$ and for every $\varepsilon < \varepsilon_1$ (Propositions S1.3 and S1.4 there). The proof for Θ_ε uses two resultants that we computed by computer algebra, and it also shows that the bound ε_1 is sharp. In the row P_4 the first sentence is the case $d = 4$ of Proposition S1.3, and the rest is an exact computation of the law at the values of N listed. A well-separated community wins for a long range of τ, such as a triangle of Θ_ε from $\tau \approx 1$ to $\tau \approx 9/\varepsilon$. The mode need not grow around its first vertex. It leaves the leaf of the paw, and on the house and on K_4 plus a vertex it jumps to a disjoint community. The second row explains the last one. The state 5 is joined to only two of the states $1, 2, 3, 4$, so by the second row the point of $\{1,2,3,4\}$ lies strictly left of the chord from $\{5\}$ to $[5]$, and the winners cannot jump from $\{5\}$ straight to $[5]$. At feasible N the switch values are approached slowly (Figure 4(b)), and on P_4 even the order of the phases differs.

8 Verification and open questions

The theorems, propositions, lemmas and corollaries of this paper are proved in the text, in the appendices or in the supplement, with two kinds of outside input. We quote results of [35] where we say so (the uniform Bessel bound (2.1), the clipped products of [35, Lemma 3.2], the growth of the polynomials $S_{a,b}$ toward the middle and the expansion of z_N). And the rows of Table 2 marked "Lean only" are proved in Lean and not in the text. The results of [36] quoted in Remark 6.3 are not used. No proof rests on a floating-point computation. Exact computer algebra enters in two places. First, Proposition S1.4 of the supplement, which proves the row Θ_ε of Table 2, rests on two exact resultants of characteristic polynomials. The script `verify_examples.py` computes them, as described in Section S1.3 of the supplement. Second, the values of c_F for the 3-arc and for C_4 after Corollary 7.1 are exact evaluations, by `verify_second_order.py`, of the formula for c_F in Proposition 4.15(ii). The proposition is proved, and only these two values are computed. The value $c_F = -(k-1)/8$ for the complete graph is computed by hand in Section S4 of the supplement, and `verify_second_order.py` also evaluates it exactly for $k = 3$ and 4. The script `verify_forest.py` lists the spanning trees one by one. It checks Lemma A.3 in exact rational arithmetic for 119 generators with two to six states. It checks Proposition 4.9(a) to (c) in floating point against Definition 4.1, and part (d) for the directed 3-cycle. The script `verify_rate_flows.py` checks the statements of Remark 4.12 in floating point on 480 random faces with two to five states. The script `verify_weak_field.py` checks Proposition 4.16. It checks (a) by listing all words of a chain with three states, and (b) on four faces with the continuum density of Lemma 2.4, computed by the torus integral of Lemma A.2. There the tilted face maxima are compared with $\langle \xi, \alpha_F^* \rangle$, with the term $(\gamma_F \cdot \Sigma_F \hat{\xi} + \frac{1}{2}\hat{\xi}^\top \Sigma_F \hat{\xi})/T$ and with the shift $\Sigma_F(\gamma_F + \hat{\xi})/T$ of the maximizer that completing the square suggests (the proposition states only their order $1/T$), and on a pair at $N = 20000$ with an exact dynamic program of the chain. It checks (d) with the transfer matrix at $N = 10^6$. Each check is an assertion in these scripts, and the floating-point checks use a tolerance.

The statements of Sections 5 and 6 that come from [36] are also checked by its scripts in kagey-problems/problem131/archive/paperB, which use only the Python standard library. The script `verify_simplex_threshold.py` checks the refresh form, the balancing transfers, the example at $N = 4$ and the balanced crossings at high precision. The script `verify_geometry.py` checks the exact law against a dynamic program and against all words, `verify_frontier_modes.py` checks the weak field, and `verify_vertex_completion.py` checks the identities near a vertex and the root brackets of Theorem 6.2. The script `verify_general_start.py` checks the law with a start against all words, and it computes the phases of Example 5.12 at $N = 90$ and the ends of the window of Remark 5.13 at $N \geq 3000$, by bisection. The script `verify_B_tie_repair.py` runs dynamic programs over all compositions at $N = 300$ and 1002. None of these checks is interval-certified.

The other numbers in the paper are of three kinds. Some are closed forms. Some are computations of the law at a finite N, namely Figure 2, the finite-N row of P_4 in Table 2, Figure 4(b), the crossing of the 3-arc at $N = 10^{14}$ in Section 7, the phases after Example 5.12 and the widths in Remark 5.13. These are done in two independent ways where possible (for example by the run expansion and by a torus integral, or by a dynamic program and an exact formula). They illustrate the theorems, and no statement is deduced from them.

Many finite and algebraic steps, and the rows of Table 2 marked Lean, are proved in Lean 4 in the folders kagey-problems/problem131/lean/PaperC and kagey-problems/problem131/lean/PaperB. The file lean/PaperC/MAP.md matches each statement of this

paper to its Lean theorems, including the statements of Sections 5 and 6 whose Lean files are in lean/PaperB. The asymptotic results are not in Lean.

The file SUPPLEMENT.md in the folder kagey-problems/problem131 lists each check, the two methods behind it and the predictions that we wrote down before most of the computations.

Data and code availability. All code, data and Lean files supporting this paper are in the folder kagey-problems/problem131 of https://github.com/apemm/Kagey-Problems. The scripts of this paper are in the subfolder paperC and need numpy, scipy, sympy and mpmath, and the historical scripts of [36] are in archive/paperB. The verification code supporting this version is available at source revision `4c18775`. That revision includes the shared routines in common and the archived scripts in archive/paperB. To rerun the checks at that revision, run `python -B verify_all.py --paper c` and `python -B verify_all.py --paper b` in the folder problem131. The largest cases run only with the option `--full`.

Open questions. We list them in the order of the paper.

The local law at the boundary. Theorem 4.4 holds on compact subsets of the open face, and a local law up to the boundary is open. The same formula cannot hold there. Take a star with center d and $\ell = d - 1 \geq 3$ leaves, and a start of full support. Then $P_N(n) = 0$ whenever $\operatorname{supp} n = [d]$ and $n_d < \ell - 1$, since a path visits the center between any two runs at the leaves, although n/N lies in the open face. This zero set is checked in Lean. Near a vertex of the complete graph the leading crossing is known uniformly for rare counts of total $o(b)$ (Theorem 6.2), but neither the next term uniformly in that range nor the case of a nonuniform start is known.

Reducible faces. For a reducible face, Theorem 3.3(ii) places the power of T in M_F between $k_F - k^*$ and $k_F - 1$, and heuristically it is $(k^* - 1)/2 + k_F - k^*$. We do not prove this.

Uniqueness and constants. Theorem 4.7 puts all crossings of two faces in an interval of length $O(1/L)$, but uniqueness is proved only in special cases, such as the two-state walk [35], the balanced compositions of the complete graph (Theorem 5.4), the pairs of Proposition 5.1. Proposition 4.9 gives a tree formula for K_F on every irreducible face, but we have a closed form for c_F only for symmetric pairs and for the complete graph. At finite N the order of the phases can differ from first order when a face lies close to the hull, as on the path P_4 (Table 2).

The complete graph at finite N. Corollary 5.6 has no effective N_0, and by Remark 5.7 it does not hold for every N. The discrete correction term $-\frac{(k-1)^2}{2} z^2/N$, with $z = Nu = T/p$ the normalized switch coordinate, and the second-order ends of the window in Remark 5.13 are not proved, and neither is the failure of balancing at every large N for a start that is not constant on the face. A number of states that grows with N is also open. The two-state crossing value is $z_N = N(1 - p_N)$ in Corollary 5.2.

Linear functions of the composition. A linear image can add the probabilities of many compositions. For example, the origin of a planar walk collects a line of occupation vectors, rather than one vector. Theorem 3.3 concerns individual vectors, and we do not prove a general support-selection rule for such sums.

A Proofs of the local law and of the first-order bounds

This appendix proves Proposition 4.2(a), Lemma 3.2, Theorem 4.4, Lemma 4.5 and Theorem 3.3(ii). Its last section proves Lemma A.3, which is used for Proposition 4.9. Except in Section A.6, F is a face with $k \geq 2$ states, numbered $1, \ldots, k$. For $x \in \mathbb{C}^F$

we write $x' = (x_2, \dots, x_k)$. We write i for the imaginary unit, since i denotes a state, and $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$. We use two standard facts [1]. First, let $B \geq 0$ be square and S diagonal with positive diagonal. Then $\varrho(B - S) < 0$ if and only if $\varrho(S^{-1}B) < 1$, and also if and only if $(S - B)v > 0$ for some $v > 0$. In that case $(S - B)^{-1} = \sum_{j \geq 0} (S^{-1}B)^j S^{-1} \geq 0$. Second, if M is an irreducible Metzler matrix, then $\varrho(M)$ is larger than the Perron root of each proper principal submatrix of M, and every nonnegative eigenvector of M is positive and belongs to $\varrho(M)$.

A.1 The Perron root under a diagonal perturbation

An irreducible generator on F is a Metzler matrix G with $G\mathbf{1} = 0$ whose graph is strongly connected. It has a unique stationary law $\pi > 0$. The argument after Definition 4.1, with (G, π) in place of $(\widehat{Q}_\alpha, \alpha)$, shows that $Z = (\mathbf{1}\pi^\top - G)^{-1} - \mathbf{1}\pi^\top$ exists and satisfies (4.2). Put $D = \operatorname{diag} \pi$ and $\Gamma = DZ + Z^\top D$.

Lemma A.1 (the Perron root)**.** *Let G, π, Z and Γ be as above. Then $\kappa(\eta) = \varrho(G + \operatorname{diag} \eta)$ is real-analytic near 0, with $\kappa(0) = 0$, $\nabla\kappa(0) = \pi$ and $\nabla^2\kappa(0) = \Gamma$. For $\eta \in \mathbb{R}^F$ and $u = Z\eta$,*

$$\eta^\top \Gamma \eta = \sum_{i \neq j} \pi_i G_{ij} (u_i - u_j)^2. \tag{A.1}$$

So Γ is positive semidefinite with kernel $\mathbb{R}\mathbf{1}$, and Σ, which is Γ without the row and column of state 1, is positive definite.

Proof. We differentiate the eigenvalue equation twice at $\eta = 0$. Since $G + \operatorname{diag} \eta$ is irreducible, its Perron root is a simple eigenvalue. So by the implicit function theorem, κ and the right eigenvector $v(\eta)$ with $\pi^\top v = 1$ are real-analytic near 0, with $\kappa(0) = 0$ and $v(0) = \mathbf{1}$. We write subscripts for derivatives at $\eta = 0$. Differentiating $(G + \operatorname{diag} \eta)v = \kappa v$ in η_j gives $Gv_j + e_j = \kappa_j \mathbf{1}$, and multiplying by $\pi^\top$ gives $\kappa_j = \pi_j$, since $\pi^\top G = 0$. Then $v_j = Ze_j$, because by (4.2) both vectors solve $Gx = \pi_j \mathbf{1} - e_j$ with $\pi^\top x = 0$, and the kernel of G is $\mathbb{R}\mathbf{1}$. Differentiating again, in η_i, gives $Gv_{ij} + \operatorname{diag}(e_i)v_j + \operatorname{diag}(e_j)v_i = \kappa_{ij}\mathbf{1} + \kappa_i v_j + \kappa_j v_i$. We multiply by $\pi^\top$ again. The terms with G, κ_i and κ_j vanish, and we get $\kappa_{ij} = \pi_i Z_{ij} + \pi_j Z_{ji} = \Gamma_{ij}$.

For (A.1) let $\bar{\eta} = \eta - (\pi \cdot \eta)\mathbf{1}$. By (4.2), $-Gu = \bar{\eta}$ and $\pi \cdot u = 0$, so $\eta^\top \Gamma \eta = 2 \sum_i \pi_i \eta_i u_i = 2 \sum_i \pi_i \bar{\eta}_i u_i = -2 \sum_i \pi_i u_i (Gu)_i$. Now expand the squares in (A.1). By $G\mathbf{1} = 0$ and $\pi^\top G = 0$ the terms with u_i^2 and with u_j^2 each sum to $-\sum_i \pi_i G_{ii} u_i^2$, and the cross terms sum to $-2 \sum_i \pi_i u_i (Gu)_i + 2 \sum_i \pi_i G_{ii} u_i^2$. So both sides of (A.1) agree. Also $\Gamma\mathbf{1} = 0$ by (4.2). If $\eta^\top \Gamma \eta = 0$, then u is constant along every edge of the strongly connected graph of G, so u is constant. Then $u = 0$ because $\pi \cdot u = 0$, so $\bar{\eta} = 0$ and $\eta \in \mathbb{R}\mathbf{1}$. A nonzero vector that vanishes at state 1 is not in $\mathbb{R}\mathbf{1}$, which gives the claim about Σ. □

Idea of the proof of Proposition 4.2(a). The gradient of $\Lambda(\vartheta) = \varrho(Q_F + \operatorname{diag} \vartheta)$ is the stationary law of the chain tilted by ϑ, and its Hessian is the matrix Γ of that chain (Lemma A.1). So $\nabla\Lambda$ maps the tilt to the fractions α. We show that ∇I_F is the map back, so the two Hessians are inverse to each other.

Proof of Proposition 4.2(a). Step 1: the derivatives of the Perron root. For $\vartheta \in \mathbb{R}^F$ let $\Lambda(\vartheta) = \varrho(Q_F + \operatorname{diag} \vartheta)$, so that $\Lambda(\vartheta + c\mathbf{1}) = \Lambda(\vartheta) + c$ (the shift property). It is a simple eigenvalue with positive right and left eigenvectors $\hat{r}, \hat{l}$, normalized by $\sum_i \hat{r}_i = 1$ and $\hat{l} \cdot \hat{r} = 1$, and by the implicit function theorem Λ, $\hat{r}$ and $\hat{l}$ are real-analytic in ϑ. Put $\pi = \hat{l} \circ \hat{r}$ and $G = \hat{R}^{-1}(Q_F + \operatorname{diag} \vartheta - \Lambda(\vartheta) I)\hat{R}$ with $\hat{R} = \operatorname{diag} \hat{r}$. Then $G\mathbf{1} = 0$ and $\pi^\top G = 0$ by the two eigenvector equations, so G is an irreducible generator with stationary law π. Since $\hat{R}$ commutes with diagonal matrices, $\Lambda(\vartheta + \eta) = \Lambda(\vartheta) + \varrho(G + \operatorname{diag} \eta)$. So by Lemma A.1, $\nabla\Lambda(\vartheta) = \pi$ and $\nabla^2\Lambda(\vartheta) = \Gamma$, with Γ built from (G, π).

Step 2: the gradient can be inverted. Let $\Lambda_0(\beta) = \Lambda(0, \beta)$ for $\beta \in \mathbb{R}^{k-1}$. Its Hessian is Γ without the row and column of state 1, which is positive definite by Lemma A.1. So Λ_0 is strictly convex, $\nabla\Lambda_0$ is injective, and by the inverse function theorem it has a real-analytic inverse on its open image.

Step 3: the tilt that belongs to α. Fix $\alpha \in \Delta_F^\circ$ and put $\vartheta(\alpha) = q - \theta(\alpha)$. Then $Q_F + \operatorname{diag}\vartheta(\alpha) = A_F - \operatorname{diag}\theta(\alpha)$ has the positive right and left eigenvectors r_α and l_α for the eigenvalue 0, by Definition 4.1 and (4.1). So its Perron root is 0, and at $\vartheta(\alpha)$ we have $\pi = \alpha$, $G = \widehat{Q}_\alpha$ and $\Gamma = \Gamma_\alpha$. Let $\beta(\alpha)$ be the last $k-1$ coordinates of $\vartheta(\alpha) - \vartheta_1(\alpha)\mathbf{1}$. A shift along $\mathbf{1}$ does not change the derivatives of Λ, so $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$ and $\nabla^2\Lambda_0(\beta(\alpha)) = \Sigma_\alpha$. Hence Σ_α is positive definite, and $\beta(\alpha) = (\nabla\Lambda_0)^{-1}(\alpha')$ is real-analytic. The shift property and $\Lambda(\vartheta(\alpha)) = 0$ give $\vartheta_1(\alpha) = -\Lambda_0(\beta(\alpha))$. So $\theta(\alpha)$ is real-analytic, and then so are its Perron vectors and Σ_α.

Step 4: the Hessian of I_F. Since $\vartheta(\alpha) = \vartheta_1(\alpha)\mathbf{1} + (0, \beta(\alpha))$ and $\sum_i \alpha_i = 1$, we have $I_F(\alpha) = \langle \vartheta(\alpha), \alpha\rangle = \langle \beta(\alpha), \alpha'\rangle - \Lambda_0(\beta(\alpha))$. We differentiate in α'. The terms with the derivative of β cancel because $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$, and $\nabla I_F = \beta$ remains. So $\nabla^2 I_F$ is the Jacobian of $(\nabla\Lambda_0)^{-1}$ at α', which is Σ_α^{-1}. The formula for C_F follows from Definition 4.1. □

A.2 The torus representation

We split

$$A_F = \begin{pmatrix} 0 & a_1^\top \\ b_1 & A' \end{pmatrix}, \qquad \mu_F = (\mu_1, \mu'),$$

so $a_1 = (q_{1j})_{j\ge 2}$, $b_1 = (q_{i1})_{i\ge 2}$ and $A' = (q_{ij})_{i,j\ge 2}$. For $z' \in \mathbb{C}^{k-1}$ let $\mathcal{Z}' = \operatorname{diag}(z')$. Where $\mathcal{Z}' - A'$ is invertible we put $X(z') = (\mathcal{Z}' - A')^{-1}$, $g(z') = a_1^\top X(z') b_1$ and

$$b(z') = \big(\mu_1 + \mu'^\top X(z') b_1\big)\big(1 + a_1^\top X(z')\mathbf{1}\big).$$

Lemma A.2 (torus representation)**.** *Let $s' \in (0,\infty)^{k-1}$ satisfy $\varrho(A' - \operatorname{diag} s') < 0$. On the set of $z' \in \mathbb{C}^{k-1}$ with $|z_i| \ge s_i$ for every $i \ge 2$, the matrix $\mathcal{Z}' - A'$ is invertible, and X, g and b are given by absolutely convergent power series in $1/z_2, \dots, 1/z_k$ with nonnegative coefficients. Hence $|g(z')| \le g(s')$ and $|b(z')| \le b(s')$ there. For every $t \in (0,\infty)^F$,*

$$H(t) = \frac{1}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \exp\Big(t_1 g(z') + \sum_{i\ge 2} t_i z_i\Big)\, b(z') \prod_{i\ge 2} z_i \, d\varphi, \qquad z_i = s_i e^{\mathrm{i}\varphi_i}.$$

In words, $H(t)$ is a Cauchy integral over $k-1$ circles, one for each state of F except state 1. The circle of state 1 has already been integrated out, and it leaves the factor $e^{t_1 g(z')}$.

Idea of the proof. By Cauchy's formula the factor $\prod_i t_i^{m_i - 1}/(m_i - 1)!$ of a word in $H(t)$ is the integral of $e^{\langle t, z\rangle} \prod_i z_i^{1-m_i}$ over k circles. The sum of $\prod_i z_i^{-m_i}$ over all words, with their weights, is a geometric series, namely $\mu_F^\top(\mathcal{Z} - A_F)^{-1}\mathbf{1}$ with $\mathcal{Z} = \operatorname{diag}(z)$. The variable z_1 enters this resolvent only through the simple pole $1/(z_1 - g(z'))$, so the integral over the first circle is a residue.

Proof. Step 1: the series. By the first fact, $\varrho(\operatorname{diag}(s')^{-1} A') < 1$. So the series $X(z') = \sum_{j\ge 0} (\mathcal{Z}'^{-1} A')^j \mathcal{Z}'^{-1}$ converges absolutely for $|z_i| \ge s_i$, since its terms are bounded entrywise in modulus by those at $z' = s'$. As a_1, b_1 and μ are nonnegative, this gives the claims about X, g and b.

Step 2: a radius for state 1. Choose $s_1 > g(s')$ and put $S = \operatorname{diag}(s_1, s')$. We show that $\varrho(S^{-1} A_F) < 1$. For $\varepsilon > 0$ let $v = (1, v')$ with $v' = X(s')(b_1 + \varepsilon\mathbf{1}) > 0$. Then $((S - A_F)v)_i = \varepsilon$ for $i \ge 2$, and $((S - A_F)v)_1 = s_1 - g(s') - \varepsilon a_1^\top X(s')\mathbf{1}$. For small ε both are positive, and the first fact gives the claim.

Step 3: k circles. Let $\mathcal{Z} = \operatorname{diag}(z)$ with $|z_i| = s_i$ for every $i \in F$. Since $\varrho(A_F S^{-1}) = \varrho(S^{-1}A_F) < 1$ (the two matrices are similar), expanding $(\mathcal{Z} - A_F)^{-1} = \mathcal{Z}^{-1}(I - A_F\mathcal{Z}^{-1})^{-1}$ gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1}\mathbf{1} = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}} \prod_{i \in F} z_i^{-m_i(c)},$$

absolutely and uniformly on the torus. Here $c = c_1 \cdots c_M$ runs over the words in F with $M \geq 1$ and $c_j \neq c_{j+1}$, and $m_i(c)$ is the number of letters i in c. For an integer m and $t_i > 0$, Cauchy's formula gives $\frac{1}{2\pi}\int_0^{2\pi} e^{t_i z_i} z_i^{1-m}\, d\varphi_i = t_i^{m-1}/(m-1)!$ if $m \geq 1$ and 0 if $m \leq 0$. We multiply the series by $e^{\langle t,z\rangle}\prod_i z_i$ and integrate term by term. The words that miss a state of F drop out, and grouping the others by $m(c)$ gives

$$H(t) = \frac{1}{(2\pi)^k}\int_{\mathbb{T}^k} e^{\langle t,z\rangle}\, \mu_F^\top (\mathcal{Z} - A_F)^{-1}\mathbf{1} \prod_{i\in F} z_i\, d\varphi. \tag{A.2}$$

Step 4: the first circle. The Schur complement $z_1 - g(z')$ of $\mathcal{Z}' - A'$ in $\mathcal{Z} - A_F$ is not 0, since $|g(z')| \leq g(s') < s_1$. Block inversion gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1}\mathbf{1} = \frac{b(z')}{z_1 - g(z')} + \mu'^\top X(z')\mathbf{1}.$$

We integrate over φ_1 in (A.2) and divide by 2π. The second term does not depend on z_1, so it contributes $\frac{1}{2\pi}\int_0^{2\pi} e^{t_1 z_1} z_1\, d\varphi_1 = 0$. In the first term $g(z')$ lies inside the circle $|z_1| = s_1$, so Cauchy's formula gives $e^{t_1 g(z')} b(z')$. The factors without z_1 are unchanged, and this proves the formula. □

All coefficients are nonnegative, so the integrand has its largest modulus at $\varphi = 0$. This gives Lemma 3.2, and a Laplace expansion around $\varphi = 0$ gives Theorem 4.4.

A.3 Proof of Lemma 3.2

We bound the sum of Lemma 2.3 term by term by a continuum sum H_σ with slightly larger rates, and H_σ by the largest modulus of the integrand of Lemma A.2.

Step 1: the radii. Let F be irreducible and $\theta \in V_F$. Then $A_F - \operatorname{diag}\theta$ is an irreducible Metzler matrix with Perron root 0. So it has a right eigenvector $u > 0$ for 0, and $\theta_i = (A_F u)_i / u_i$. Each row of A_F has a positive entry, since G_F is strongly connected and $k \geq 2$, so $\theta > 0$. By the second fact, $\varrho(A' - \operatorname{diag}\theta') < 0$. The rows $i \geq 2$ of $(A_F - \operatorname{diag}\theta)u = 0$ give $u' = u_1 X(\theta')b_1$, and then row 1, which reads $a_1^\top u' = \theta_1 u_1$, gives $\theta_1 = g(\theta')$. Hence

$$\theta > 0, \qquad \varrho(A' - \operatorname{diag}\theta') < 0, \qquad g(\theta') = \theta_1 \qquad \text{for every } \theta \in V_F. \tag{A.3}$$

Step 2: a continuum sum. Fix n with support F and h with $hq_{\max} \leq \frac{1}{2}$, and put $|n| = \sum_i n_i$, $t = hn$ and $\sigma = e^{hq_{\max}}$. Lemma 2.3 with $|n|$ in place of N gives the probability $P_{|n|}(n)$ in the lemma. Only terms with $m_i \leq n_i$ for every i are nonzero, and in them $h^{m_i - 1}\binom{n_i - 1}{m_i - 1} \leq t_i^{m_i-1}/(m_i-1)!$ and $(1 - hq_i)^{n_i - m_i} \leq e^{-hq_i(n_i - m_i)} \leq e^{-q_i t_i}\sigma^{m_i}$. Note that $h^{M-1} = h^{k-1}\prod_i h^{m_i - 1}$ and $\sigma^M W(m) = \sigma W_\sigma(m)$, where W_σ is W for σA_F in place of A_F (a word with M letters has $M - 1$ switches). So

$$P_{|n|}(n) \leq \sigma\, h^{k-1} e^{-\langle q,t\rangle} H_\sigma(t), \tag{A.4}$$

where H_σ is H for σA_F.

Step 3: the bound for H_σ. We use Lemma A.2 for σA_F with the radii $\sigma\theta'$, which is allowed since $\varrho(\sigma A' - \operatorname{diag}\sigma\theta') = \sigma\varrho(A' - \operatorname{diag}\theta') < 0$. The functions of Lemma A.2 for σA_F satisfy $X_\sigma(\sigma z') = \sigma^{-1}X(z')$, so $g_\sigma(\sigma z') = \sigma g(z')$ and $b_\sigma(\sigma z') = b(z')$. So on the torus

with radii $\sigma\theta'$ we have $|g_\sigma| \le \sigma g(\theta') = \sigma\theta_1$ by (A.3), $|b_\sigma| \le b(\theta')$ and $|z_i| = \sigma\theta_i$. Hence the integrand has modulus at most $e^{\sigma\langle t,\theta\rangle} b(\theta')\sigma^{k-1}\prod_{i\ge2}\theta_i$, and for every $\sigma > 0$

$$H_\sigma(t) \le \sigma^{k-1} b(\theta') \prod_{i\ge2} \theta_i \, e^{\sigma\langle t,\theta\rangle}. \tag{A.5}$$

With (A.4) this gives $P_{|n|}(n) \le \sigma^k b(\theta')\prod_{i\ge2}\theta_i\, h^{k-1}\exp(-h\langle n, q-\theta\rangle + (\sigma-1)h\langle n,\theta\rangle)$. Finally $\sigma \le e^{1/2}$ and $\sigma - 1 \le 2hq_{\max}$, so the lemma holds with

$$C_\theta = \max\Big(e^{k/2} b(\theta') \prod_{i\ge2}\theta_i,\ 2q_{\max}\max_i \theta_i\Big).$$

The bound after the lemma is the case $\theta = q - \lambda_F \mathbf{1}$. □

A.4 Proof of Theorem 4.4

For $k = 1$ the formula is part of Lemma 2.4. Let $k \ge 2$ and $\alpha \in \mathcal{K}$, and write $\theta = \theta(\alpha)$, $r = r_\alpha$, $l = l_\alpha$ and $\Theta' = \operatorname{diag}(\theta')$. By (A.3) we may use Lemma A.2 with the radii θ' and $t = T\alpha$. With Lemma 2.4 this gives

$$f_T(\alpha) = \frac{T^{k-1} e^{-T\langle q,\alpha\rangle}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} e^{T\Psi(\alpha,\varphi)} \chi(\alpha,\varphi)\, d\varphi, \tag{A.6}$$

where $z_i = \theta_i e^{\mathrm{i}\varphi_i}$, $\Psi(\alpha,\varphi) = \alpha_1 g(z') + \sum_{i\ge2}\alpha_i z_i$ and $\chi(\alpha,\varphi) = b(z')\prod_{i\ge2} z_i$. We check the hypotheses of Lemma B.1 with $m = k-1$.

Hypothesis (iii). By (A.3), $\Psi(\alpha,0) = \langle\alpha,\theta\rangle$ is real. By Lemma A.2, $|g(z')| \le g(\theta') = \theta_1$ on the torus, so

$$\operatorname{Re}\Psi(\alpha,\varphi) \le \Psi(\alpha,0) - \sum_{i\ge2}\alpha_i\theta_i(1-\cos\varphi_i).$$

All α_i and θ_i are positive, so the sum is positive unless $\varphi = 0$.

Hypotheses (i) *and* (ii). We need the first two derivatives of g at θ'. We get them by differentiating a relation $\lambda(g(z'),z') = 0$ for a Perron root λ, whose derivatives are known from Lemma A.1. For $z \in \mathbb{R}^F$ near θ let $\lambda(z) = \varrho(A_F - \operatorname{diag} z)$. With $R = \operatorname{diag} r$ we have $A_F - \operatorname{diag}(\theta-\eta) = R(\widehat{Q}_\alpha + \operatorname{diag}\eta)R^{-1}$, and $\widehat{Q}_\alpha$ is an irreducible generator with stationary law α. So $\lambda(\theta-\eta) = \varrho(\widehat{Q}_\alpha + \operatorname{diag}\eta)$, and Lemma A.1 gives $\nabla\lambda(\theta) = -\alpha$ and $\nabla^2\lambda(\theta) = \Gamma_\alpha$. For real z' near θ' we have $\varrho(A' - \operatorname{diag} z') < 0$, so $(1, X(z')b_1) \ge 0$ by the first fact. This vector lies in the kernel of $A_F - \operatorname{diag}(g(z'),z')$ (rows $i \ge 2$ by the definition of X, and row 1 by the definition of g). By the second fact a nonnegative eigenvector belongs to the Perron root, so $\lambda(g(z'),z') = 0$.

Write g_j, g_{ij} for the derivatives of g at θ' and λ_j, λ_{ij} for those of λ at θ. Differentiating $\lambda(g(z'),z') = 0$ in z_j gives $\lambda_1 g_j + \lambda_j = 0$, and differentiating again in z_i gives $\lambda_1 g_{ij} + \lambda_{11}g_ig_j + \lambda_{1j}g_i + \lambda_{i1}g_j + \lambda_{ij} = 0$. Put $w_j = e_j - (\alpha_j/\alpha_1)e_1 \in \mathbb{R}^F$ for $j \ge 2$. Then the first equation is $g_j = -\alpha_j/\alpha_1$, so $w_j = e_j + g_j e_1$, and the last four terms of the second equation are $w_i^\top \Gamma_\alpha w_j$. So

$$g_j = -\frac{\alpha_j}{\alpha_1}, \qquad \alpha_1 g_{ij} = w_i^\top \Gamma_\alpha w_j.$$

Hence $\partial\Psi/\partial z_j = \alpha_1 g_j + \alpha_j = 0$ at θ', and the Hessian $\mathcal{H}$ of Ψ in z' at θ' is $V^\top\Gamma_\alpha V$, where V has columns $w_2,\dots,w_k$. Since $\partial/\partial\varphi_j = \mathrm{i}z_j\,\partial/\partial z_j$ and the first derivatives vanish, $\nabla_\varphi\Psi(\alpha,0) = 0$. We saw above that $\Psi(\alpha,0)$ is real, so this is hypothesis (i). Also $-\nabla^2_\varphi\Psi(\alpha,0) = \Theta'\mathcal{H}\Theta'$, which is real and symmetric, as (ii) requires.

For the rest of (ii), namely that this matrix is positive definite, let $B = I + \mathbf{1}\alpha'^\top/\alpha_1$. For $y \in \mathbb{R}^{k-1}$ the vector Vy has the entry $-\alpha'^\top y/\alpha_1$ at state 1 and the entries y elsewhere, so $Vy + (\alpha'^\top y/\alpha_1)\mathbf{1} = (0, By)$. Since $\Gamma_\alpha\mathbf{1} = 0$, we may replace Vy by $(0,By)$ in $y^\top V^\top\Gamma_\alpha V y$,

which gives $\mathcal{H} = B^\top \Sigma_\alpha B$. The matrix determinant lemma gives $\det B = 1 + \alpha'^\top \mathbf{1}/\alpha_1 = 1/\alpha_1$. By Lemma A.1, Σ_α is positive definite, and hence so are $\mathcal{H}$ and $\Theta'\mathcal{H}\Theta'$. Also $\det \mathcal{H} = \det \Sigma_\alpha / \alpha_1^2$.

Regularity. By Proposition 4.2(a), $\theta(\alpha)$ is real-analytic. Fix $\alpha_0 \in \mathcal{K}$. By (A.3) and continuity there is $\delta \in (0, \min_i \theta_i(\alpha_0))$ with $\varrho(A' - \operatorname{diag}(\theta'(\alpha_0) - \delta\mathbf{1})) < 0$. By Lemma A.2, X, g and b are holomorphic where $|z_i| > \theta_i(\alpha_0) - \delta$ for every $i \geq 2$, and this holds on the torus of radii $\theta'(\alpha)$ for α near α_0. So Ψ and χ are real-analytic near $\{\alpha_0\} \times \mathbb{T}^{k-1}$. Covering $\mathcal{K}$ by finitely many such sets shows that their φ-derivatives are jointly continuous on $\mathcal{K} \times \mathbb{T}^{k-1}$.

The constant. The matrix $\mathcal{J} = \Theta'\mathcal{H}\Theta'$ of Lemma B.1 has $\det \mathcal{J} = \prod_{i \geq 2} \theta_i^2 \, \det \Sigma_\alpha / \alpha_1^2$. So the lemma gives, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^{k-1}} e^{T\Psi} \chi \, d\varphi = e^{T\langle \alpha, \theta \rangle} \Big(\frac{2\pi}{T}\Big)^{\frac{k-1}{2}} \frac{\alpha_1}{\prod_{i \geq 2} \theta_i \sqrt{\det \Sigma_\alpha}} \Big(\chi(\alpha, 0) + O\Big(\frac{1}{T}\Big)\Big).$$

Here $\chi(\alpha, 0) = b(\theta') \prod_{i \geq 2} \theta_i$. As in the derivation of (A.3), $r' = r_1 X(\theta') b_1$. In the same way the columns $j \geq 2$ of $l^\top (A_F - \operatorname{diag}\theta) = 0$, which is (4.1), give $l' = l_1 X(\theta')^\top a_1$. Hence $\mu \cdot r = r_1(\mu_1 + \mu'^\top X(\theta') b_1)$ and $l \cdot \mathbf{1} = l_1(1 + a_1^\top X(\theta')\mathbf{1})$. Since $l_1 r_1 = \alpha_1$,

$$b(\theta') = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{\alpha_1}.$$

As $r_\alpha, l_\alpha > 0$ and $\mu(F) > 0$ by (H), $\chi(\alpha, 0)$ is positive and continuous, hence bounded below on $\mathcal{K}$. So the last bracket is $\chi(\alpha, 0)(1 + O(1/T))$. Putting this into (A.6) and using $I_F(\alpha) = \langle q - \theta, \alpha \rangle$ gives $f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-T I_F(\alpha)} (1 + O(1/T))$, uniformly on $\mathcal{K}$. □

A.5 Proof of Lemma 4.5

Idea of the proof. The discrete law of Lemma 2.3 and the continuum density of Lemma 2.4 are sums over the same vectors m of run counts, and we compare them term by term. The two terms for m differ by a factor κ_m that is close to 1 when the number of runs M is small compared with $\sqrt{N}$, and under the continuum weights a path has of order T runs.

Step 1: the ratio of the terms. Let $t = hn$ and $\alpha = n/N \in \mathcal{K}$. Lemma 2.4 gives the identity $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q, t \rangle} H(t) = \sum_m v_m$ with

$$v_m = W(m)\, h^{k-1} \prod_{i \in F} e^{-q_i t_i} \frac{t_i^{m_i - 1}}{(m_i - 1)!}, \qquad m \in \mathbb{Z}^F_{\geq 1}.$$

Since $h^{m_i - 1} \binom{n_i - 1}{m_i - 1} = \frac{t_i^{m_i - 1}}{(m_i - 1)!} \prod_{1 \leq j < m_i} (1 - j/n_i)$, the m-th term of Lemma 2.3 is $\kappa_m v_m$ with

$$\kappa_m = \prod_{i \in F} \Big[\prod_{1 \leq j < m_i} \Big(1 - \frac{j}{n_i}\Big) \Big] (1 - hq_i)^{n_i - m_i} e^{h q_i n_i}.$$

Let $w_m = v_m / \sum_{m'} v_{m'}$, a probability on the vectors m since $H(t) > 0$ under (H). We write $\mathbb{E}_w$ and $\mathbb{P}_w$ for it and regard $M = \sum_i m_i$ as a random variable. By Lemma 2.3 the ratio in the lemma is $\sum_m \kappa_m v_m / \sum_m v_m = \mathbb{E}_w \kappa_m$.

Step 2: bounds for κ_m. Let $n_{\min} = \min_i n_i$. We claim that for every m

$$\Big(1 - \frac{M^2}{2 n_{\min}}\Big) e^{-q_{\max}^2 T^2 / N} \leq \kappa_m \leq e^{h q_{\max} M}. \tag{A.7}$$

First let $m_i \leq n_i$ for every i. The factors $1 - j/n_i$ are at most 1, and $(1 - hq_i)^{n_i - m_i} \leq e^{-h q_i (n_i - m_i)}$, so $\kappa_m \leq e^{h q_{\max} M}$. For the lower bound, $\log(1 - x) \geq -x - x^2$ for $0 \leq x \leq \frac{1}{2}$

gives $(1-hq_i)^{n_i-m_i}e^{hq_in_i} \geq e^{-n_ih^2q_i^2}$, and the product of these over i is at least $e^{-q_{\max}^2T^2/N}$. The factors $1-j/n_i$ lie in $[0,1]$, so their product is at least 1 minus the sum of the j/n_i, and hence at least $1-\sum_i m_i^2/(2n_i) \geq 1-M^2/(2n_{\min})$. Now let $m_i > n_i$ for some i. Then $\kappa_m = 0$, while $M \geq n_{\min}+1$ gives $M^2/(2n_{\min}) \geq 2$, so the left side of (A.7) is negative.

By (A.7) it remains to bound $\mathbb{E}_w e^{hq_{\max}M}$ and $\mathbb{E}_w M^2$ from above.

Step 3: the number of runs under w. Since W for σA_F is $\sigma^{M-1}W(m)$, we have $\mathbb{E}_w\sigma^{M-1} = H_\sigma(t)/H(t)$. So the function $\Upsilon(c) = \log \mathbb{E}_w e^{c(M-1)} = \log H_{e^c}(t) - \log H(t)$ is finite. It is convex with $\Upsilon(0)=0$, as the logarithm of a moment generating function, and $\Upsilon(1) \geq 0$ since $M \geq 1$. So $\Upsilon(c) \leq c\,\Upsilon(1)$ for $0 \leq c \leq 1$. We show that $\Upsilon(1) \leq C_{\mathcal{K}}T$, with an upper bound for H_e and a lower bound for H at $t = T\alpha$. By (A.5) with $\sigma = e$ and $\theta = \theta(\alpha)$, $H_e(T\alpha) \leq e^{k-1}b(\theta')\prod_{i\geq 2}\theta_i\, e^{eT\langle\alpha,\theta\rangle}$. For the lower bound note that $H(T\alpha)e^{-T\langle\alpha,\theta\rangle} = T^{-(k-1)}e^{TI_F(\alpha)}f_T(\alpha)$, by the definition of f_T and $I_F(\alpha) = \langle q-\theta,\alpha\rangle$. By Theorem 4.4 this is at least $\frac{1}{2}T^{-(k-1)/2}C_F(\alpha)$ for $T \geq T_1$, and for $1 \leq T \leq T_1$ it is continuous and positive on $[1,T_1]\times\mathcal{K}$. So in both cases $H(T\alpha) \geq c_{\mathcal{K}}T^{-(k-1)/2}e^{T\langle\alpha,\theta\rangle}$. Since $\theta(\alpha)$ and $b(\theta'(\alpha))$ are bounded on $\mathcal{K}$, dividing the two bounds gives $\Upsilon(1) \leq C_{\mathcal{K}}T$.

Step 4: the two averages. Let $x = hq_{\max} \leq \frac{1}{2}$. Then $\mathbb{E}_w e^{xM} = e^{x+\Upsilon(x)} \leq e^{x(1+C_{\mathcal{K}}T)} = 1+O(T^2/N)$, because $x \leq q_{\max}T/N$ and $T^2/N \leq 1$. For the second moment we use the Chernoff bound $\mathbb{P}_w(M-1 \geq y) \leq e^{\Upsilon(1)-y}$ for $y \geq \Upsilon(1)$ and the bound 1 for $y < \Upsilon(1)$. This gives

$$\mathbb{E}_w(M-1)^2 = \int_0^\infty 2y\,\mathbb{P}_w(M-1\geq y)\,dy \leq \Upsilon(1)^2 + \int_{\Upsilon(1)}^\infty 2ye^{\Upsilon(1)-y}\,dy = \Upsilon(1)^2+2\Upsilon(1)+2,$$

so $\mathbb{E}_w M^2 = O(T^2)$. Finally $n_{\min} \geq \varepsilon_{\mathcal{K}}N$ with $\varepsilon_{\mathcal{K}} = \min_{\alpha\in\mathcal{K}}\min_i \alpha_i > 0$. So by (A.7), $\mathbb{E}_w\kappa_m$ lies between $e^{-q_{\max}^2T^2/N}(1-\mathbb{E}_wM^2/(2\varepsilon_{\mathcal{K}}N))$ and $\mathbb{E}_we^{hq_{\max}M}$, and both are $1+O(T^2/N)$, uniformly for $\alpha\in\mathcal{K}$ and $1 \leq T \leq N^{1/2}$. □

A.6 Proof of Theorem 3.3(ii)

Let F be admissible, with components $C_1,\dots,C_r$. If F is irreducible, then $r=1$, and the argument below uses only Lemma 3.2 and an explicit family of paths, not the local law.

Idea of the proof. A path with support F visits the components one after the other, in r blocks, and each passage to the next block is one switch. For the upper bound we apply Lemma 3.2 to every block. For the lower bound we choose the block lengths ourselves. A component of least exit rate gets almost all the time, and every other component C gets only about N/T steps, the time scale of a single switch. Such a short block has probability bounded below, and it has only about $(N/T)^{|C|-1}$ compositions instead of $N^{|C|-1}$.

If $Q=0$, only singletons with positive start mass are admissible, and the bounds follow from $M_{\{i\}} = \mu_i$. Otherwise let a be the least positive rate q_{ij}. Since $T \leq N^{1/3}$, we have $h \leq N^{-2/3}$ and $T^2/N \leq 1$, and we take N_0 so large that h is as small as we need. As in the proof of Lemma 2.2, a path of positive probability with support F visits each component in one block of consecutive times, and it passes from one block to the next by one switch. We call the order of the blocks, the first state and these switches the pattern of the path. There are finitely many patterns.

Upper bound. Fix a pattern whose j-th block lies in the component D_j and starts at y_j, and whose switch out of block j is $x_j \to y_{j+1}$ for $j<r$. So block j ends at x_j for $j<r$, and the pattern does not fix the end of the last block. Let $\operatorname{supp} n = F$. Block j has length $N_j = \sum_{i\in D_j} n_i$. By the Markov property the paths with this pattern and composition n have total probability $\mu_{y_1}\prod_{j<r}hq_{x_jy_{j+1}}\prod_{j\leq r}\varpi_j$. Here ϖ_j is the probability that the chain started at y_j stays in D_j for N_j visits, with the composition that n prescribes on D_j, and for $j<r$ also ends at x_j. We drop the condition on the end. Then ϖ_j is at most

the probability that $X_1, \dots, X_{N_j}$ have that composition for the chain started at y_j. If $|D_j| \geq 2$, Lemma 3.2 with start y_j and $\theta = q - \lambda_{D_j}\mathbf{1}$ gives $\varpi_j \leq \kappa_1 h^{|D_j|-1} e^{-h\lambda_{D_j} N_j + \kappa_1 h^2 N_j}$, where κ_1 is the largest of these finitely many constants. If $D_j = \{i\}$, then $\lambda_{D_j} = q_i$ and $\varpi_j = (1-hq_i)^{N_j - 1} \leq e^{1/2 - h\lambda_{D_j} N_j}$. Now we multiply over j. The powers of h add up to $\sum_j (|D_j| - 1) + r - 1 = k_F - 1$. In the exponent, $\lambda_{D_j} \geq \lambda_F$ and $\sum_j N_j = N$ give $h \sum_j \lambda_{D_j} N_j \geq T\lambda_F$ and $\kappa_1 h^2 \sum_j N_j = \kappa_1 T^2/N \leq \kappa_1$. Summing over the patterns gives $P_N(n) \leq \kappa_2 (T/N)^{k_F - 1} e^{-T\lambda_F}$.

Lower bound. We write down a family of paths with support F, bound its probability from below and count its compositions. By Lemma 2.2(a) we can number the components so that there are $y_1 \in C_1$ with $\mu_{y_1} > 0$ and $x_j \in C_j$, $y_{j+1} \in C_{j+1}$ with $q_{x_j y_{j+1}} > 0$ for $j < r$. Take any $x_r \in C_r$, choose j^* with $\lambda_{C_{j^*}} = \lambda_F$ and $|C_{j^*}| = k^*$, and put $m = \lceil N/T \rceil$, so that $1 \leq hm \leq 2$.

Claim A. There is $c_A > 0$ such that for every component C and all $y, x \in C$, the chain started at y stays in C for its first m visits, visits every state of C and has $X_m = x$ with probability at least c_A. Take a walk $y = v_0, \dots, v_J = x$ through every state of C with $q_{v_s v_{s+1}} > 0$ and $J \leq |C|^2$. Consider the paths $X_1, \dots, X_m$ that make exactly these J switches, in this order, at any J of the $m-1$ transition times. There are $\binom{m-1}{J}$ of them, and each has probability at least $(ha)^J (1 - hq_{\max})^m$. For large N we have $m > |C|^2 \geq J$. Then $\binom{m-1}{J} h^J \geq (hm - Jh)^J / J! \geq (1 - Jh)^J / J!$ and $(1 - hq_{\max})^m \geq e^{-2hq_{\max} m} \geq e^{-4q_{\max}}$, since $1 \leq hm \leq 2$. So their total probability is at least $(1 - Jh)^J a^J e^{-4q_{\max}} / J!$, and $(1 - Jh)^J \geq 2^{-J}$ for small h.

Claim B. There is $c_B > 0$ such that for all $y, x \in C_{j^*}$ and $\ell \geq 2m$, the event of Claim A, with C_{j^*} in place of C and ℓ visits in place of m, has probability at least $c_B e^{-h\lambda_F \ell - h^2 \lambda_F^2 \ell}$. Let $w > 0$ satisfy $Q_{C_{j^*}} w = -\lambda_F w$ and let $B = I + hQ_{C_{j^*}} \geq 0$. Then $(B^s)_{zz'}$ is the probability of going from z to z' in s steps inside C_{j^*}. Since $B^s w = (1 - h\lambda_F)^s w$, the probability of staying in C_{j^*} for s steps from z is $\sum_{z'} (B^s)_{zz'} \geq (B^s w)_z / \max w \geq (1 - h\lambda_F)^s \min w / \max w$. We use Claim A for the first m visits, from y back to y, then $s = \ell - 2m + 1$ steps inside C_{j^*} to any state z', then Claim A for the last m visits, from z' to x. By the Markov property this gives at least $c_A^2 (\min w / \max w)(1 - h\lambda_F)^\ell$. Also $\lambda_F \leq q_{\max}$, because the Perron root of $Q_{C_{j^*}}$ is at least each diagonal entry. So $h\lambda_F \leq \frac{1}{2}$ and $\log(1 - h\lambda_F) \geq -h\lambda_F - h^2\lambda_F^2$.

The family of paths. Give block $j \neq j^*$ the length m and block j^* the length $\ell = N - (r-1)m$, which is at least $2m$ for $T \geq T_0 = r + 2$ and $N \geq N_0$. Consider the paths that go from y_j to x_j in block j as in the claims and pass from block j to block $j+1$ by the switch $x_j \to y_{j+1}$. They all have support F. By the Markov property, and since $h\ell \leq T$ and $h^2 \ell \leq T^2/N \leq 1$, their total probability is at least $\mu_{y_1} (ha)^{r-1} c_A^{r-1} c_B e^{-T\lambda_F - q_{\max}^2}$. The composition of such a path is fixed by its restrictions to the blocks. On the block in C_j the restriction is a vector of $|C_j|$ positive integers whose sum is m for $j \neq j^*$ and ℓ for $j = j^*$. A vector of positive integers with a given sum is fixed by all but one of its entries, and each entry is at most the sum. Also $\ell \leq N$. So these paths have at most $(N+1)^{k^* - 1} \prod_{j \neq j^*} (m+1)^{|C_j| - 1}$ compositions, and $m + 1 \leq 3N/T$. Since $\sum_{j \neq j^*} (|C_j| - 1) = k_F - k^* - (r-1)$ and $h = T/N$, dividing the total probability by the number of compositions gives $M_F \geq c\, N^{-(k_F - 1)} T^{k_F - k^*} e^{-T\lambda_F}$. □

A.7 A forest formula for the covariance

This lemma is used for Proposition 4.9. It writes $\det \Sigma$ through spanning trees, with no reversibility.

For weights $f_{ij} \geq 0$ on the ordered pairs $i \neq j$ of F and a state $i \in F$ let

$$\operatorname{arb}_i(f) = \sum_{\mathcal{T}} \prod_{a \neq i} f_{a\, p_{\mathcal{T}}(a)},$$

the sum over the spanning trees $\mathcal{T}$ of the complete graph on F, where $p_{\mathcal{T}}(a)$ is the neighbor of a on the path from a to i in $\mathcal{T}$. So $\mathrm{arb}_i(f)$ is the weighted count of the spanning trees oriented toward the root i. If f is symmetric, then $\mathrm{arb}_i(f) = \mathrm{tree}(f)$ for every i.

Lemma A.3 (a forest formula for the covariance)**.** *Let G be an irreducible generator on F with stationary law π, where $k = |F| \geq 2$, and let D, Z and Γ be as at the start of this appendix, and Σ as in Lemma A.1 (namely Γ without the row and column of state 1). Put $f_{ij} = \pi_i G_{ij}$ for $i \neq j$, the stationary flow from i to j.*

(i) The number $\mathrm{arb}_i(f)$ does not depend on i. We write $\mathrm{arb}(f)$ for it. Also $\det(-G)_{(i)} = \pi_i \,\mathrm{arb}(f)/\prod_j \pi_j$ for every i, where (i) means that row and column i are deleted, and

$$\det(\mathbf{1}\pi^\top - G) = \sum_{i \in F} \det(-G)_{(i)} = \frac{\mathrm{arb}(f)}{\prod_j \pi_j}.$$

(ii) Let $S = -(DG + G^\top D)$. Then S is the Laplacian of the conductances $f_{ij} + f_{ji}$, and $\Gamma = Z^\top S Z$.

(iii) With $\mathrm{tree}(f + f^\top)$ the weighted count of spanning trees for the conductances $f_{ij} + f_{ji}$,

$$\det \Sigma = \frac{\mathrm{tree}(f + f^\top)}{\det(\mathbf{1}\pi^\top - G)^2} = \frac{(\prod_j \pi_j)^2 \,\mathrm{tree}(f + f^\top)}{\mathrm{arb}(f)^2}.$$

In words, the determinant of the covariance of the occupation fractions is the square of the product of the stationary probabilities, times a count of undirected spanning trees for the symmetrized stationary flow, divided by the square of a count of directed spanning trees for the flow itself. No reversibility is needed. For two states with rates a and b the flow is $ab/(a+b)$ in each direction, the two counts are $2ab/(a+b)$ and $ab/(a+b)$, and $\det \Sigma = 2ab/(a+b)^3$, which is $\frac{1}{4}$ for $a = b = 1$.

Idea of the proof. The matrix $\Gamma = DZ + Z^\top D$ factors as $Z^\top S Z$, where S is twice the symmetric part of the Laplacian $-DG$ of the stationary flow. Each of the three factors has rank $k - 1$, so the determinant of the product is 0 and tells us nothing. The adjugate is still multiplicative, and the adjugate of a matrix of rank $k - 1$ is a rank one matrix that we know up to one scalar from the two kernels. For S the scalar is a count of spanning trees by the matrix-tree theorem. For Z the scalar is $1/\det(\mathbf{1}\pi^\top - G)$, and the directed matrix-tree theorem turns this determinant into a count of oriented spanning trees.

Proof. We use three facts about the adjugate $\mathrm{adj}\, M$ of a $k \times k$ matrix M. First, $\mathrm{adj}(AB) = \mathrm{adj}(B)\,\mathrm{adj}(A)$ for all square matrices A and B, by the Cauchy–Binet formula. Second, $\det(M + uv^\top) = \det M + v^\top \mathrm{adj}(M)\, u$ for all vectors u and v. Third, suppose that M has rank $k - 1$ and that $Mu = 0$ and $v^\top M = 0$ with u and v nonzero. Then $\mathrm{adj}\, M = m\, uv^\top$ for some number m. Indeed $M \,\mathrm{adj}\, M = \mathrm{adj}(M) M = (\det M) I = 0$, so every column of $\mathrm{adj}\, M$ lies in the kernel $\mathbb{R}u$ of M, and every row lies in the left kernel $\mathbb{R}v^\top$. A matrix with these two properties is a multiple of $uv^\top$.

(i) Let $\mathcal{L} = -DG$. Its off-diagonal entries are $-f_{ij}$, its row sums vanish because $G\mathbf{1} = 0$, and its column sums vanish because $\pi^\top G = 0$. The kernel of G is $\mathbb{R}\mathbf{1}$ because G is irreducible, so $\mathcal{L}$ has rank $k - 1$. By the third fact with $u = v = \mathbf{1}$, all entries of $\mathrm{adj}\,\mathcal{L}$ are equal. By the matrix-tree theorem for directed graphs [39, 9], the diagonal entry $\det \mathcal{L}_{(i)}$ of $\mathrm{adj}\,\mathcal{L}$ is $\mathrm{arb}_i(f)$. So $\mathrm{arb}_i(f)$ does not depend on i. Deleting row and column i from $-G = D^{-1}\mathcal{L}$ gives $\det(-G)_{(i)} = \mathrm{arb}(f)/\prod_{j \neq i} \pi_j = \pi_i \,\mathrm{arb}(f)/\prod_j \pi_j$, and the sum over i is $\mathrm{arb}(f)/\prod_j \pi_j$ because $\sum_i \pi_i = 1$. Next, $-G$ has rank $k - 1$, right kernel $\mathbb{R}\mathbf{1}$ and left kernel $\mathbb{R}\pi^\top$. So $\mathrm{adj}(-G) = g\,\mathbf{1}\pi^\top$ for some number g by the third fact. The trace of

$\mathbf{1}\pi^\top$ is 1, so g is the trace of $\operatorname{adj}(-G)$, which is $\sum_i \det(-G)_{(i)}$. Since $\det(-G) = 0$, the second fact gives

$$\det(\mathbf{1}\pi^\top - G) = \pi^\top \operatorname{adj}(-G)\,\mathbf{1} = g\,(\pi^\top \mathbf{1})(\pi^\top \mathbf{1}) = g.$$

(ii) The matrix S is symmetric. For $i \neq j$ its entry is $-(\pi_i G_{ij} + \pi_j G_{ji}) = -(f_{ij} + f_{ji})$, and $S\mathbf{1} = -DG\mathbf{1} - G^\top \pi = 0$. So S is the Laplacian of the conductances $f_{ij} + f_{ji}$. By (4.2), $-GZ = I - \mathbf{1}\pi^\top$ and $\pi^\top Z = 0$. Hence

$$Z^\top(-DG)Z = Z^\top D(I - \mathbf{1}\pi^\top) = Z^\top D - (\pi^\top Z)^\top \pi^\top = Z^\top D,$$

where we used $D\mathbf{1} = \pi$. The transpose of this identity is $Z^\top(-G^\top D)Z = DZ$. Adding the two gives $Z^\top S Z = Z^\top D + DZ = \Gamma$.

(iii) Put $w = \det(\mathbf{1}\pi^\top - G)$, which is not 0 because Z exists. By the matrix-tree theorem [5] every diagonal entry of $\operatorname{adj} S$ is $\operatorname{tree}(f + f^\top)$. This number is positive because the graph of G is strongly connected. So S has rank $k-1$, and the third fact with $u = v = \mathbf{1}$ gives $\operatorname{adj} S = \operatorname{tree}(f + f^\top)\,\mathbf{1}\mathbf{1}^\top$. Now we turn to Z. By (4.2), $Z\mathbf{1} = 0$ and $\pi^\top Z = 0$. The matrix $Z + \mathbf{1}\pi^\top = (\mathbf{1}\pi^\top - G)^{-1}$ is invertible, so Z has rank $k-1$, and $\operatorname{adj} Z = z\,\mathbf{1}\pi^\top$ for some number z by the third fact. Since $\det Z = 0$, the second fact gives

$$\frac{1}{w} = \det(Z + \mathbf{1}\pi^\top) = \pi^\top \operatorname{adj}(Z)\,\mathbf{1} = z.$$

Also $\operatorname{adj}(Z^\top) = (\operatorname{adj} Z)^\top = z\,\pi\mathbf{1}^\top$. By (ii) and the first fact,

$$\operatorname{adj}\Gamma = \operatorname{adj}(Z)\operatorname{adj}(S)\operatorname{adj}(Z^\top) = z^2\,\operatorname{tree}(f + f^\top)\,\mathbf{1}(\pi^\top\mathbf{1})(\mathbf{1}^\top\pi)\mathbf{1}^\top = \frac{\operatorname{tree}(f + f^\top)}{w^2}\,\mathbf{1}\mathbf{1}^\top.$$

The entry of $\operatorname{adj}\Gamma$ at state 1 is $\det\Sigma$. This is the first formula in (iii), and the second follows from (i). □

B A Laplace lemma with a parameter, and the next term

This appendix proves the Laplace lemma that Appendix A uses for Theorem 4.4. It then carries the lemma one step further (Lemma B.2) and proves Proposition 4.15. On $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$ we use $\|\varphi\| = \max_i \operatorname{dist}(\varphi_i, 2\pi\mathbb{Z})$. For $x \in \mathbb{R}^m$ we write $\|x\| = \max_i |x_i|$ and $|x|$ for the Euclidean norm. Repeated indices are summed from 1 to m.

Lemma B.1 (Laplace integrals with a parameter)**.** *Let $\mathcal{K}$ be a compact metric space and $m \geq 1$. Let $\Psi, a : \mathcal{K} \times \mathbb{T}^m \to \mathbb{C}$ be such that the φ-derivatives of Ψ up to order 4 and of a up to order 2 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Assume that for every $\alpha \in \mathcal{K}$*

(i) the value $\Psi(\alpha, 0)$ is real and $\nabla_\varphi \Psi(\alpha, 0) = 0$,

(ii) the matrix $\mathcal{J}(\alpha) = -\nabla^2_\varphi \Psi(\alpha, 0)$ is real, symmetric and positive definite, and

(iii) the inequality $\operatorname{Re}\Psi(\alpha, \varphi) < \Psi(\alpha, 0)$ holds for every $\varphi \neq 0$ in $\mathbb{T}^m$.

Then, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha,\varphi)} a(\alpha, \varphi)\, d\varphi = e^{T\Psi(\alpha,0)} \Big(\frac{2\pi}{T}\Big)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \Big(a(\alpha, 0) + O\Big(\frac{1}{T}\Big)\Big).$$

In words, the integral is the value of the integrand at $\varphi = 0$ times the volume of a Gaussian peak, which is $(2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. The error is uniform in the parameter α.

Idea of the proof. By (iii) and compactness the part of the torus away from $\varphi = 0$ is exponentially small. Near 0 we substitute $\varphi = x/\sqrt{T}$ and expand. The integrand becomes a Gaussian weight times $a(\alpha, 0)$, plus a term of order $T^{-1/2}$ that is odd in x and integrates to 0, plus a remainder of order $1/T$. All constants depend on α only through bounds that hold on the whole of $\mathcal{K}$.

Proof. By compactness $B_\Psi = \max_{|\beta|\le 4} \sup |\partial_\varphi^\beta \Psi|$ and $B_a = \max_{|\beta|\le 2} \sup |\partial_\varphi^\beta a|$ are finite, the least eigenvalue of $\mathcal{J}(\alpha)$ has a positive minimum $\sigma_{\min}$ over $\mathcal{K}$, and $\det \mathcal{J}(\alpha) \le (mB_\Psi)^m$. Below, C depends only on m, $\sigma_{\min}$, B_Ψ, B_a and the numbers ε and η chosen below, and it may change from line to line. So every bound below is uniform on $\mathcal{K}$.

Localization. Choose $\varepsilon \in (0,1)$ with $m^3 B_\Psi \varepsilon \le \frac{3}{2}\sigma_{\min}$. The set of (α,φ) with $\|\varphi\| \ge \varepsilon$ is compact, and by (iii) the continuous function $\operatorname{Re}\Psi(\alpha,\varphi) - \Psi(\alpha,0)$ is negative on it. So it is at most $-\eta$ there for some $\eta > 0$, and the integral over this set has modulus at most $(2\pi)^m B_a e^{T(\Psi(\alpha,0)-\eta)} \le CT^{-1-m/2} e^{T\Psi(\alpha,0)}$.

Scaling. Reduction modulo 2π identifies the cube $\|x\| \le \varepsilon$ in $\mathbb{R}^m$ with the set $\|\varphi\| \le \varepsilon$, and it preserves measure. Let $R(\alpha,\varphi) = \Psi(\alpha,\varphi) - \Psi(\alpha,0) + \frac{1}{2}\varphi^\top \mathcal{J}(\alpha)\varphi$. By (i) and (ii), R and its derivatives of order 1 and 2 vanish at $\varphi = 0$. Taylor's theorem with integral remainder, which holds for complex values, gives on the cube $R = \frac{1}{6}\Psi_{ijk}\varphi^i\varphi^j\varphi^k + \omega_0$ with $|\omega_0| \le C\|\varphi\|^4$, where Ψ_{ijk} are the third derivatives at $(\alpha,0)$. It also gives $|R| \le \frac{m^3}{6} B_\Psi \|\varphi\|^3 \le \frac{1}{4}\sigma_{\min}|\varphi|^2$, since $\sum_{|\beta|=3} 1/\beta! = m^3/6$ and $\|\varphi\|^3 \le \varepsilon|\varphi|^2$. We substitute $\varphi = x/\sqrt{T}$ and put $w(x) = TR(\alpha, x/\sqrt{T})$. The integral over the cube is

$$T^{-m/2} e^{T\Psi(\alpha,0)} \int_{\|x\| \le \varepsilon\sqrt{T}} e^{-\frac{1}{2}x^\top \mathcal{J} x}\, e^{w(x)} a(\alpha, x/\sqrt{T})\, dx,$$

and $|w(x)| \le \frac{1}{4}\sigma_{\min}|x|^2$.

Taylor expansion. Let $P(x) = \frac{1}{6}\Psi_{ijk}x^i x^j x^k$, $a_0 = a(\alpha,0)$ and $a_i = \partial_{\varphi_i} a(\alpha,0)$. Then $w = T^{-1/2}P + \omega_1$ with $|\omega_1| \le CT^{-1}|x|^4$, and $|w| \le CT^{-1/2}|x|^3$. Since $|e^w - 1 - w| \le \frac{1}{2}|w|^2 e^{|w|}$, this gives $|e^w - 1 - T^{-1/2}P| \le CT^{-1}(|x|^4 + |x|^6)e^{\sigma_{\min}|x|^2/4}$. Taylor's theorem for a to orders 1 and 2 gives $|a(\alpha, x/\sqrt{T}) - a_0| \le CT^{-1/2}|x|$ and $|a(\alpha, x/\sqrt{T}) - a_0 - T^{-1/2}a_i x^i| \le CT^{-1}|x|^2$. Multiplying out,

$$e^w a(\alpha, x/\sqrt{T}) = a_0 + T^{-1/2}\big(a_0 P(x) + a_i x^i\big) + \omega(x), \qquad |\omega(x)| \le CT^{-1}(1+|x|)^6 e^{\sigma_{\min}|x|^2/4}.$$

Integration. Since $x^\top \mathcal{J} x \ge \sigma_{\min}|x|^2$, the term ω contributes at most CT^{-1} to the integral, because $(1+|x|)^6 e^{-\sigma_{\min}|x|^2/4}$ is integrable. Outside the cube $|x| > \varepsilon\sqrt{T}$, so extending the integrals of the other two terms to $\mathbb{R}^m$ changes them by at most $Ce^{-\sigma_{\min}\varepsilon^2 T/4} \le CT^{-1}$. On $\mathbb{R}^m$ we have $\int e^{-\frac{1}{2}x^\top \mathcal{J} x} dx = (2\pi)^{m/2} \det \mathcal{J}^{-1/2}$, and the odd polynomial $a_0 P(x) + a_i x^i$ integrates to 0 against this weight. This is why the error is $O(1/T)$ and not $O(T^{-1/2})$. So the integral over the cube is $e^{T\Psi(\alpha,0)} T^{-m/2}\big[(2\pi)^{m/2} \det \mathcal{J}^{-1/2} a_0 + O(T^{-1})\big]$. Since $\det \mathcal{J} \le (mB_\Psi)^m$, this error and the one from the localization are $O(1/T)$ relative to $e^{T\Psi(\alpha,0)}(2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. □

B.1 The next term of the Laplace lemma

With two more derivatives the same proof gives the term of order $1/T$ and an error $O(T^{-2})$.

Lemma B.2 (the next term)**.** *Let $\mathcal{K}$, m, Ψ and a satisfy the hypotheses of Lemma B.1. Assume also that the φ-derivatives of Ψ up to order 6 and of a up to order 4 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Write $a_0 = a(\alpha,0)$, write $\mathcal{J}^{ij}$ for the entries of $\mathcal{J}(\alpha)^{-1}$, and write Ψ_{ijk}, Ψ_{ijkl}, a_i and a_{ij} for the φ-derivatives at $(\alpha,0)$. Put*

$$\begin{aligned} c_1(\alpha) = {} & \tfrac{1}{2} a_{ij}\mathcal{J}^{ij} + \tfrac{1}{2} a_i \Psi_{jkl}\mathcal{J}^{ij}\mathcal{J}^{kl} \\ & + a_0\Big[\tfrac{1}{8}\Psi_{ijkl}\mathcal{J}^{ij}\mathcal{J}^{kl} + \tfrac{1}{12}\Psi_{ijk}\Psi_{lrs}\mathcal{J}^{il}\mathcal{J}^{jr}\mathcal{J}^{ks} + \tfrac{1}{8}\Psi_{ijk}\Psi_{lrs}\mathcal{J}^{ij}\mathcal{J}^{kl}\mathcal{J}^{rs}\Big]. \end{aligned} \tag{B.1}$$

Then c_1 is continuous on $\mathcal{K}$, and uniformly for $\alpha \in \mathcal{K}$ and $T \ge 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha,\varphi)} a(\alpha,\varphi)\, d\varphi = e^{T\Psi(\alpha,0)} \Big(\frac{2\pi}{T}\Big)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \Big(a_0 + \frac{c_1(\alpha)}{T} + O\Big(\frac{1}{T^2}\Big)\Big).$$

The proof, including the Gaussian moment calculation, is in Section S3 of the supplement.

B.2 The coefficient in Proposition 4.15

For the phase Ψ and amplitude χ of (A.6), apply (B.1) with $a = \chi$ and $\mathcal{J}(\alpha) = -\nabla^2_\varphi \Psi(\alpha, 0)$. Define

$$c(\alpha) = \frac{c_1(\alpha)}{\chi(\alpha, 0)}, \qquad \chi(\alpha, 0) = b(\theta'(\alpha)) \prod_{i \geq 2} \theta_i(\alpha) > 0. \tag{B.2}$$

This gives the coefficient of the local law in Proposition 4.15(i). Maximizing near α_F^* adds $\frac{1}{2}\gamma_F^\top \Sigma_F \gamma_F$ to give c_F. Section S4 of the supplement proves all four parts of the proposition, including the all-size hand evaluation $c_F = -(k-1)/8$ on the complete graph and the independent Bessel check for a symmetric pair.

C Proofs for the complete graph

This appendix proves Theorems 5.4 and 5.5 and Corollary 5.6. We use the notation of Section 5.2, so $Q = J - dI$, $\sigma = p - h$, $u = h/p$ and $v = h/\sigma$. The first lemma writes the law in a second form, with blocks between refreshes in place of runs.

Lemma C.1 (the refresh form)**.** *Let $N \geq 1$ and let n be a composition with support F. Sums over m run over the integer vectors with $1 \leq m_i \leq n_i$ for $i \in F$, and $M = \sum_i m_i$.*

(a) With $0^0 = 1$,

$$P_N(n) = \sum_m \Big(\sum_{i \in F} \mu_i m_i \Big) \frac{(M-1)!}{\prod_i m_i!}\, h^{M-1} \sigma^{N-M} \prod_{i \in F} \binom{n_i - 1}{m_i - 1}.$$

(b) Let $1/d < p < 1$, so that $0 < u < 1$ and $v > 0$. Then

$$S_n(u) = (1-u)^{N-1} \sum_m M!\, v^{M-1} \prod_{i \in F} \frac{1}{m_i!} \binom{n_i - 1}{m_i - 1}, \tag{C.1}$$

which is a sum of positive terms, and

$$S_n(u) = \frac{(1-u)^{N-1}}{v} \int_0^\infty e^{-t} \prod_{i \in F} B_{n_i}(vt)\, dt, \tag{C.2}$$

where B_n is defined in (5.4)*.*

Proof. (a) The transition matrix is $I + hQ = \sigma I + hJ$, so a word $x_1 \cdots x_N$ has probability $\mu_{x_1} \prod_{s<N} (\sigma \mathbf{1}\{x_s = x_{s+1}\} + h)$. We expand each factor into its two terms and call the steps where the term h is chosen the refreshes. If there are $M - 1$ refreshes, they cut the word into M blocks, and a term of the expansion is not zero only if the word is constant on each block. So $P_N(n)$ is the sum, over the sequences of M block labels and over the block lengths, of $\mu_{\ell_1} h^{M-1} \sigma^{N-M}$, where ℓ_1 is the label of the first block. If m_i blocks carry the label i, there are $m_i (M-1)! / \prod_j m_j!$ sequences of labels that start with i. Consecutive labels may be equal. Given the labels, the lengths of the m_i blocks of the label i form an ordered sum of n_i with m_i positive parts, in $\binom{n_i - 1}{m_i - 1}$ ways. Multiplying and summing gives (a).

(b) For the uniform start $\sum_i \mu_i m_i = M/d$. By (5.1) we divide (a) by $V_N = p^{N-1}/d$, and $h^{M-1}\sigma^{N-M}/p^{N-1} = u^{M-1}(1-u)^{N-M} = (1-u)^{N-1} v^{M-1}$, since $\sigma/p = 1 - u$. This gives (C.1). We write $M! = \int_0^\infty e^{-t} t^M\, dt$ and sum over each m_i separately. All terms are nonnegative, so the sum and the integral can be exchanged, and this gives (C.2). □

When $\sigma \geq 0$, part (a) has a direct reading. The chain keeps its state with probability σ and otherwise draws a fresh state uniformly from all d states, possibly the same one. The proof does not need $\sigma \geq 0$. At $p = 1/d$ we have $\sigma = 0$ and only $M = N$ survives, which gives the multinomial law for the uniform start.

C.1 Proof of Theorem 5.4

Let F be a face of k states and n a balanced composition of F, so that $P_N(n)/V_N = S_n(u)$ by (5.1), with $u = (1-p)/((d-1)p)$.

Part (i), one crossing. To count $W_n(j)$, choose a sequence of $j+1$ run colors in which successive colors differ. If color i appears m_i times, its run lengths can be chosen in $\binom{n_i-1}{m_i-1}$ ways. So every coefficient of S_n is nonnegative, and the first nonzero one is $W_n(k-1) = k!$, from the $k!$ orders of k single runs. Hence S_n increases strictly from 0 to ∞ on $u > 0$ and equals 1 exactly once. As p increases over $(0,1)$, u decreases over $(0,\infty)$. So $P_N(n) = V_N$ at exactly one $p_{N,k} \in (0,1)$, and the composition is more likely than a vertex below $p_{N,k}$ and less likely above it.

Part (i), monotonicity in N. Choose the balanced compositions for increasing N by adding 1 to one of the smallest coordinates. Each coefficient $W_n(j)$ is then nondecreasing. To see this, map a word to the word with an extra copy of the incremented color just before its first occurrence. This keeps the number of switches, and it is injective, since deleting the first letter of the first run of that color inverts it. The coefficient of u^k is

$$W_n(k) = \frac{k!(k-1)}{2} \sum_i (n_i - 1) = \frac{k!(k-1)}{2}(N-k).$$

To see this, note that exactly one run color appears twice. For each choice of it there are $(k+1)!/2 - k! = k!(k-1)/2$ sequences of run colors in which successive colors differ (all orders, minus those with the two copies adjacent), and $n_i - 1$ ways to split its n_i visits into two runs. This coefficient increases strictly with N. So at every $u > 0$ the value $S_n(u)$ increases strictly with N, the root of $S_n(u) = 1$ decreases strictly with N, and $p = 1/(1+(d-1)u)$ increases.

Part (i), the first crossing. If $N = k$, every coordinate is 1 and $S_n(u) = k!u^{k-1}$, which gives $p_{k,k}$. Since the root decreases with N, every root has $u \leq (k!)^{-1/(k-1)} < 1$, and $u < 1$ means $h < p$, that is, $p > 1/d$.

Part (i), the face maximum. For $p \geq 1/d$, Theorem 5.5 gives $M_F = P_N(n)$. For $p < 1/d$ we have $u > 1$, which exceeds the root of $S_n(u) = 1$, so $M_F \geq P_N(n) > V_N$. Hence $M_F = V_N$ exactly at $p_{N,k}$, with the same two inequalities.

Part (ii). By Proposition 4.15(ii) with $K_F = D_k/d$, $\lambda_F = d-k$ and $c_F = -(k-1)/8$ (Corollary 5.3), for large N and uniformly for $1 \leq T \leq N^{1/3}$,

$$\log M_F = -(k-1)L + \tfrac{k-1}{2}\log T + \log(D_k/d) - (d-k)T - \frac{k-1}{8T} + O(T^{-2} + T^2/N).$$

By (2.2), $\log V_N = -\log d - (d-1)T + O(T^2/N)$. Subtracting, and using $\log D_k = -(k-1)a_k$, gives the display. With $T = L - \frac{1}{2}\log L + t$ and t in a compact set, $T - L + \frac{1}{2}\log T = t + \frac{1}{2}\log(T/L) = t + O(\log L/L)$, and the terms $1/(8T)$ and the error tend to 0.

Part (iii). Let G be a vertex, so that $M_G = V_N$. By Theorem 4.7(i) the function $\log(M_F/M_G)$ has a zero in $[L/2, 2L]$, since $\Delta = \delta = k-1$, and by (i) this zero is $T_{N,k}$. By Corollary 5.3, $\log(K_F/K_G) = \log D_k = -(k-1)a_k$ and $c_F - c_G = -(k-1)/8$. So (4.8) reads $T(k-1)(L - \frac{1}{2}\log T - T + a_k) = -\frac{k-1}{8} + O(1/L)$ at $T = T_{N,k}$, and dividing by $(k-1)T$ gives the implicit equation. For the explicit form put $z_* = L - \frac{1}{2}\log L + a_k + e_L/L$ with $e_L = \frac{1}{4}\log L - \frac{1}{2}a_k + \frac{1}{8}$. Since $\log z_* = \log L - \frac{\log L}{2L} + \frac{a_k}{L} + O(\frac{(\log L)^2}{L^2})$, we get $z_* + \frac{1}{2}\log z_* - 1/(8z_*) = L + a_k + O((\log L)^2/L^2)$. The derivative of $z + \frac{1}{2}\log z - 1/(8z)$ is at least 1 for $z > 0$, so $T_{N,k} - z_* = O((\log L)^2/L^2)$, which is (5.3).

Part (iv). We have $a_k = \frac{1}{2}\log(4\pi) - f(k)$ with $f(x) = (x+\frac{1}{2})\log x/(x-1)$. For $x \geq 2$ the sign of $f'(x)$ is the sign of $\Delta(x) = 2x^2 - x - 1 - 3x\log x$. Here $\Delta(2) = 5 - 6\log 2 > 0$, and $\Delta'(x) = 4x - 4 - 3\log x > 0$ for $x \geq 2$, since $\Delta'(2) > 0$ and $\Delta''(x) = 4 - 3/x > 0$. So f increases strictly and a_k decreases strictly. This order is checked in Lean. □

C.2 Proof of Theorem 5.5

Let $\mu_a = \mu_0$ for $a \in F$. By (2.3), $P_N(n) = \mu_0 p^{N-1} S_n(u)$ for every composition with support F, so it is enough to show that the move strictly increases $S_n(u)$. The polynomial S_n is symmetric in the coordinates of n, so all balanced compositions of F have the same value, and repeated moves reach a balanced composition from any composition with support F. So the move gives the theorem. At $p = 1/d$ we have $u = 1$ and $S_n(1) = N!/\prod_i n_i!$, the number of words with the counts n, and the move multiplies it by $n_b/(n_a+1) > 1$. Let $1/d < p < 1$, so that $\sigma > 0$ and $v > 0$.

Log-concavity. We show that for each $w > 0$ the sequence $B_n(w)$, $n \geq 1$, is strictly log-concave. Write

$$\frac{B_n(w)}{w} = b_{n-1}, \qquad b_j = \sum_{i=0}^{j} \binom{j}{i} \rho_i, \quad \rho_i = \frac{w^i}{(i+1)!}, \qquad c_j = \sum_{i=0}^{j} \binom{j}{i} \rho_{i+1}.$$

Pascal's identity gives $b_{j+1} = b_j + c_j$. The ratio c_j/b_j is the mean of the strictly decreasing sequence $f_i = \rho_{i+1}/\rho_i = w/(i+2)$ under the weights $\omega_i = \binom{j}{i}\rho_i$, $0 \leq i \leq j$, normalized. Let $\omega'_i = \binom{j+1}{i}\rho_i$, $0 \leq i \leq j+1$, be the weights for $j+1$, and put $\omega_{j+1} = 0$. On $0 \leq i \leq j$ the ratio $\omega'_i/\omega_i = (j+1)/(j+1-i)$ increases in i. So for $i < l$ we have $\omega'_i\omega_l - \omega'_l\omega_i \leq 0$, with strict inequality when $l = j+1$. Hence

$$\sum_i \omega'_i f_i \sum_l \omega_l - \sum_i \omega'_i \sum_l \omega_l f_l = \sum_{i<l} (\omega'_i\omega_l - \omega'_l\omega_i)(f_i - f_l) < 0,$$

since $f_i > f_l$ for $i < l$. So the mean decreases strictly, $c_{j+1}/b_{j+1} < c_j/b_j$, and $b_{j+1}/b_j = 1 + c_j/b_j$ decreases strictly with j. This is the strict log-concavity. By the Laguerre form in (5.4), this is a Turán-type inequality for the Laguerre polynomials.

The transfer. Let $n_b \geq n_a + 2$. By log-concavity, $B_{n_a+1}(w)/B_{n_a}(w) > B_{n_b}(w)/B_{n_b-1}(w)$, that is, $B_{n_a+1}(w)B_{n_b-1}(w) > B_{n_a}(w)B_{n_b}(w)$. We apply this for $w = vt > 0$ in (C.2) and integrate. So the move strictly increases $S_n(u)$. □

C.3 Proof of Corollary 5.6

For large N, $T \leq CL$ forces $h \to 0$ and so $p > 1/d$. By Theorem 5.5 every global mode is a vertex or a balanced composition, and all balanced compositions with the same number of positive coordinates have the same probability. So we compare the ratios M_F/V_N, $2 \leq |F| \leq d$, which equal $S_n(u)$ at the balanced compositions n, with 1 and with each other. They depend on F only through $k = |F|$. Fix $0 < c < 1$.

Case $T \leq cL$. The ratio $S_n(u)$ increases with u, and u increases with T. So it is at most its value at $T = cL$, which is $\exp(-(k-1)(1-c)L + O(\log L)) \to 0$ by Theorem 5.4(ii). So the vertices are the only modes, and $p > p_{N,d}$.

Case $cL \leq T \leq CL$. Theorem 5.4(ii) gives $\log(M_F/V_N) = (k-1)(y - a_k) + o(1)$ for $|F| = k$, with $y = T - L + \frac{1}{2}\log T$, uniformly. The number y is the same for every k. Fix $\delta > 0$, with $\delta < (a_{d-1} - a_d)/2$ when $d \geq 3$. This is possible since $a_2 > \cdots > a_d$. If $y \leq a_d - \delta$, every face ratio is at most $e^{-(k-1)\delta + o(1)} < 1$, so only the vertices are modes. If $y \geq a_d + \delta$, the full center beats the vertices, and it beats a balanced composition with $2 \leq k < d$ positive coordinates by

$$(d-k)(y - a_d) + (k-1)(a_k - a_d) + o(1) > 0.$$

If $|y-a_d|<\delta$, a proper face with $k\ge 2$ states has $(k-1)(y-a_k)+o(1)\le -(k-1)\delta+o(1)<0$, since $a_k-a_d>2\delta$. So it loses to the vertices, and the comparison of the vertices with the full centers is decided by the unique crossing $p_{N,d}$ of Theorem 5.4(i). In each subcase the conclusion agrees with (i) to (iii).

The last claim follows from (5.3) and $p_{N,k}=1-(d-1)T_{N,k}/N$, since $a_2>\cdots>a_d$. □

D Proofs near a vertex

We use the notation of Section 6 and prove Lemma 6.1 and Theorem 6.2. After the proof of the lemma, the first lemma below collects what we need about the polynomial $\mathcal{B}$ of the rare counts, and the second writes $S_{(b,a)}$ exactly as an average of $\mathcal{B}$.

Proof of Lemma 6.1. By definition, $B_a(w)/w=\sum_{j=0}^{a-1}\binom{a-1}{j}w^j/(j+1)!$. So the coefficient of $(aw)^j/[j!(j+1)!]$ in $B_a(w)/w$ is $\binom{a-1}{j}j!/a^j=\prod_{l=1}^{j}(1-l/a)_+$, also for $j\ge a$, where it is 0. By the clipped products of [35, Lemma 3.2(b)], its deficit from 1 lies between 0 and $j(j+1)/(2a)$. Also $\sum_{j\ge1}j(j+1)x^j/[j!(j+1)!]=\sum_{j\ge1}x^j/[(j-1)!\,j!]=x\,\mathcal{I}(x)$. So

$$0\le \mathcal{I}(aw)-\frac{B_a(w)}{w}\le \frac{1}{2a}\sum_{j\ge0}\frac{j(j+1)(aw)^j}{j!(j+1)!}=\frac{w}{2}\,\mathcal{I}(aw). \qquad \square$$

Lemma D.1 (elasticity and concavity of the rare-count polynomial)**.** *The polynomial $\mathcal{B}$ is convex, $\log\mathcal{B}$ is concave on $(0,\infty)$, and κ is nondecreasing. Also*

$$\frac{1}{|a|+1}\le y_a\le 1,\qquad \kappa(y)\le \ell+\sqrt{\ell|a|y}\quad (y>0). \tag{D.1}$$

For $0<y\le 1$ we also have $\kappa(y)\ge\max\{\ell,\tfrac12\sqrt{|a|y}\}$.

Proof. Step 1. One factor as a law. The polynomial $\mathcal{B}$ has nonnegative coefficients, so it is convex on $(0,\infty)$. For one factor B_a and $y>0$, let J take values in $\{0,\dots,a-1\}$ with $\mathbb{P}(J=j)$ proportional to $\binom{a-1}{j}y^j/(j+1)!$. Since $B_a(y)=\sum_j y\binom{a-1}{j}y^j/(j+1)!$, the elasticity of B_a is $\kappa_a=yB_a'/B_a=\mathbb{E}[J+1]$. The ratio

$$\mathcal{R}(j)=\frac{(j+1)\mathbb{P}(J=j+1)}{\mathbb{P}(J=j)}=\frac{y(a-1-j)}{j+2},\qquad \mathcal{R}(a-1)=0,$$

is nonincreasing in j. Summing the ratio identity gives $\mathbb{E}J=\mathbb{E}\mathcal{R}(J)$ and $\mathbb{E}J^2=\mathbb{E}[(J+1)\mathcal{R}(J)]$. So

$$\operatorname{Var}J=\mathbb{E}J+\operatorname{Cov}(J,\mathcal{R}(J))\le \mathbb{E}J.$$

The covariance is nonpositive because, for an independent copy J', twice the covariance equals $\mathbb{E}[(J-J')(\mathcal{R}(J)-\mathcal{R}(J'))]\le 0$.

Step 2. Concavity and monotonicity. Since $y^2B_a''/B_a=\mathbb{E}[J(J+1)]$, we get

$$(\log B_a)''=\frac{\operatorname{Var}J-\mathbb{E}[J+1]}{y^2}<0,\qquad y\kappa_a'=\operatorname{Var}J\ge 0.$$

Summing over the factors proves the concavity of $\log\mathcal{B}$ and the monotonicity of $\kappa=\sum_i\kappa_{a_i}$.

Step 3. The elasticity bounds. The differential equation $yB_a''+yB_a'=aB_a$, which we check on the coefficients, gives

$$ay=\kappa_a^2+\operatorname{Var}J+(y-1)\kappa_a.$$

Using $0\le\operatorname{Var}J\le\kappa_a-1$ we get

$$\kappa_a\le 1+\sqrt{ay},\qquad \kappa_a^2+y\kappa_a\ge ay+1.$$

For $0 < y \le 1$ the second inequality gives $(\kappa_a + \frac{1}{2})^2 > ay$, and $\kappa_a \ge 1$. So $\kappa_a \ge \max\{1, \sqrt{ay} - 1/2\} \ge \frac{1}{2}\sqrt{ay}$. Summing over the factors proves the elasticity bounds, using the Cauchy–Schwarz inequality $\sum_i \sqrt{a_i} \le \sqrt{\ell |a|}$ for the upper bound and $\sqrt{\sum_i x_i} \le \sum_i \sqrt{x_i}$ for the lower one.

Step 4. The root. Finally, $B_a(1) \ge 1$, from its term $m = 1$, and Lemma 6.1 gives

$$B_a(y) \le y\,\mathcal{I}(ay) \le y e^{ay/2},$$

where $(j+1)! \ge 2^j$ proves the last inequality term by term. So $\mathcal{B}(1) \ge 1$, while, since $|a| + 1 \ge 2$, $\mathcal{B}(1/(|a|+1)) \le (|a|+1)^{-\ell} e^{|a|/(2(|a|+1))} \le \frac{1}{2} e^{1/2} < 1$. □

Lemma D.2 (an exact compound-binomial representation)**.** *For $0 < u < 1$, put $v = u/(1-u)$. Let Γ_2 have the gamma distribution with shape 2 and rate 1, let $E_1, \dots, E_{b-1}$ be independent rate-one exponential variables, and let $Z_1, \dots, Z_{b-1}$ be independent Bernoulli variables with parameter u. Take all these variables independent and set*

$$\mathcal{X} = \Gamma_2 + \sum_{j=1}^{b-1} Z_j E_j.$$

Then

$$S_{(b,a)}(u) = (1-u)^{|a|}\mathbb{E}[\mathcal{B}(v\mathcal{X})], \qquad \mathbb{E}e^{s\mathcal{X}} = (1-s)^{-2}\left(1 + \frac{us}{1-s}\right)^{b-1} \quad (s < 1). \tag{D.2}$$

In particular, set $y = bu^2$ and $\kappa = \kappa(y)$. For every $0 < u < 1$,

$$(1-u)^{|a|}\mathcal{B}(y) \le S_{(b,a)}(u). \tag{D.3}$$

If also $s = \kappa/[bu(1-u)] < 1$, then

$$S_{(b,a)}(u) \le (1-u)^{|a|}\mathcal{B}(y) e^{-\kappa}(1-s)^{-2}\left(1 + \frac{us}{1-s}\right)^{b-1}. \tag{D.4}$$

Proof. For $0 < u < 1$ the stay probability $p = 1/(1 + (d-1)u)$ lies in $(1/d, 1)$, so by (C.2) with $N = b + |a|$,

$$S_{(b,a)}(u) = \frac{(1-u)^{b+|a|-1}}{v}\int_0^\infty e^{-t} B_b(vt)\,\mathcal{B}(vt)\,dt.$$

The integral of $e^{-t}B_b(vt)$ is $\sum_m \binom{b-1}{m-1} v^m = v(1+v)^{b-1}$, and $(1-u)^{b-1}(1+v)^{b-1} = 1$. After normalization its density is $\sum_{j=0}^{b-1}\binom{b-1}{j} u^j (1-u)^{b-1-j}\, t^{j+1} e^{-t}/(j+1)!$, a mixture of gamma densities with shape $J+2$, where J has the binomial law with parameters $b-1$ and u. This is exactly the law of $\mathcal{X}$, which proves the identity and its moment generating function. Since $\mathbb{E}\mathcal{X} = 2 + (b-1)u$, we have $v\mathbb{E}\mathcal{X} - bu^2 = u(2 - u + bu^2)/(1-u) > 0$, so $v\mathbb{E}\mathcal{X} \ge y$. Jensen's inequality for the convex function $\mathcal{B}$, and the monotonicity of $\mathcal{B}$, prove (D.3). Concavity of $\log \mathcal{B}$ gives, for $w > 0$,

$$\mathcal{B}(w) \le \mathcal{B}(y)\exp\left(\frac{\kappa}{y}(w - y)\right).$$

Insert $w = v\mathcal{X}$, note that $\kappa v/y = s$, and use the moment generating function to prove (D.4). □

Proof of Theorem 6.2. Write $S = S_{(b,a)}$. Put $u_0 = \sqrt{y_a/b}$ and $\kappa_0 = \kappa(y_a)$. Since S increases on $u > 0$, it suffices to show $\log S(u_+) > 0$ and $\log S(u_-) < 0$ at $u_\pm = (1 \pm \delta)u_0$. Note that $\delta \le 1/2$ forces $\varepsilon \le 1/(32c_\ell) \le 1/96$, and that $u_0 \le \sqrt{1/b} \le \varepsilon$ since $y_a \le 1$.

Step 1. The upper point. We have $u_+ \le \frac{3}{2}\varepsilon < 1/4$. Since $\log \mathcal{B}(y_a) = 0$ and $(\log \mathcal{B})' = \kappa/y$, the monotonicity of κ gives $\log \mathcal{B}(bu_+^2) \ge \kappa_0 \log(bu_+^2/y_a) = 2\kappa_0 \log(1+\delta)$. The lower bound in Lemma D.1, applied at $y_a \le 1$, gives $|a|u_0/\kappa_0 \le 2\sqrt{|a|/b} \le 2\varepsilon$. Hence, by (D.3), which holds for all $0 < u < 1$, and $-\log(1-x) \le x/(1-x)$,

$$\log S(u_+) \ge 2\kappa_0 \log(1+\delta) - \frac{|a|u_+}{1-u_+} \ge \kappa_0 \left(\frac{4}{3}\delta - 4\varepsilon\right) > 0.$$

Here we used $2\log(1+\delta) \ge \frac{4}{3}\delta$ for $\delta \le \frac{1}{2}$, and $|a|u_+/(1-u_+) \le \frac{4}{3} \cdot \frac{3}{2}|a|u_0 \le 4\kappa_0\varepsilon$. The last inequality holds since $\frac{4}{3}\delta = \frac{64}{3}c_\ell\varepsilon > 4\varepsilon$.

Step 2. The lower point. Write $y = bu_-^2$ and $\kappa = \kappa(y)$. The inequalities $y_a \ge 1/(|a|+1)$, $\delta \le 1/2$ and the upper bound in (D.1) imply

$$u_- \le \varepsilon, \qquad bu_- \ge \frac{1}{2\varepsilon}, \qquad s = \frac{\kappa}{bu_-(1-u_-)} \le 2(2\ell + \sqrt{\ell})\varepsilon \le 4c_\ell\varepsilon \le \frac{1}{8}.$$

For the bound on s, note that $\kappa/(bu_-) \le \ell/(bu_-) + \sqrt{\ell|a|y}/(bu_-) \le 2\ell\varepsilon + \sqrt{\ell|a|/b} \le (2\ell+\sqrt{\ell})\varepsilon$, and $1 - u_- \ge \frac{1}{2}$. We use $\log(1+x) \le x$, $-\log(1-s) \le s/(1-s)$ and $u_-, s \le 1/4$ in (D.4), and we drop the nonpositive term $|a|\log(1-u_-)$. This gives

$$\begin{aligned}
\log \frac{S(u_-)}{\mathcal{B}(y)} &\le -\kappa + \frac{2s}{1-s} + \frac{bu_-s}{1-s} \\
&= \kappa\left(\frac{1}{(1-u_-)(1-s)} - 1\right) + \frac{2s}{1-s} \\
&\le 5\kappa(u_- + s) \le 25c_\ell\kappa\varepsilon.
\end{aligned}$$

Here the equality uses $bu_-s = \kappa/(1-u_-)$. The next inequality uses $(1-u_-)(1-s) \ge \frac{21}{32}$, so the first term is at most $\frac{32}{21}\kappa(u_- + s)$, and $\kappa \ge \ell \ge 1$, so the second is at most $\frac{16}{7}\kappa s$. The last one uses $u_- + s \le 5c_\ell\varepsilon$. Again by the monotonicity of κ,

$$\log \mathcal{B}(y) = -\int_y^{y_a} \frac{\kappa(x)}{x}\,dx \le 2\kappa\log(1-\delta) \le -2\kappa\delta = -32c_\ell\kappa\varepsilon.$$

Hence $\log S(u_-) \le \kappa(-32c_\ell + 25c_\ell)\varepsilon < 0$. Since S is increasing on $u > 0$, the two strict inequalities bracket the crossing. □

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A note on the use of AI

Most results of Sections 5 and 6 come from the preprint [36]. For that preprint I brainstormed conjectures with Claude Opus 5.5 (Anthropic, through Claude Code), and several results came from it together with proofs. Of these, this paper contains the tree form of the constants (Example 4.11 and the constant $D(\alpha)$ in (5.2)), the global modes with a start (Corollary 5.10) and the example in which the start decides which face wins (Example 5.12). I rechecked and rederived these proofs by hand. I also checked the proofs with GPT-6 Astra (OpenAI), which found nothing beyond the earlier check. The Lean 4 files in kagey-problems/problem131/lean/PaperC and

kagey-problems/problem131/lean/PaperB were generated by the model, and those in lean/PaperB were then revised and checked by hand against the corresponding results (see MAP.md in each folder). Parts of the Python verification scripts of this paper and of [36] were also written by the model, which cross-checked them against each other. I also used it for literature search, namely to find and verify sources beyond the ones I had first checked on my own.

I later used OpenAI Codex for another mathematical and editorial review. That revision corrected hypotheses and normalization statements, clarified proof exposition, moved the complete higher-order proofs to the supplement, and added the K_4 envelope figure and literature comparisons. Codex also reorganized the introduction and checked the revised arguments against primary sources. I checked the resulting revisions. This review does not replace external peer review.

All mathematical claims, proofs and computations in this paper are my responsibility.

Supplementary Material

Supplement to "Face selection for slowly switching Markov chains". Section S1 proves the rows P_d and Θ_ε of Table 2, and Section S2 gives the full proof of Proposition 4.14. Sections S3 and S4 give the higher-order Laplace proof and the full proof of Proposition 4.15. The supplement is the file supplement.tex in the folder kagey-problems/problem131/paperC of https://github.com/apemm/Kagey-Problems.

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graph	winners in order, with the value of τ where each one starts	status
K_d	$\lambda_F = d - \lvert F\rvert$. Vertices for $\tau < 1$ and $[d]$ for $\tau > 1$. Every face ties at $\tau = 1$, and Section 5 resolves the tie at second order.	proved here, and in Lean
symmetric Q	$\lambda_{[d]\setminus\{x\}} \le q_x/(d-1)$, with equality exactly when $q_{yx} = q_x/(d-1)$ for all $y \neq x$. With unit rates on a connected graph, the winners go from vertices straight to $[d]$ exactly when the graph is complete.	proved here
C_d, $d \ge 4$	Arcs of $k = 1, \dots, k_d$ states with $k_d = \lfloor 2d/3 \rfloor$, then C_d. The arc of $k \ge 2$ states wins from $1/(\lambda_{k-1} - \lambda_k)$ with $\lambda_k = 2 - 2\cos\frac{\pi}{k+1}$, and C_d from $\lceil d/3 \rceil/\lambda_{k_d}$. These values are 1, $1+\sqrt{2}$, 4.906 and 8.771 for $k = 2, \dots, 5$, and they grow like $k^3/(2\pi^2)$. Arcs of $k \ge 3$ states win exactly when $d \ge \lceil 3k/2 \rceil$.	Lean only (computer-verified), asymptotics by hand
P_d, $d \ge 3$	End-arcs of $1, \dots, \lfloor 2d/3 \rfloor$ states, then P_d. The end-arc of k states has exit rate $2 - 2\cos\frac{\pi}{2k+1}$, and the first switch is at the golden ratio $(1+\sqrt{5})/2$.	supplement, Proposition S1.3
P_4 at finite N	The end-arc of three states never wins, and its exit rate is only 0.0071 too large to win. Yet with a uniform start it is the exact mode for $T \in (9.381, 12.502)$ at $N = 40$, and on a window at every N tested up to 300.	supplement, Proposition S1.3, finite N numerical
Θ_ε	Triangles $\{1,2,3\}$ and $\{4,5,6\}$ joined by a link 3–4 of rate $\varepsilon < \varepsilon_1 = (64+\sqrt{46})/60$. The states $1, 2, 5, 6$, then $\{1,2\}$ and $\{5,6\}$ from 1, the triangles from $1 + \varepsilon/3 + O(\varepsilon^2)$, and $[6]$ from $9/\varepsilon + 2 + O(\varepsilon)$.	supplement, Proposition S1.4, with two exact resultants
paw	A triangle $\{1,2,3\}$ with a leaf 4 at 1. First $\{4\}$, then $\{1,4\}$ from $1+\sqrt{2}$, $\{1,2,3\}$ from $\sqrt{2}+\sqrt{3}$, and all four states from $2+\sqrt{3}$.	Lean only (computer-verified)
house	The square $1,2,3,4$ with a roof state 5 joined to $1,2$. First $\{3\}, \{4\}, \{5\}$, then $\{3,4\}$ from 1, $\{1,2,5\}$ from $1+\sqrt{2}$, and all five states from $2+\sqrt{2}$.	Lean only (computer-verified)
K_4 plus a vertex	A state 5 joined to $1,2$ of K_4. First $\{5\}$, then $\{1,2,3,4\}$ from $3(\sqrt{17}+1)/8 = 1.921$, and all five states from $(5+\sqrt{17})/4 = 2.281$.	Lean only (computer-verified)

Table 2: First-order winners on some graphs with unit rates and a start of full support (the link of Θ_ε has rate ε, and the second row allows any symmetric Q). The last column says where each row is proved (Remark 7.2).